

A 3D visualization of a quantum spin chain, showing a lattice of blue and purple spheres with arrows indicating spin states. Red wavy lines represent excitations or entanglement. Text overlays include "IBM Quantum" and "Slides? will also post on BSS and zlatko-minev.com/blog". Social media icons and handles are at the bottom.

Zlatko K. Minev
IBM Quantum

Slides? will also post on BSS
and zlatko-minev.com/blog

Where can you find things?

Lecture Slides

Boulder School for Condensed Matter and Materials Physics

boulderschool.yale.edu/2023/boulder-school-2023-lecture-notes



Will also post on zlatko-minev.com/education

Tutorials and additional lecture notes

Twirling, Measurements and Walsh-Hadamard

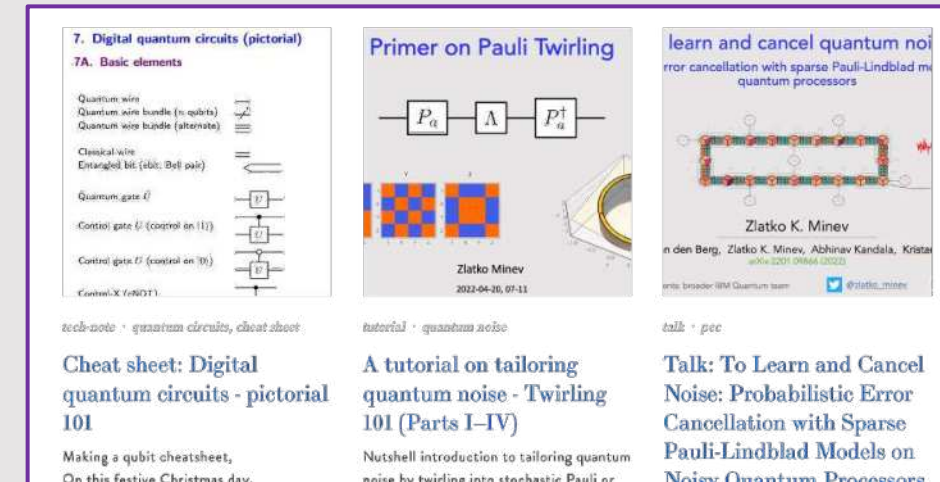
Cheat sheets, Videos, ...

zlatko-minev.com/blog

See also lectures on qiskit.org/learn

Tutorials and additional lecture notes

Latest seminar qiskit.org/events/seminar-series



What is one thing you
learned in Lecture 1?



Review of Lecture 1



Quantum Error Mitigation for Non-Equilibrium Quantum Dynamics

Lecture 1

Big picture

Why quantum computers?

Status and outlook

Why error mitigation?

Noise in quantum computers

Overview of error mitigation

Mitigation fundamentals

Probabilistic error cancelation (PEC)

Introduction

One qubit example



Quantum computers

Biggest Problem: Noise

Mitigation

My experience circa 2010

Maybe 1 or 2 qubits working some small fraction of the time in select labs

Photo with dilution fridge called Sunshine from Michel Devoret's lab at Yale during my Ph.D.

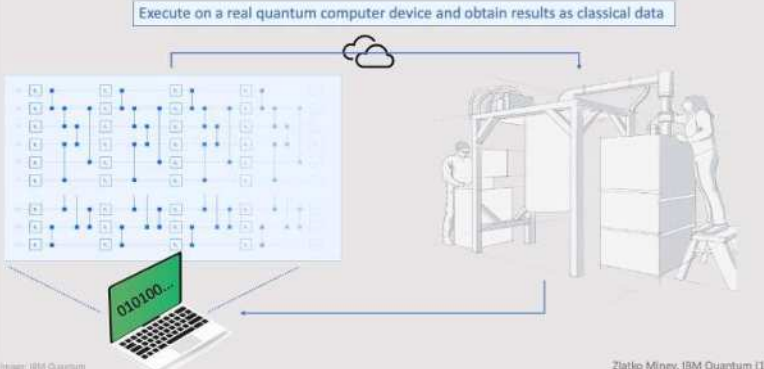


Hopes for a working qubit in here

Mineev, IBM Quantum [9]

Quantum simulation on a quantum computer

Execute on a real quantum computer device and obtain results as classical data



Images: IBM Quantum

Zlatko Mineev, IBM Quantum [17]

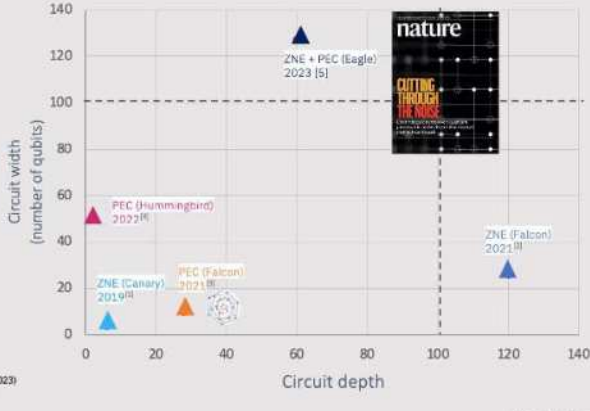
Overview of some key experimental progress in error mitigation:

Error mitigation

No matter what you do you have to chop it to this graph

PEC: Probabilistic error cancellation

ZNE: Zero-noise extrapolation



Technique	Circuit depth	Circuit width (number of qubits)
ZNE + PEC (Eagle)	~65	~125
PEC (Hummingbird)	~10	~55
ZNE (Canary)	~10	~15
PEC (Falcon)	~35	~25
ZNE (Falcon)	~120	~35

[1] Kandala, Nature (2019)
[2] Kim, Nature Phys. (2023)
[3] van den Berg, Mineev, Nature Phys. (2023)
[4] Tierney, IBM Research Blog (2022)
[5] Kim, Nature (2023)

IBM Quantum

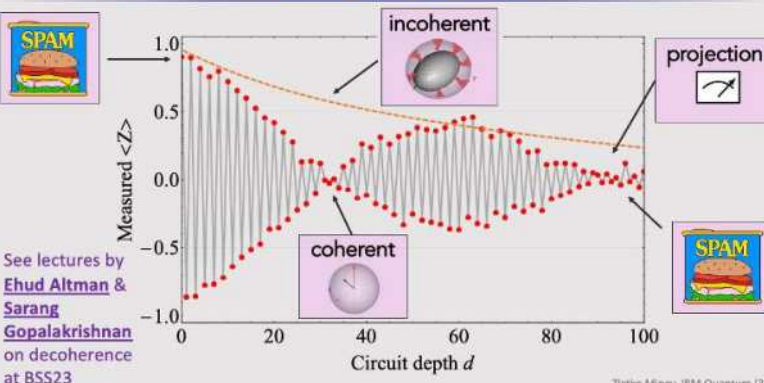
This year 2023

A 127-qubit quantum computer installed in the lobby cafeteria of a research building dutifully executing jobs almost all the time.

at Cleveland Clinic



Elements of noise



See lectures by [Ehud Altman & Sarang Gopalakrishnan](#) on decoherence at BSS23

Zlatko Mineev, IBM Quantum [33]

High-level message

Learn

accurate, efficient, scalable



Cancel

noise with noise, practical



Cost

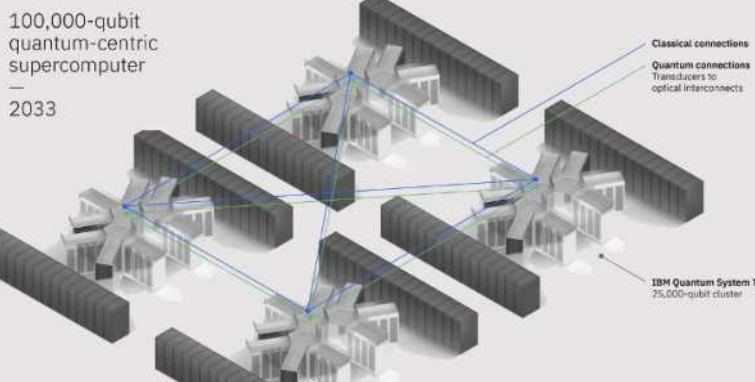
more noise more cost



IBM Quantum [48]

100,000-qubit quantum-centric supercomputer

2033

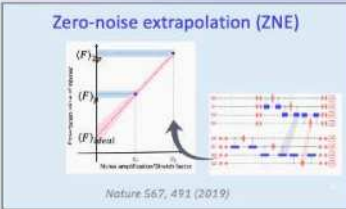


Classical connections
Quantum connections
Transducers to optical interconnects

IBM Quantum System Two 75,000-qubit cluster

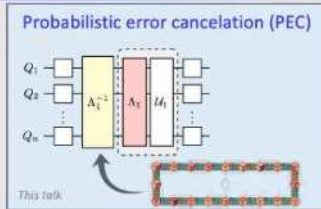
Error mitigation landscape

Zero-noise extrapolation (ZNE)



Nature 567, 491 (2019)

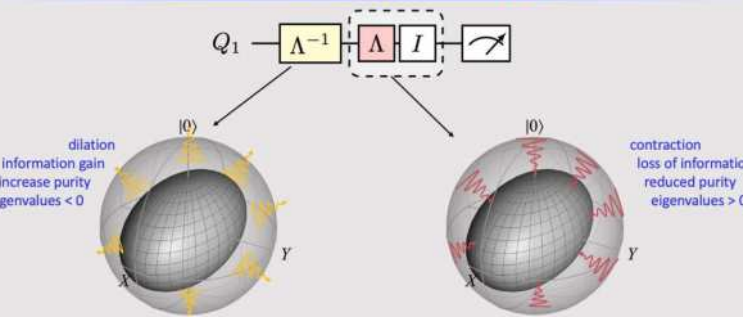
Probabilistic error cancellation (PEC)



This talk

more speed → more information, accuracy

Inverse of noise map is not physical



dilation
information gain
increase purity
eigenvalues < 0

contraction
loss of information
reduced purity
eigenvalues > 0

Quantum Error Mitigation for Non-Equilibrium Quantum Dynamics

Lecture 2

Mitigation fundamentals

Probabilistic error cancelation (PEC)

Summarize one qubit example

Analogy to random walks

Error bars & confidence

Generalize (optional)

Show unbiased estimator



Learning quantum noise

Challenge

Overcoming: sparse model

Putting it together

Experiments – Ising model

Consequences for the big picture

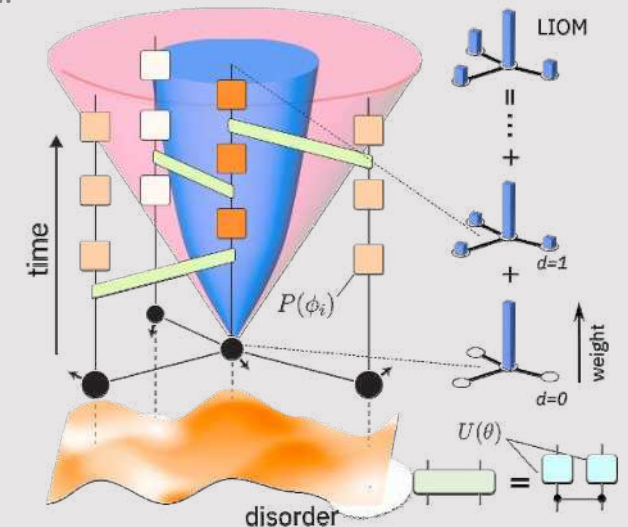
Hardware progress

Lecture 3

Wrap up of Lecture 2, Twirling, ...

State-of-art experiments at the 120Q+, depth 50+: uncovering local integrals of motion

...



Deep dive:
Probabilistic error cancellation (PEC)
To learn and cancel quantum noise



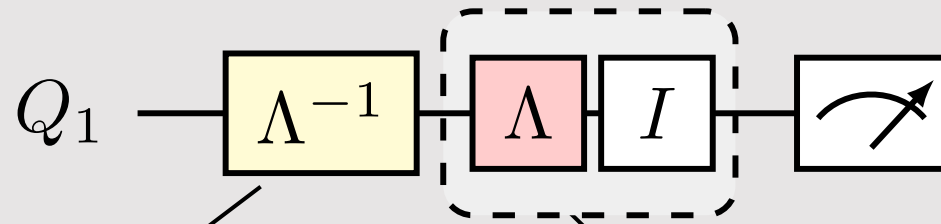
Got Slides?



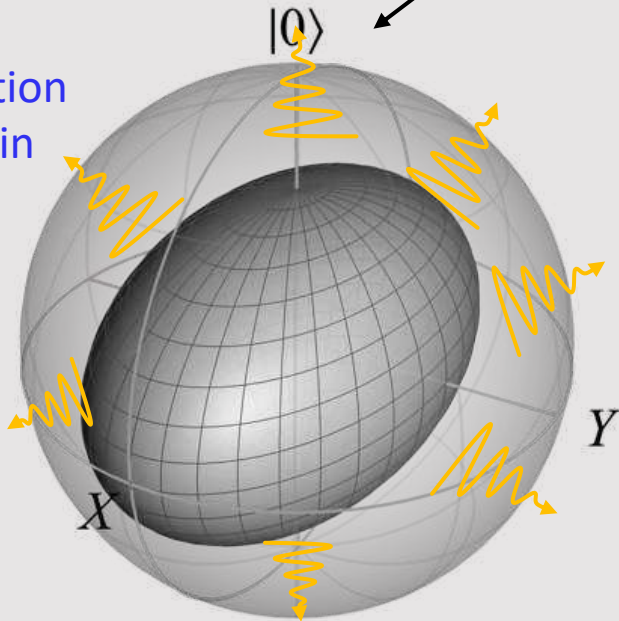
Paper: [Nature Physics \(2023\)](#)

Ewout van den Berg, Zlatko K. Mineev, Abhinav Kandala, Kristan Temme

Inverse of noise map is not physical

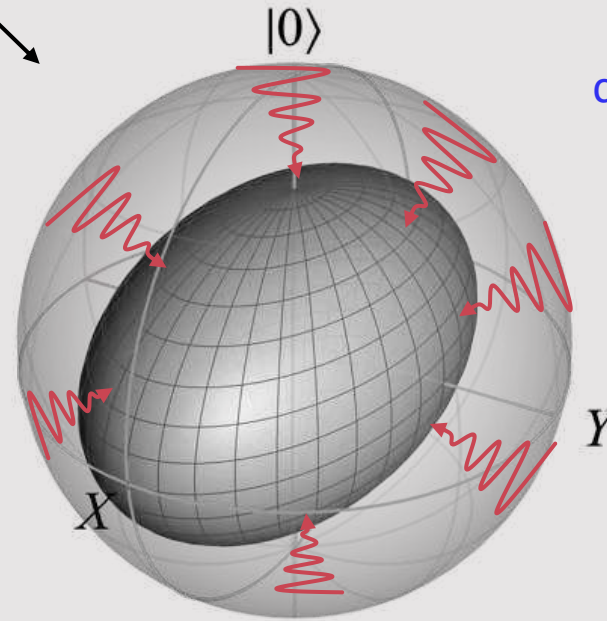


dilation
information gain
increase purity
eigenvalues < 0



$$\Lambda^{-1}(\rho) = ?$$

contraction
loss of information
reduced purity
eigenvalues > 0

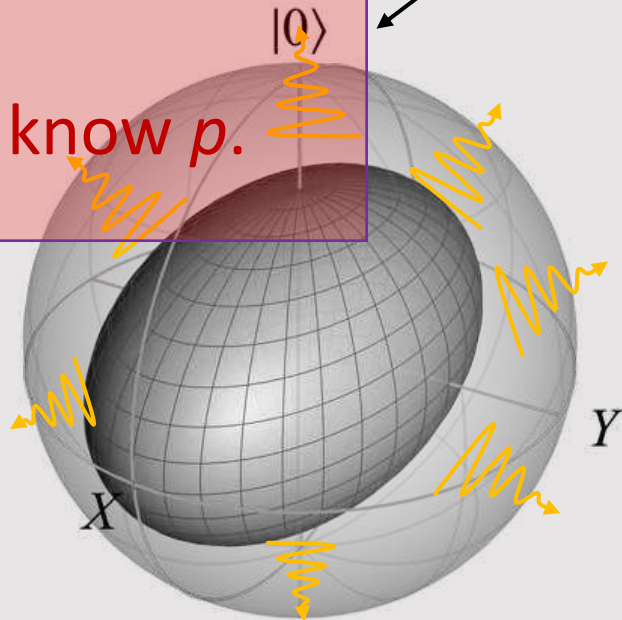


$$\Lambda(\rho)$$

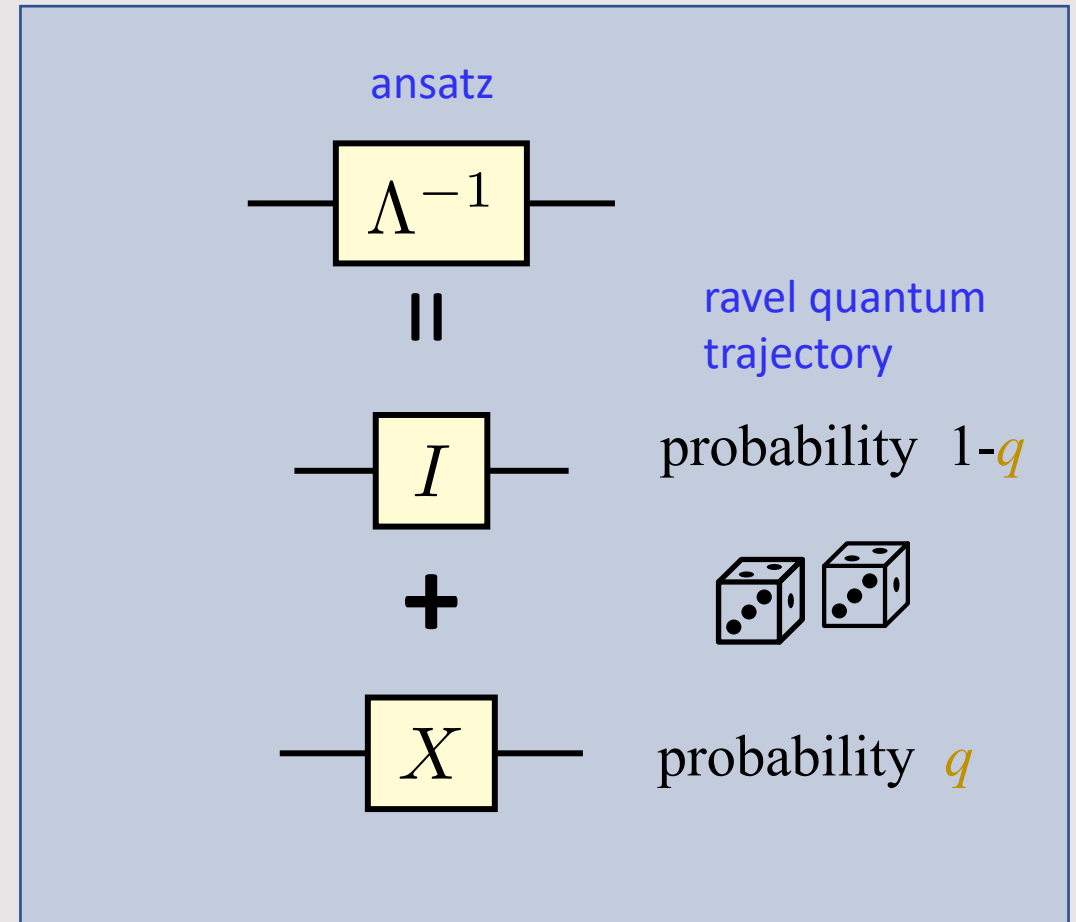
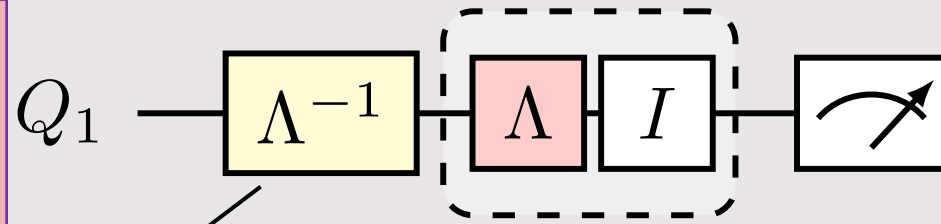
Inverse of noise map is not physical

Critically hinges on
knowing the noise
exactly!

Need to know p .



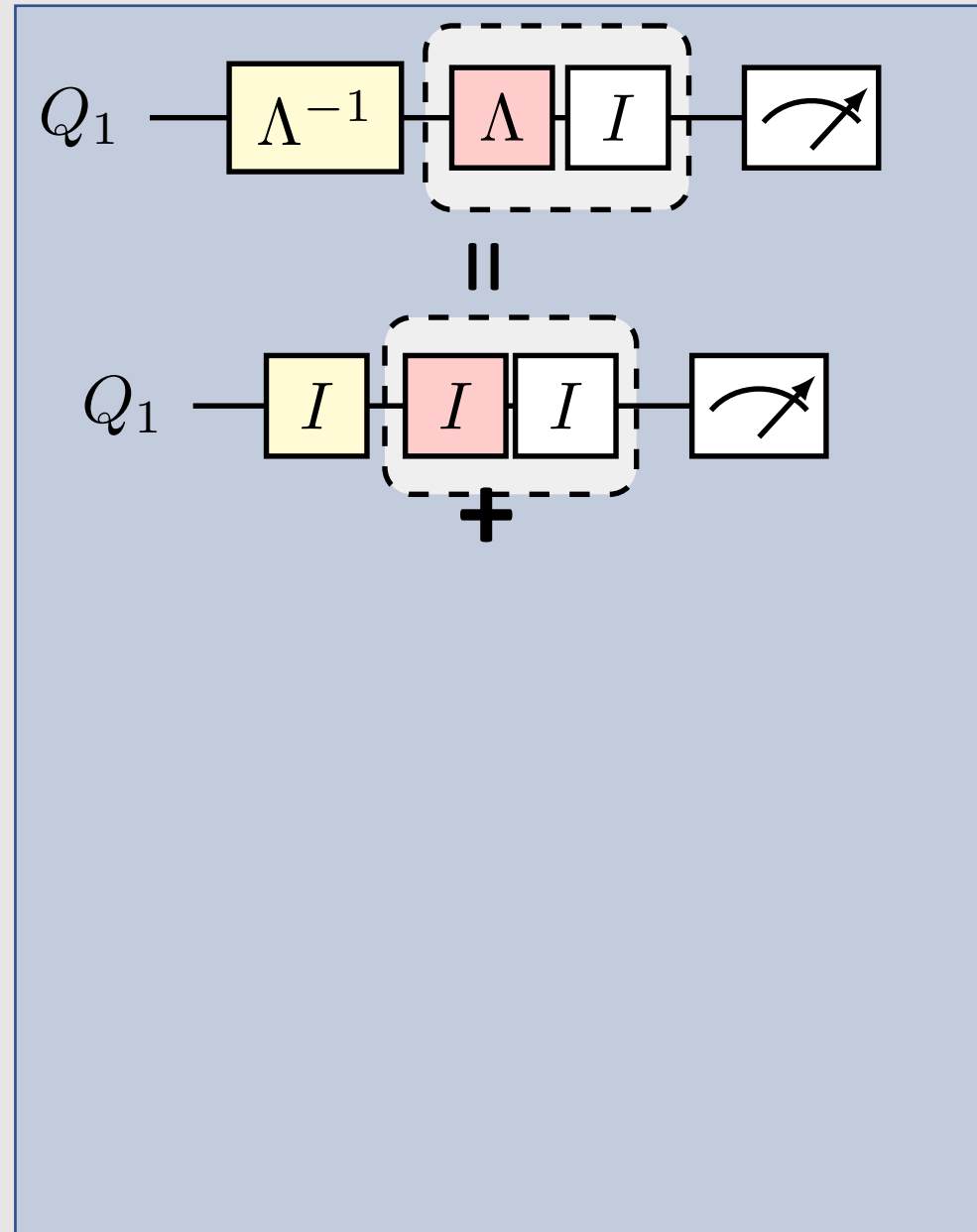
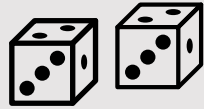
$$\Lambda^{-1}(\rho) = ?$$



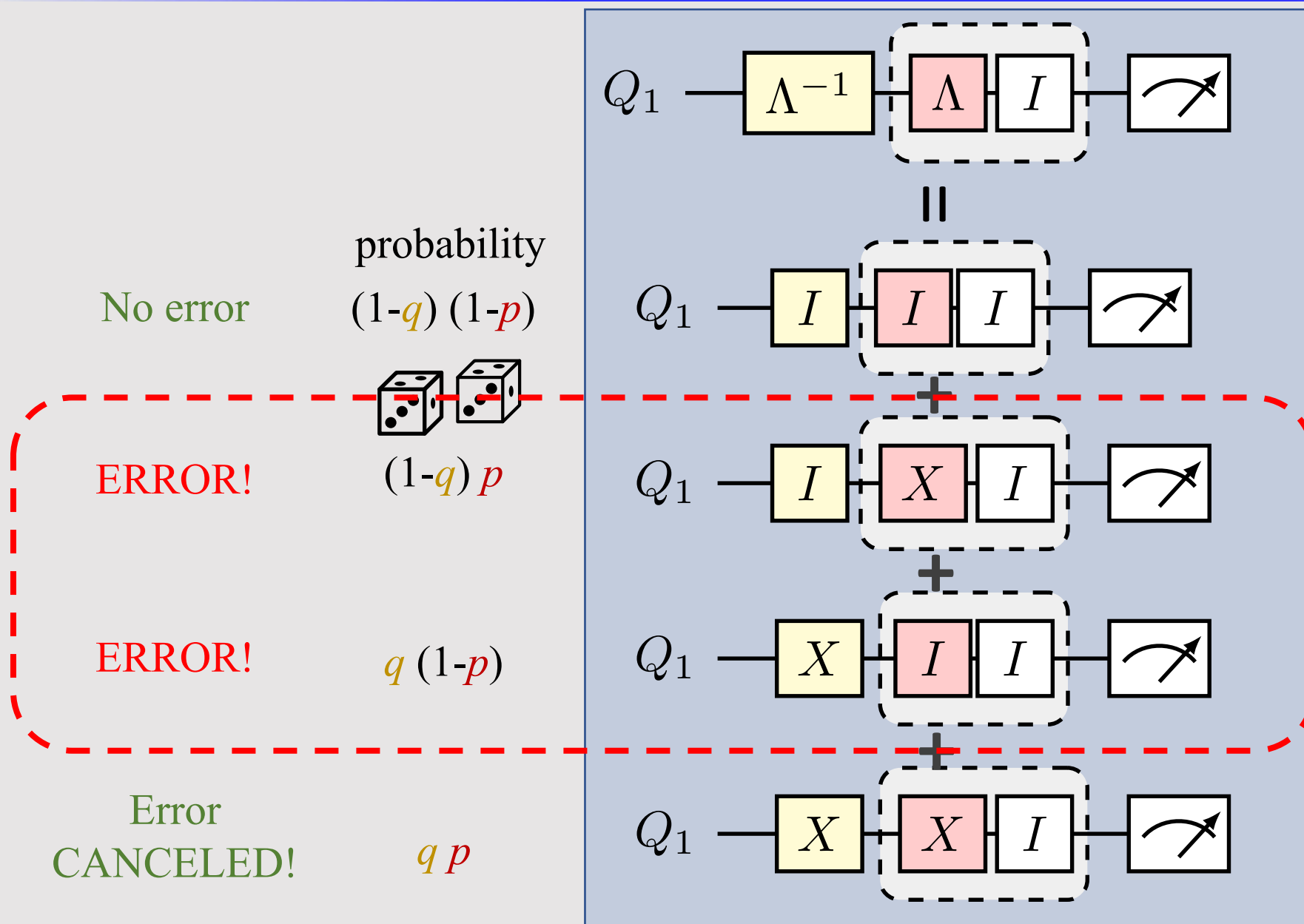
Raveling quantum trajectories to undo noise

No error

probability
 $(1-q)(1-p)$



Raveling quantum trajectories to undo noise

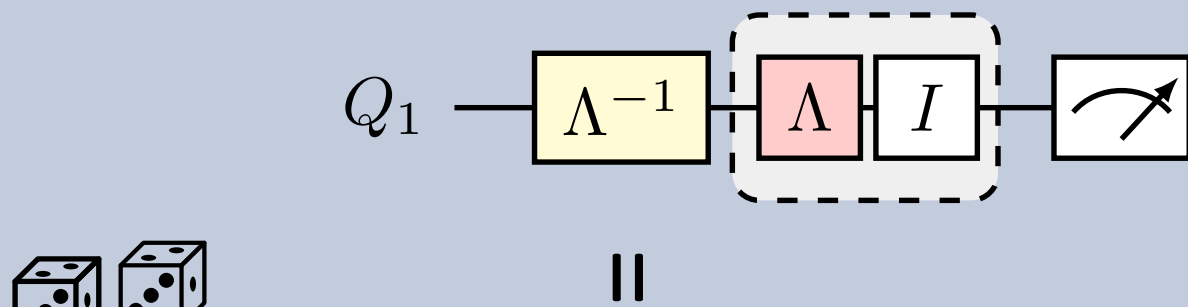


Solution to noise free!

$$q = \frac{-p}{1 - 2p}$$

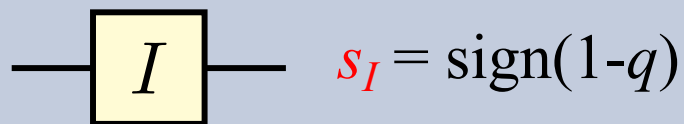
Interfere
Destructively

Raveling quantum trajectories to undo noise



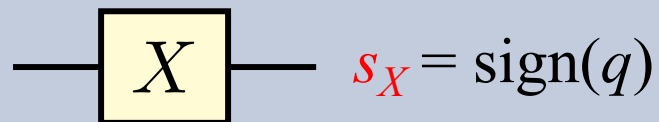
probability

$$P_I = |1 - q|/\gamma$$



+

$$P_X = |q|/\gamma$$



$$P_I + P_X = 1$$

Solution to noise free!

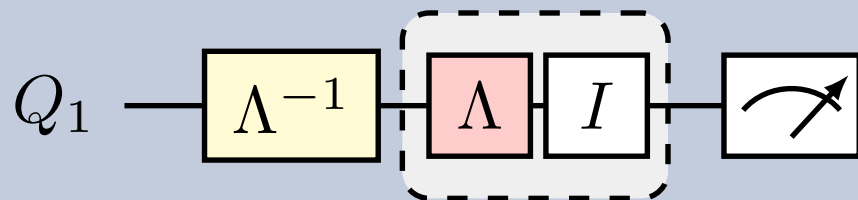
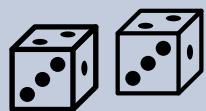
$$q = \frac{-p}{1 - 2p}$$

norm

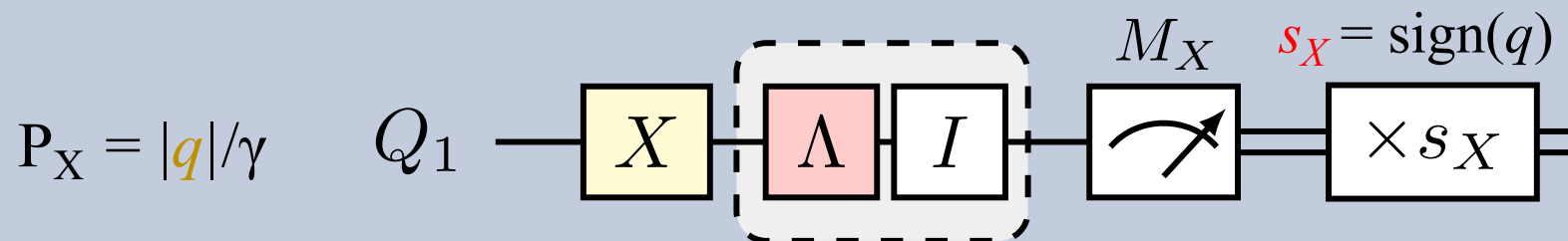
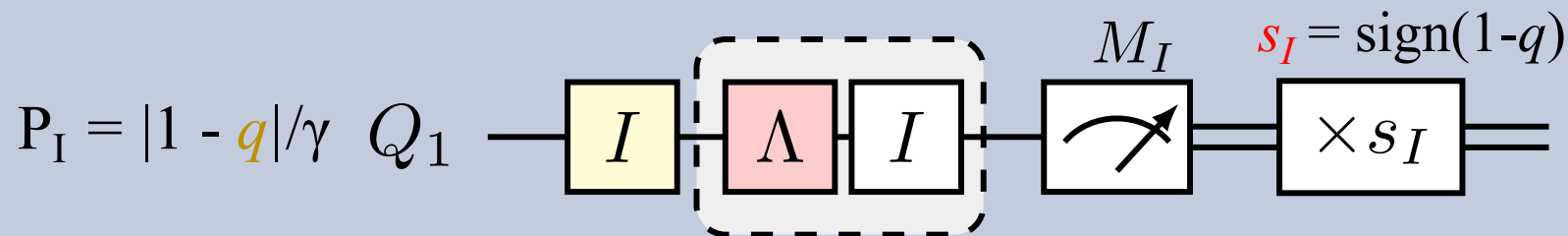
$$\gamma = |1 - q| + |q|$$

Quasi-probability

Raveling quantum trajectories to undo noise



||



mitigated expectation

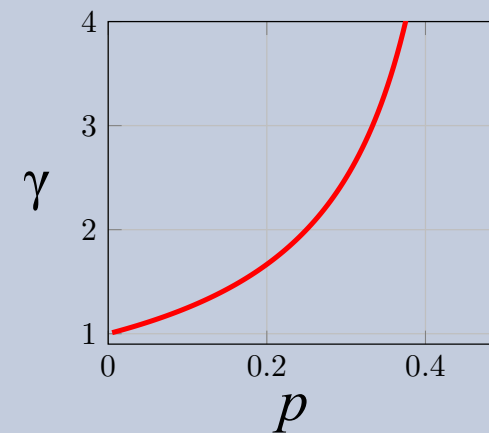
$$\langle M \rangle = \gamma (s_I P_I M_I + s_X P_X M_X)$$

Gain: Bias-free estimate!

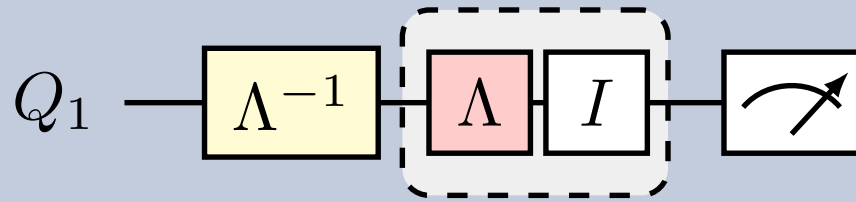
Cost: Variance

Sampling overhead

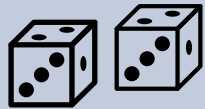
$$\gamma = |1 - q| + |q|$$



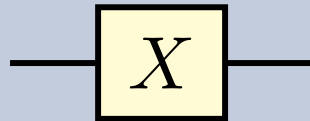
Canceling noise with noise



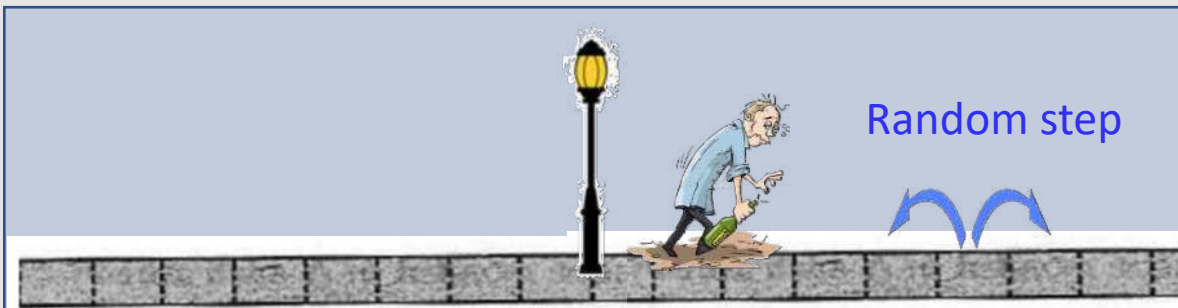
$=$



+



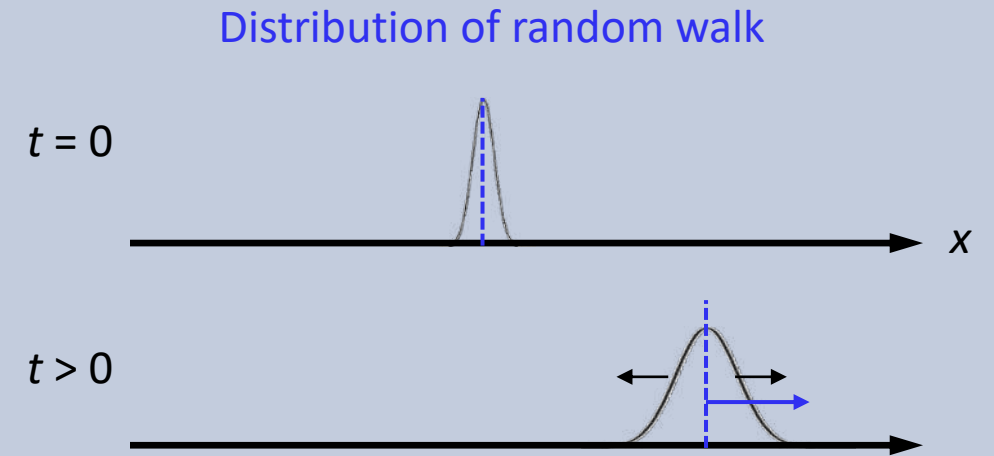
Canceling noise with noise: Drunkard's classical random walk analogy



Random step

$$P(1 \text{ step left}) = \frac{1}{2} - p$$

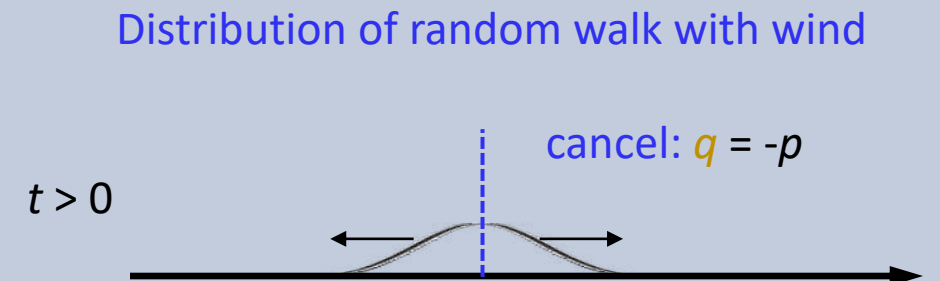
$$P(1 \text{ step right}) = \frac{1}{2} + p$$



add 2nd random process
wind blows

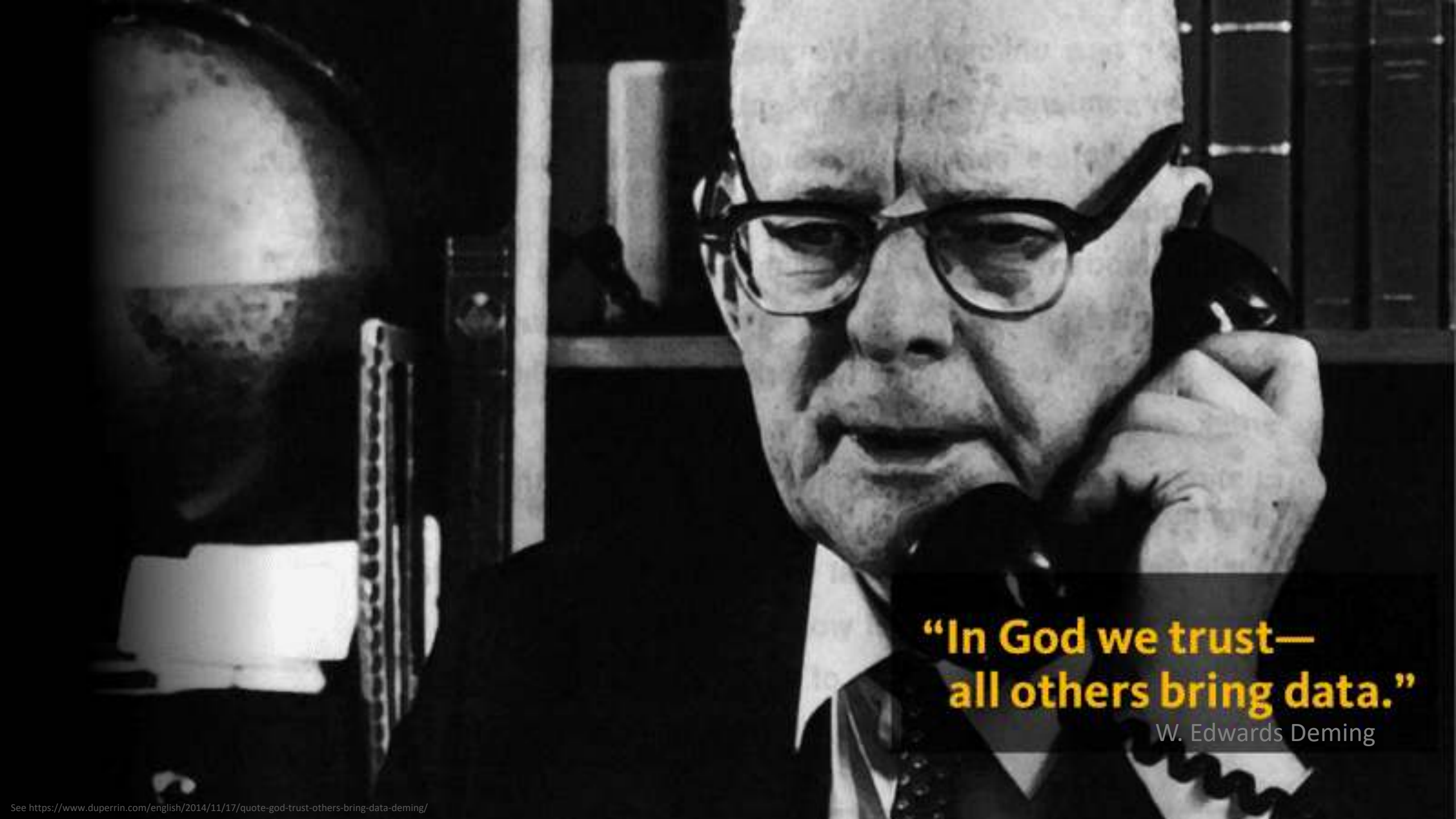
$$P(1 \text{ step left}) = \frac{1}{2} - q$$

$$P(1 \text{ step right}) = \frac{1}{2} + q$$



Gain: Bias-free estimate!

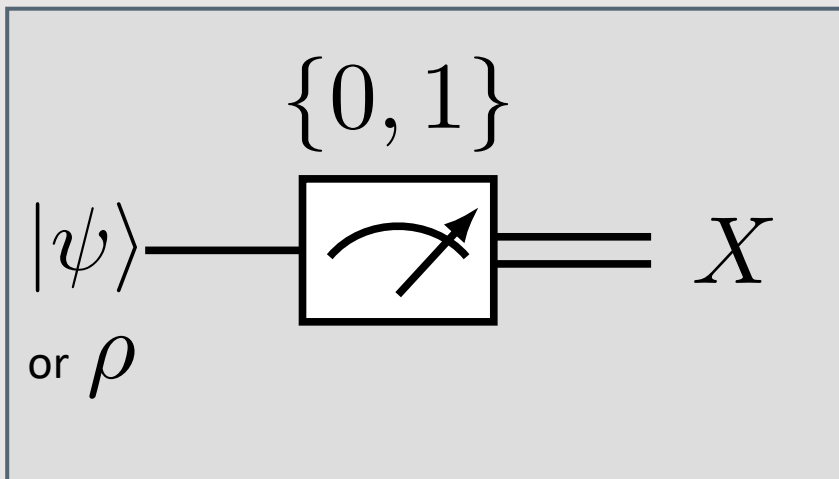
Cost: Variance

A black and white photograph of W. Edwards Deming, an older man with glasses, wearing a suit and tie, holding a rotary telephone receiver to his ear. He is looking slightly off-camera with a thoughtful expression. The background is dark and out of focus, showing what appears to be a bookshelf.

**“In God we trust—
all others bring data.”**

W. Edwards Deming

Summary: Qubit measured in the computational basis



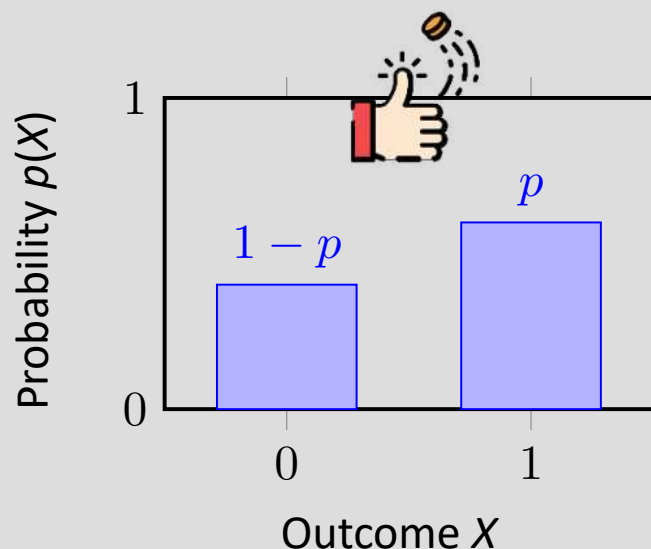
Set of possible outcomes

$$\Sigma = \{0, 1\}$$

$$X \in \Sigma$$

Measurement operators

$$\begin{aligned} x & \quad \hat{\mu}(x) \\ 0 : & \quad \hat{\mu}(0) = |0\rangle \langle 0| \\ 1 : & \quad \hat{\mu}(1) = |1\rangle \langle 1| \end{aligned}$$



Probability to measure outcome

$$\begin{cases} X = 0 : & p(X = 0) = \text{Tr}(\hat{\mu}(0)^\dagger \rho) = \text{Tr}(|0\rangle \langle 0| \rho) = \frac{1}{2} (1 + \langle Z \rangle) \\ X = 1 : & p(X = 1) = \text{Tr}(\hat{\mu}(1)^\dagger \rho) = \text{Tr}(|1\rangle \langle 1| \rho) = \frac{1}{2} (1 - \langle Z \rangle) \end{cases}$$

Bernoulli distribution. Single shot outcome follows a Bernoulli distribution:

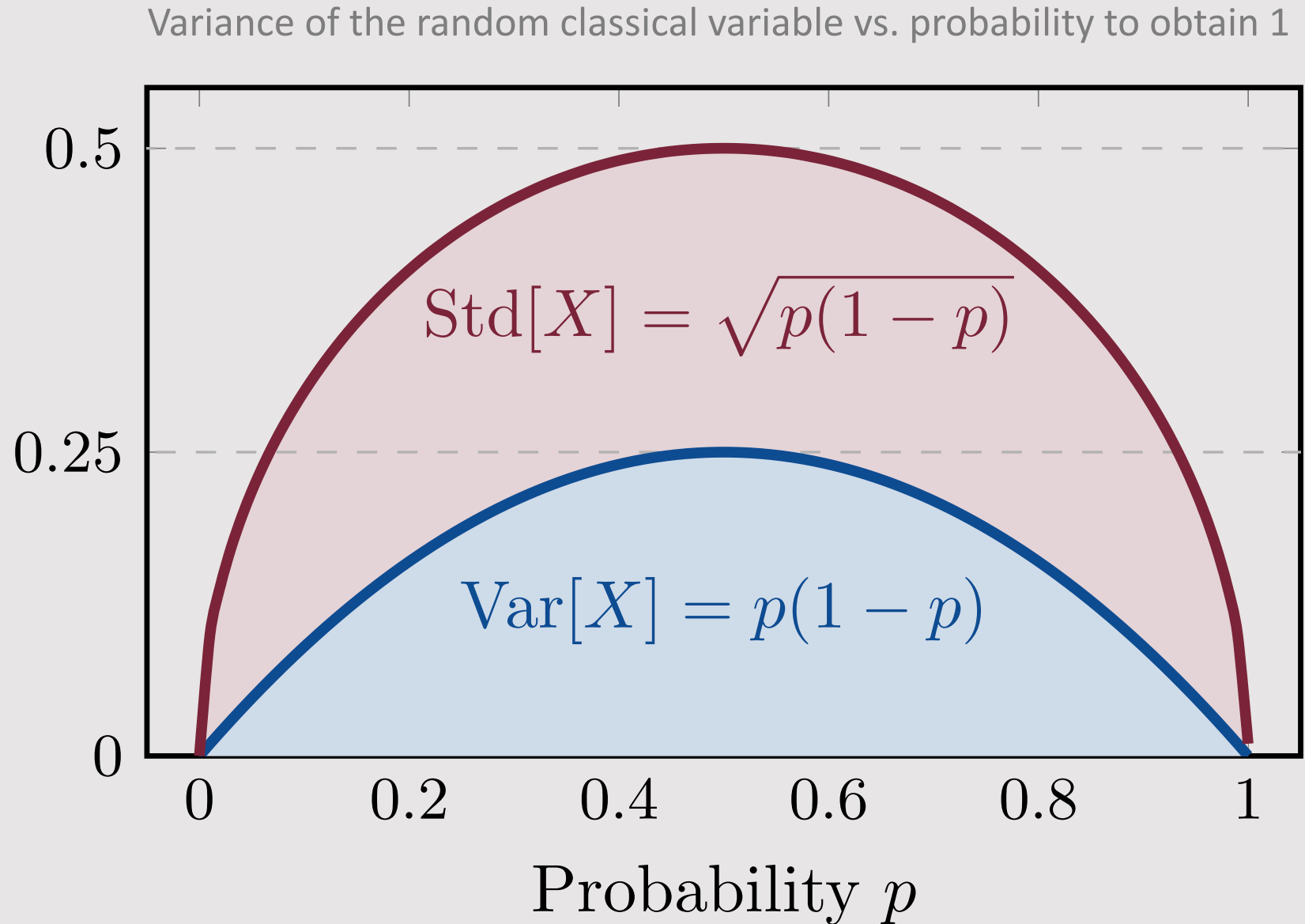
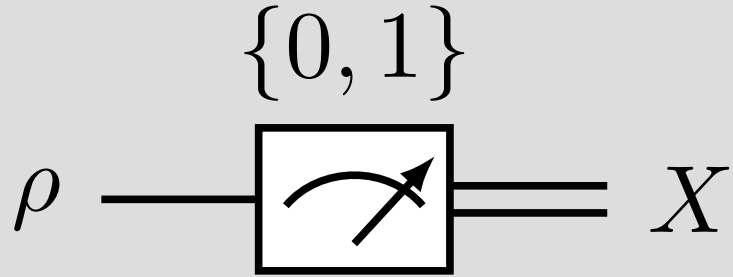
$$X \sim \text{Bernoulli}(p)$$

$$p := \text{Tr}(|1\rangle \langle 1| \rho) \in [0, 1]$$

$$\mathbb{E}[X] = p$$

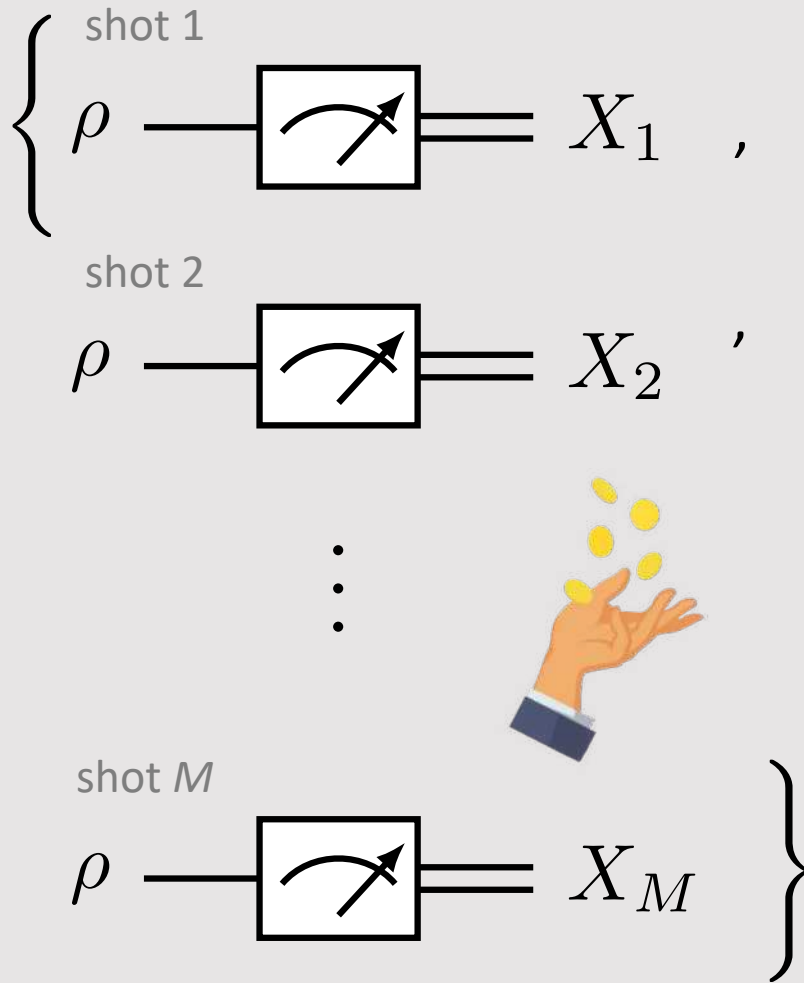
$$\text{Var}[X] = p(1 - p)$$

Quantum projection noise for a single shot





Ideal single qubit measurement with M shots



M shots with IID distribution

M outcomes: independent and identically distributed (iid) random variables

$$X_1, X_2, \dots, X_M \in \Sigma$$

$$X_1, X_2, \dots, X_M \sim \Pr[X = x] = \langle \hat{\mu}(x) \rangle$$

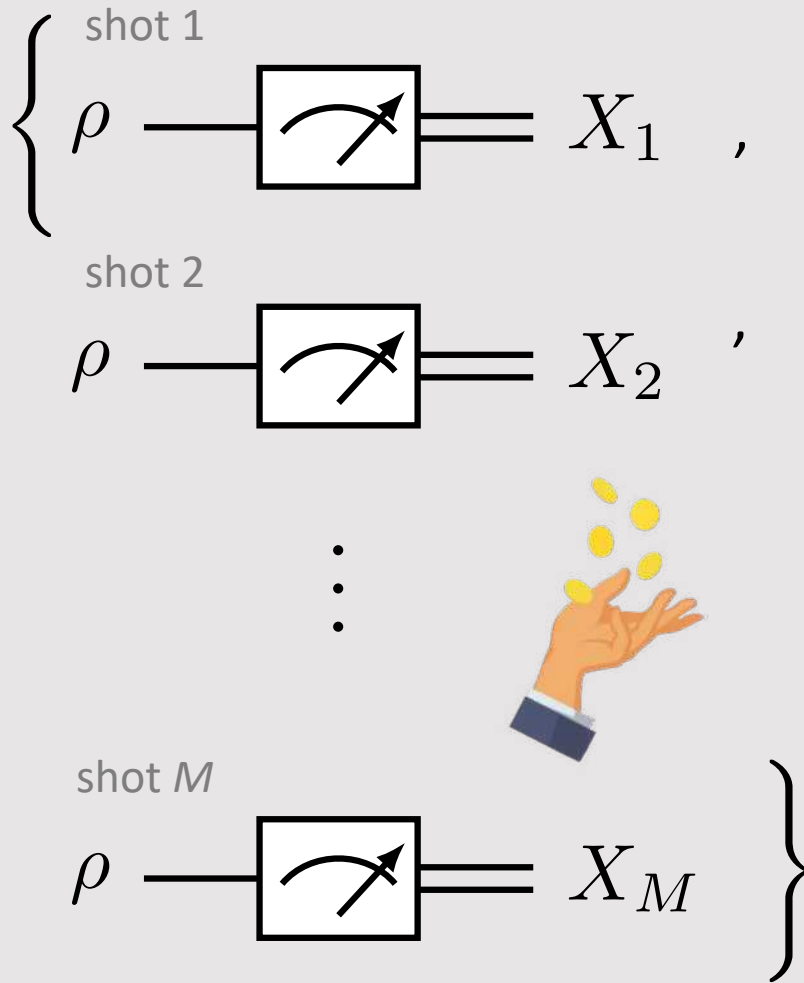
Empirical mean random variable (sample mean statistic)

$$S = \frac{1}{M} \sum_{m=1}^M X_m$$

Find the expectation value and variance of the empirical mean



Empirical mean: an unbiased estimator



$$S = \frac{1}{M} \sum_{m=1}^M X_m$$

Find the expectation value and variance of the empirical mean



$$\mathbb{E}[S] = \mathbb{E}\left[\frac{1}{M} \sum_{m=1}^M X_m\right]$$

$$\mathbb{E}[X_m] = \mathbb{E}[X] = p \quad \forall m \in \{1, \dots, M\}$$

$$\mathbb{E}[aX_m + bX_n] = a\mathbb{E}[X_m] + b\mathbb{E}[X_n]$$

$$\forall m, n, \quad a, b \in \mathbb{C}$$

$$= \frac{1}{M} \sum_{m=1}^M \mathbb{E}[X]$$

linear functional

$$= \mathbb{E}[X]$$

expectation value of a single shot

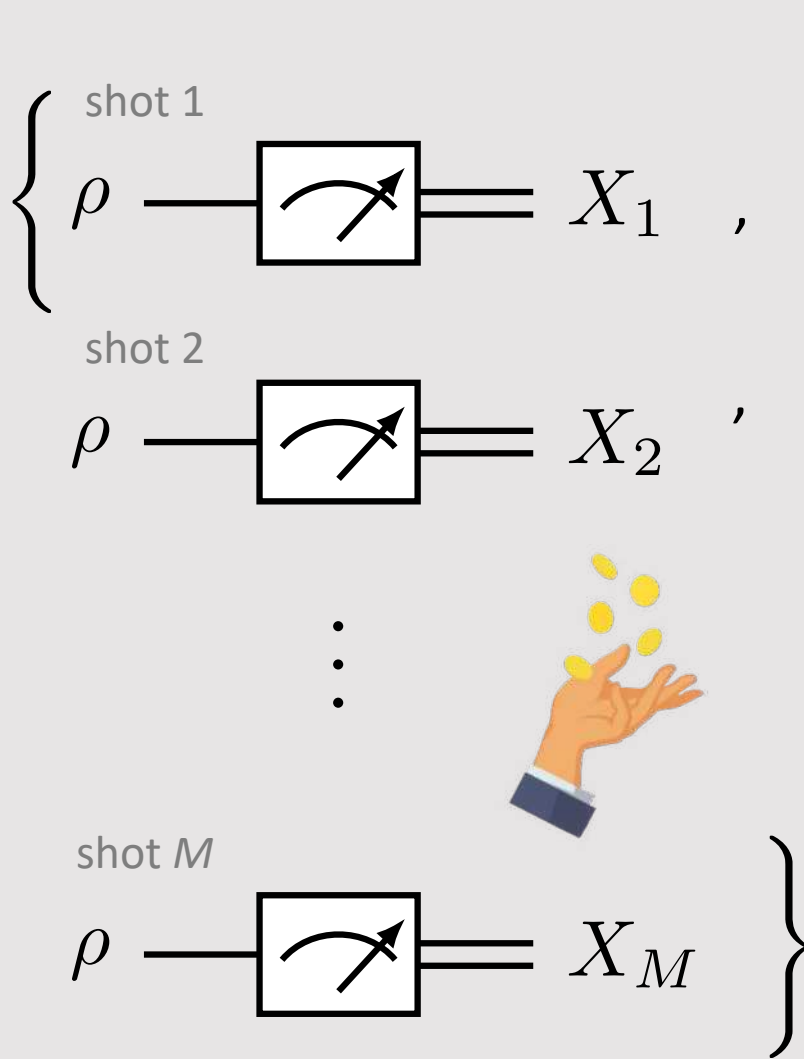
$$= \langle \hat{M} \rangle$$

relation to quantum operator we derived earlier (unbiased estimator)

$$= p$$

relation to probability to measure 1

How noisy is our estimate of the empirical mean?



$$S = \frac{1}{M} \sum_{m=1}^M X_m$$

$$\begin{aligned} \mathbb{V}[S] &= \mathbb{V}\left[\frac{1}{M} \sum_{m=1}^M X_m\right] \\ &= \frac{1}{M^2} \sum_{m=1}^M \mathbb{V}[X] \end{aligned}$$

$$\begin{aligned} &= \frac{1}{M} \mathbb{V}[X] \\ &= \frac{p(1-p)}{M} \end{aligned}$$

$$\begin{aligned} \sigma_S &= \sqrt{\mathbb{V}[S]} \\ &= \sqrt{\frac{\mathbb{V}[X]}{M}} \\ &= \sqrt{\frac{p(1-p)}{M}} \end{aligned}$$

Find the expectation value and variance of the empirical mean



Use key identity for variance

$$\mathbb{V}[aX_m + bX_n] = a^2 \mathbb{V}[X_m] + b^2 \mathbb{V}[X_n]$$

(you can derive this from the definition)

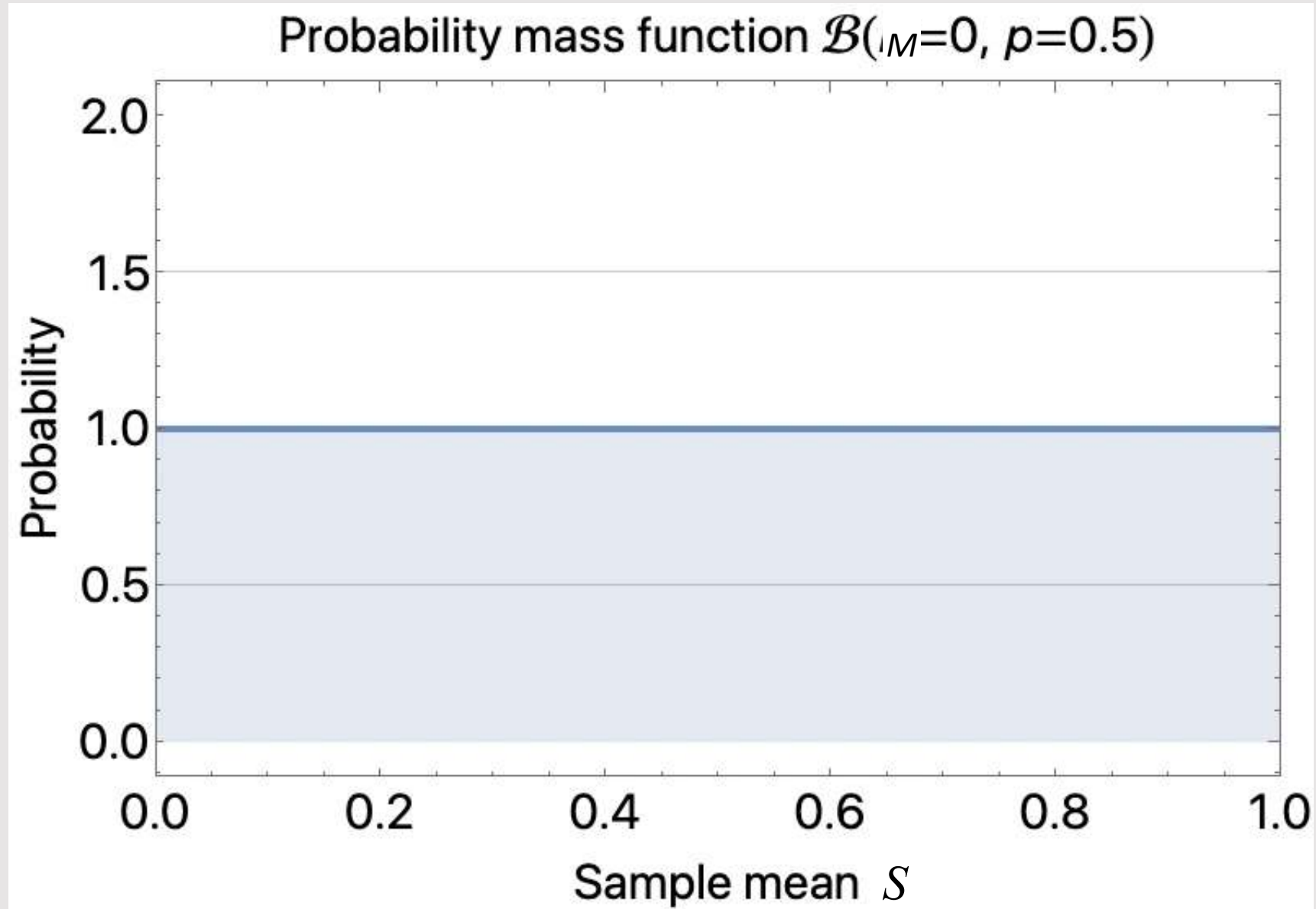
$$\mathbb{V}[X_m] = \mathbb{V}[X] = p(1-p) \quad \forall m \in \{1, \dots, M\}$$

The variance is reduced by the number of sampler we take!

Thus we can suppress the quantum projection noise with enough shots.

The standard deviation drop as one over square root of the number of shots

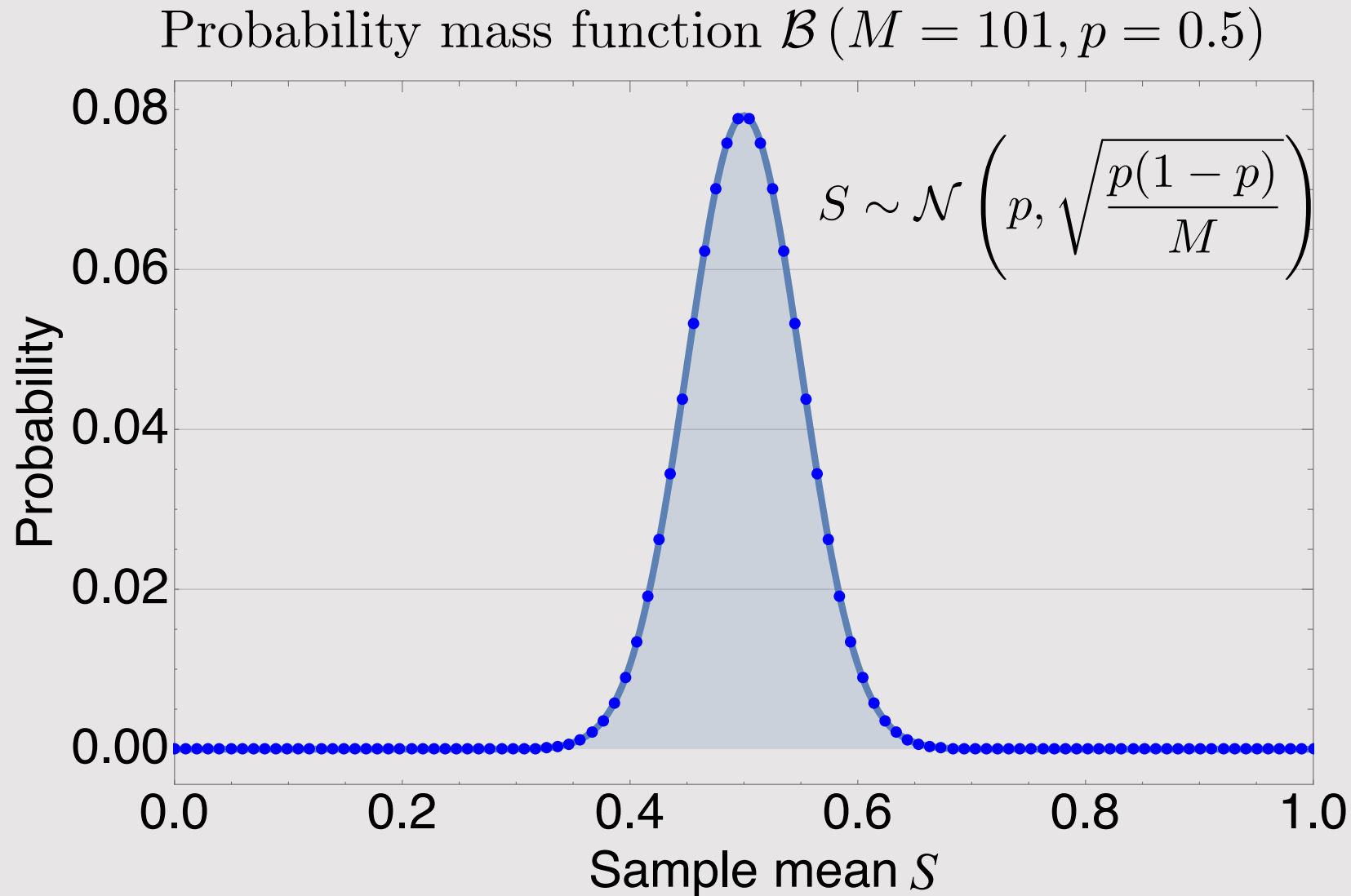
Animation of convergence of shots expectation value and mean



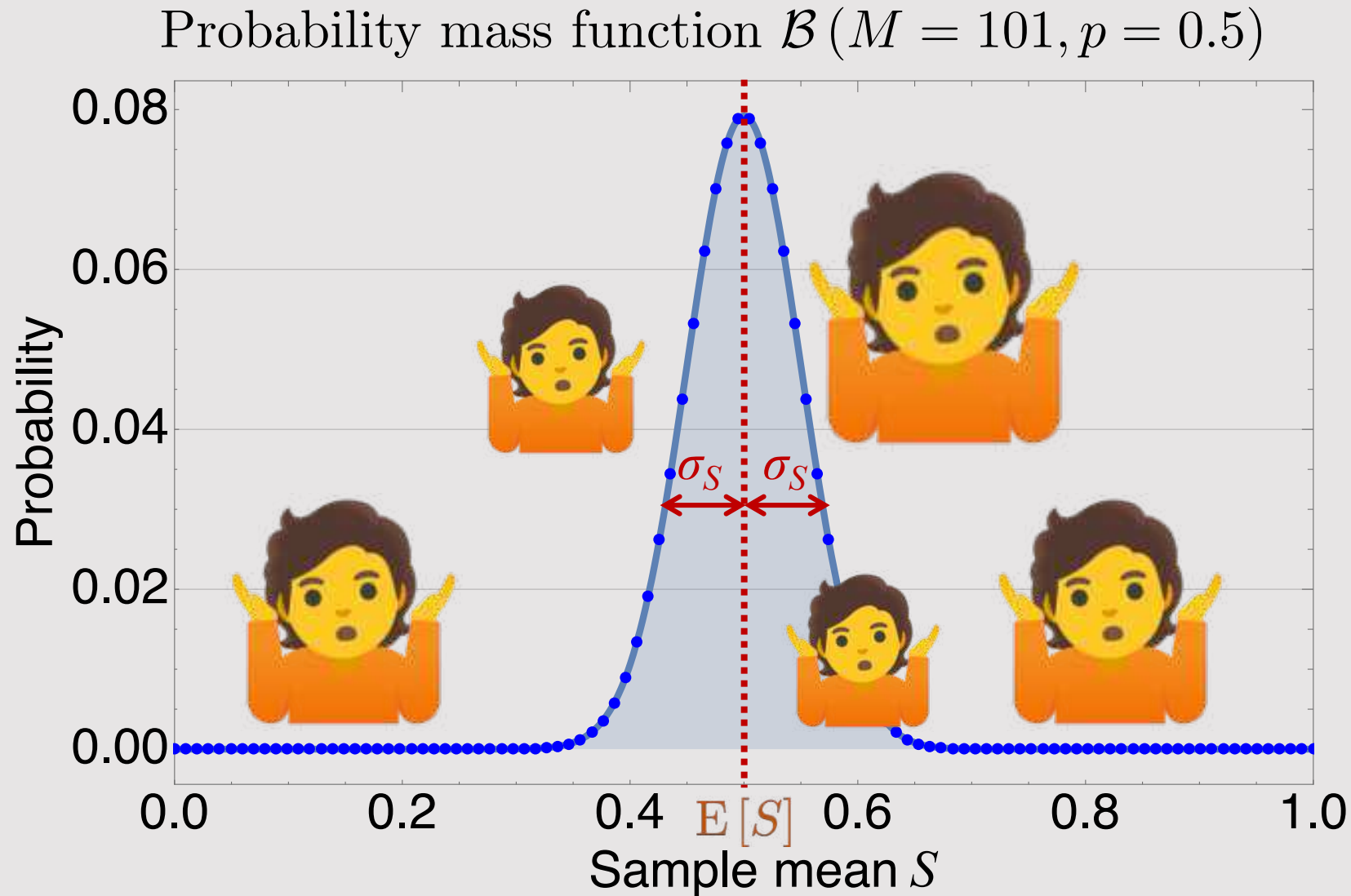
Binomial distribution

$$\binom{M}{k} p^k (1-p)^{M-k}$$

Sampled output distribution



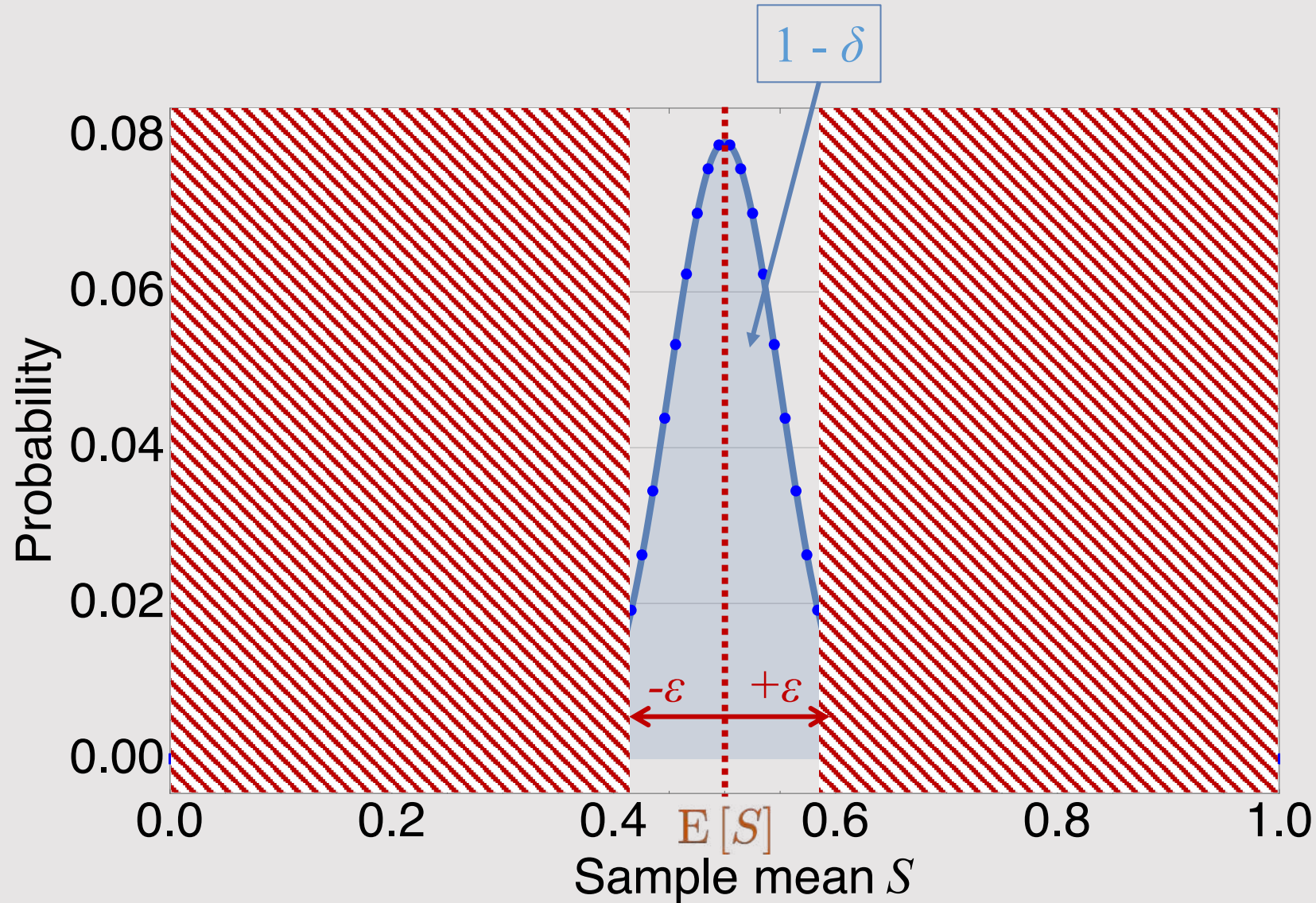
Properties of the output distribution



$$E[S] = p$$

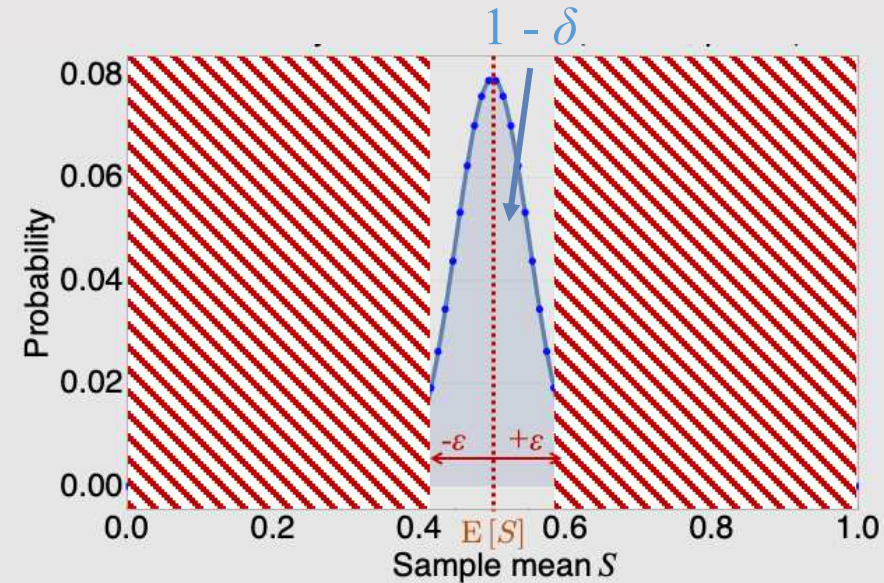
$$\begin{aligned}\sigma_S &= \sqrt{\text{Var}[S]} \\ &= \sqrt{\frac{p(1-p)}{M}}\end{aligned}$$

Concentration Measure for Sampling Expectation Values



Error Bound on Quantum Expectation Values

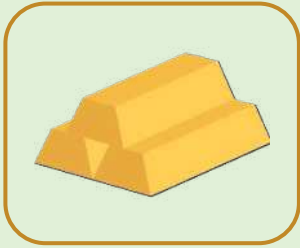
Concentration Measure for Sampling Expectation Values



Chernoff-Hoeffding two-sided tail bound, given $|O(x)| \leq 1$ for ϵ specified (additive) precision (worst case additive error) for S with success probability at least $1 - \delta$.

$$\Pr \left[\left| S - \langle \hat{O} \rangle \right| > \epsilon \right] \leq \delta := 2 \exp \left(-\frac{1}{2} M \epsilon^2 \right)$$

↑ δ specific failure probability for meeting precision ϵ empirically.



Empirical mean & sample properties

$$S = \frac{1}{M} \sum_{m=1}^M O(X_m)$$

Tf. required number of shots is at least [or with high probability (greater than 2/3)]

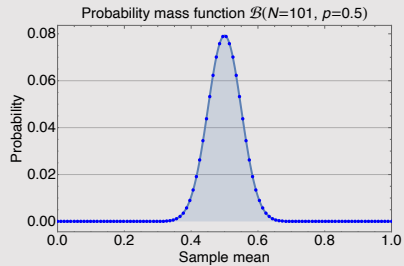
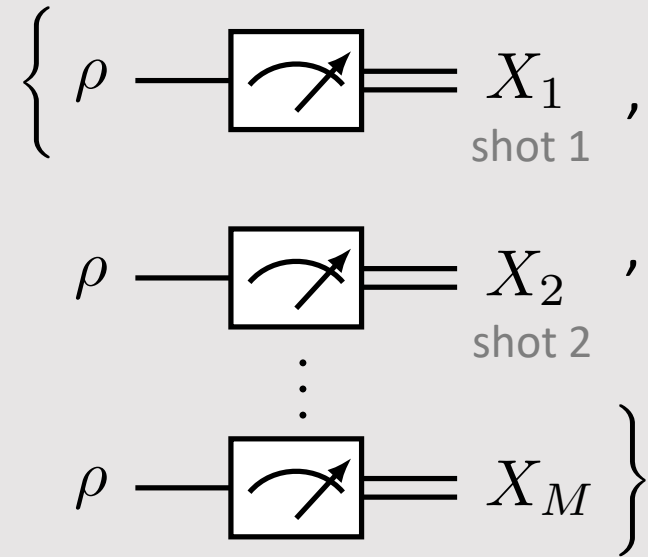
$$M \geq 2\epsilon^{-2} \log(2\delta^{-1}) \quad \left[M \gtrapprox 4\epsilon^{-2} \right] .$$

* Can find even tighter bound here owing to smaller $[0,1]$ range

Note that this scales same way (mod δ) as the variance bound with $\epsilon = \sigma$:

$$M \geq \frac{1}{4} \epsilon^{-2}$$

Ideal single qubit measurement with M shots



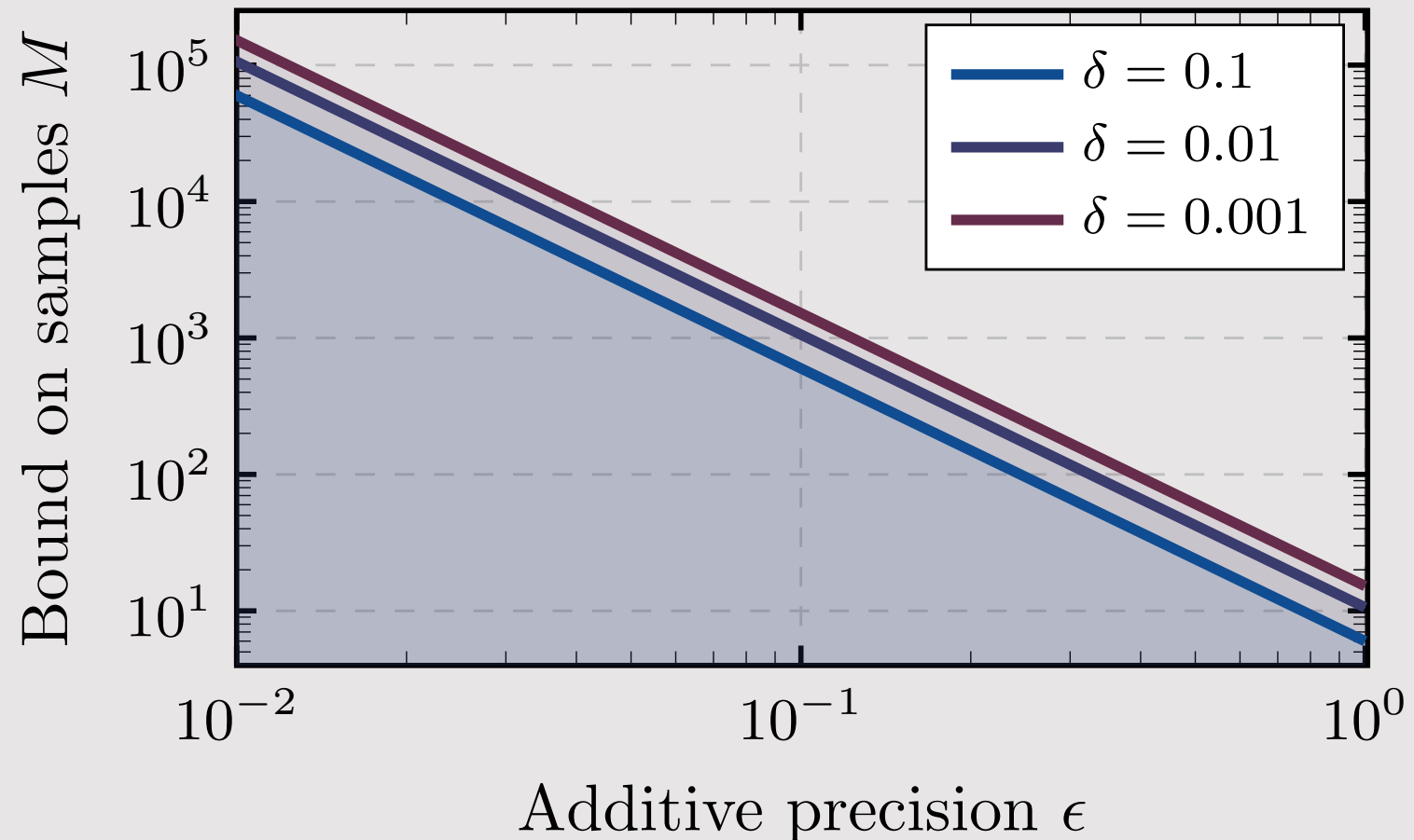
Observe that the **probability** δ is much cheaper than the **precision** ϵ .

Observe, n is not part of the equation.

“Knowing you are *not* wrong is cheaper than knowing you are right.” - Derek

Tail bound on sample complexity

$$M \geq 2\epsilon^{-2} \log(2\delta^{-1}) \quad [M \gtrsim 4\epsilon^{-2}]$$



Concentration inequalities and tail bounds



*Making a list,
checking it twice,
going to see
which inequality
is nice!*

*Markov? Hoeffding?
Jensen? Chebyshev?
Chernoff?*

<https://www.zlatko-minev.com/blog/inequalities>

1. Probability (Technical note 11.9 v0.6)

1A. Concentration inequalities and tail bounds

Unless otherwise specified, all variables are real $\mathbb{R}$. Inequalities come as one-sided $\Pr(\dots \leq \dots)$ and two-sided $\Pr(|\dots| \leq \dots)$. Notation: X is a random variable, $\mu := \mathbb{E}[X]$, $\sigma^2 := \text{Var}[X]$, $S_n := X_1 + \dots + X_n$.

Inequality	Conditions	Common form	Notes / Alternate form
Single random variable			
Markov ¹	Non-negative $X \geq 0$	$\Pr[X \geq a] \leq \frac{\mathbb{E}[X]}{a}$	$\forall a > 0$ $\Pr[X \geq k\mathbb{E}[X]] \leq \frac{1}{k}$ $k > 1$ [3, Sec. 5.1][6, Thm 1.13]
extension	+ non-negative, strictly increasing func Φ $\Phi(X) \geq 0$ increasing	$\Pr[X \geq a] = \Pr[\Phi(X) \geq \Phi(a)] \leq \frac{\mathbb{E}[\Phi(X)]}{\Phi(a)}$	$\forall a > 0$ Wiki
Reverse Markov	upper-bounded by U (can be positive) $\max X = U$	$\Pr[X \leq a] \leq \frac{U - \mathbb{E}[X]}{U - a}$	$\forall a > 0$ [1, Sec. 3.1]
Chebyshev ²	Finite mean and variance $\mathbb{E}[X]$, $\text{Var}[X]$ finite	$\Pr[X - \mathbb{E}[X] \geq a] \leq \frac{\sigma^2}{a^2}$	$\Pr[X - \mathbb{E}[X] \geq a \cdot \sigma] \leq \frac{1}{a^2}$ $\forall a > 0, \sigma^2 = \text{Var}[X]$ [1, Sec. 3.2] [3, Sec. 5.1][2, Thm 18.11]
Cantelli	Improved Chebyshev (same; but one-sided)	$\Pr[X - \mathbb{E}[X] \geq a] \leq \frac{\sigma^2}{\sigma^2 + a^2}$	$\forall a > 0, \sigma^2 = \text{Var}[X]$ Wiki
Chernoff ³	Generic	$\Pr[X \geq a] = \Pr[e^{tX} \geq e^{ta}]$	$\forall t > 0, a \in \mathbb{R}$ [1, Sec. 3.3]
Jensen	$f: \mathbb{R} \rightarrow \mathbb{R}; f$ convex	$f(\mathbb{E}[X]) \leq \mathbb{E}[f(X)]$	[3, Prob. 5.3][6, Thm 1.14]
Hoeffding's lemma	$\mathbb{E}[X] = \mu$ $a \leq X \leq b$	$\mathbb{E}[e^{\lambda X}] \leq e^{\lambda \mu + \frac{\lambda^2 (b-a)^2}{8}}$	$\lambda \in \mathbb{R}$ [1, Sec. 3.4]
Sum of random variables			
Chernoff-Hoeffding (one-sided)	n independent random vars $X_1, \dots, X_n$ indep $S_n = X_1 + \dots + X_n$ $X_i \in [a_i, b_i] \forall i$	$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq \exp\left(\frac{-2t^2 n^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$	[1, Sec. 3.5]
(two-sided) ⁴	(same as above)	$\Pr[S_n - \mathbb{E}[S_n] \geq t] \leq 2 \exp\left(\frac{-2t^2}{\sum_{i=1}^n (b_i - a_i)^2}\right)$	$\forall t \in (0, \frac{1}{2})$ [5, Thm.1.1]
(two-sided iid)	same plus iid, range, mean μ for each $X_1, \dots, X_n \in [0, 1]$ $\mathbb{E}[X_i] = \mu$ iid	$\Pr\left[\left \frac{S_n}{n} - \mu\right \geq \epsilon\right] \leq 2 \exp(-2n\epsilon^2)$	$\forall \epsilon > 0$ [6, Thm 1.16]
Thm 1.3	n independent random vars $X_1, \dots, X_n$ indep $S_n = X_1 + \dots + X_n$	$\Pr[S_n - \mathbb{E}[S_n] > \epsilon] \leq 2 \exp\left(\frac{-\epsilon^2}{4 \sum_{i=1}^n \text{Var}[X_i]}\right)$	$\epsilon \in (0, 2 \text{Var}[S_n] / (\max_i X_i - \mathbb{E}[X_i]))$ [5, Thm. 1.3]
Azuma			
Weak law of large numbers	n independent iid random vars $X_1, \dots, X_n$ indep $\mathbb{E}[X_i] = \mu$ iid	$\lim_{n \rightarrow \infty} \Pr\left[\left \frac{1}{n} S_n - \mu\right \geq \epsilon\right] = 0$	$\forall \epsilon > 0$ [3, Sec. 5.2][6, Thm 1.15]
Strong law of large numbers	(same)	(same)	$\Pr\left[\lim_{n \rightarrow \infty} \frac{1}{n} S_n = \mu\right] = 1$ [3, Sec. 5.5]
Advanced			
Bennett	n independent zero-mean $X_1, \dots, X_n$ indep $\mathbb{E}[X_i] = 0$ iid	$\Pr[S_n > \epsilon] \leq \exp\left(-n\sigma^2 h\left(\frac{\epsilon}{n\sigma^2}\right)\right)$	$\sigma^2 := \frac{1}{n} \sum_{i=1}^n \text{Var}[X_i], \forall \epsilon > 0$, $h(a) := (1+a) \log(1+a) - a$ for $a \geq 0$ [1, 4.1]
Bernstein	(same)	(same)	(same) [1, 4.2]
Efron-Stein	scalar func of vars $f: \mathcal{X}^n \rightarrow \mathbb{R}$	$X_1, \dots, X_n$ indep w/ values in set $\mathcal{X}$	$\text{Var}[Z] \leq \sum_{i=1}^n \mathbb{E}[(Z - \mathbb{E}_i[Z])^2]$ $Z := g(X_1, \dots, X_n)$ $\mathbb{E}_i[Z] := \mathbb{E}[Z X_1, \dots, X_{i-1}, X_{i+1}, \dots, X_n]$ [1, 4.3]
McDiarmid's	scalar func of vars $f: \mathcal{X}^n \rightarrow \mathbb{R}$	$X_1, \dots, X_n$ indep w/ values in set $\mathcal{X}$	$\Pr[f(X_1, \dots, X_n) - \mathbb{E}[f(X_1, \dots, X_n)] \geq \epsilon] \leq \exp\left(\frac{-2\epsilon^2}{\sum_{i=1}^n c_i^2}\right)$ condition: c -bounded difference property $\forall \epsilon > 0$ $ f(X_1, \dots, X_i, \dots, X_n) - f(X_1, \dots, X'_i, \dots, X_n) \leq c_i$ [1, 4.4]

¹Markov's inequality bounds the first moment of random variable. Use it when a constant probability bound is sufficient [1, Sec. 3.3].

²Chebyshev is derived from Markov. It bounds the second moment. It is the appropriate one when the variance σ is known. If σ is unknown, we can use the bounds of $X \in [a, b]$.

³Chernoff bound is used to bound the tails of the distribution for a sum of independent random variables. By far the most useful tool in randomized algorithms [1, Sec. 3.3].

⁴This probability can be interpreted as the level of significance ϵ (probability of making an error) for a confidence interval around the mean of size 2ϵ . Therefore, we require at least $\log(2\alpha)/2\epsilon^2$ samples to acquire $1 - \alpha$ confidence interval $\mathbb{E}[X] \pm t$.

Scaling PEC to n qubits and larger circuits (ADVANCED – OPTINAL)

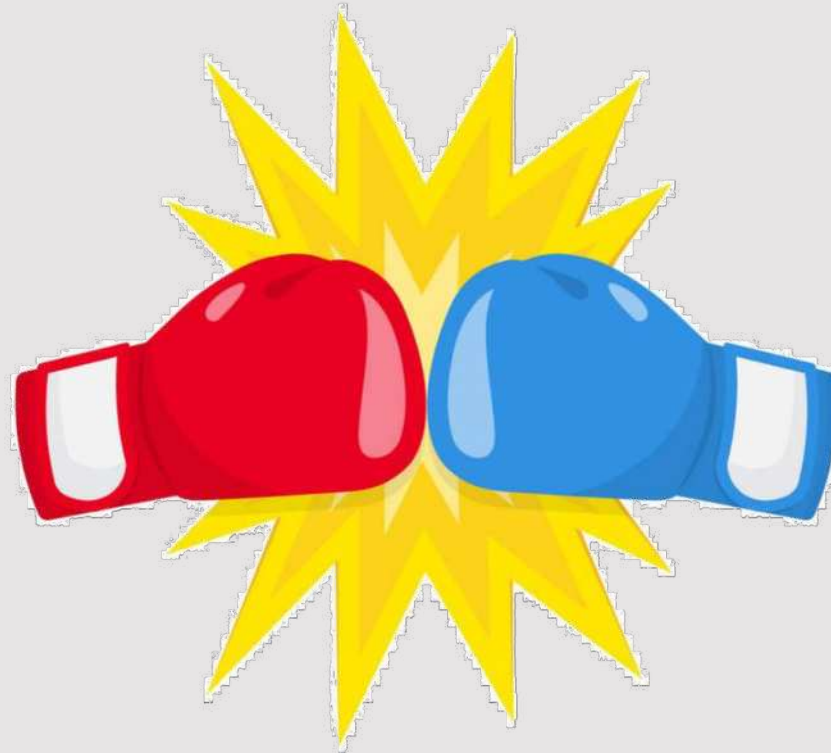


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Language of errors and error mitigation

$$\frac{d}{dx} \frac{1}{x} = -\frac{1}{x^2}$$

@formulas_for_your_comfort

Notation: Super bra ket

$$\rho \in L(\mathcal{H})$$

$$\hat{A}, \hat{B} \in L(\mathcal{H})$$

$$\mathcal{S} \in L(L(\mathcal{H}))$$

$$\begin{aligned} |\hat{A}\rangle\rangle &\leftrightarrow \hat{A} \\ \mathcal{S}|\hat{A}\rangle\rangle &\leftrightarrow \mathcal{S}(\hat{A}) \end{aligned}$$

Vectorization isomorphism

$$\begin{aligned} \langle\langle \hat{A} | \hat{B} \rangle\rangle &\leftrightarrow \langle \hat{A}, \hat{B} \rangle \\ &= \text{Tr}(\hat{B} \hat{A}^\dagger) \end{aligned}$$

$$\begin{aligned} \langle\langle \hat{A} | \mathcal{S} | \hat{B} \rangle\rangle &= \langle\langle \hat{A} | \mathcal{S}(\hat{B}) \rangle\rangle \\ &= \langle \hat{A}, \mathcal{S}(\hat{B}) \rangle \\ &= \text{Tr}(\hat{A}^\dagger \mathcal{S}(\hat{B})) \end{aligned}$$

$$\begin{aligned} \langle\langle \hat{A} | \cdot \rangle\rangle &\leftrightarrow \langle \hat{A}, \cdot \rangle \\ &= \text{Tr}(\hat{A}^\dagger \cdot) \end{aligned}$$

$$\begin{aligned} |\hat{A}\rangle\rangle \langle\langle \hat{B} | &\leftrightarrow \hat{A} \langle \hat{B}, \cdot \rangle \\ &\leftrightarrow \hat{A} \text{Tr}(\hat{B}^\dagger \cdot) \end{aligned}$$

Familiar ideas revisited with super notation

$$\langle \hat{P}_a | \hat{P}_b \rangle = \text{Tr} \left(\hat{P}_a^\dagger \hat{P}_b \right) = d \delta_{ab}$$

$$\Gamma = \{I, X, Y, Z\}^{\otimes n}$$

$$|\Gamma| = 4^n = d^2, \quad d = 2^n$$

$$\mathcal{I} = \sum_{a \in \Gamma} \frac{|\hat{P}_a\rangle \langle \hat{P}_a|}{\langle \hat{P}_a | \hat{P}_a \rangle}$$

$$a, b \in \Gamma$$

$$\Lambda = \mathcal{I} \Lambda \mathcal{I} = \sum_{a, b \in \Gamma} \frac{\langle \hat{P}_a | \Lambda | \hat{P}_b \rangle}{\langle \hat{P}_a | \hat{P}_a \rangle \langle \hat{P}_b | \hat{P}_b \rangle} |\hat{P}_a\rangle \langle \hat{P}_b| = \frac{1}{d} \sum_{a, b \in \Gamma} \Lambda_{ab} |\hat{P}_a\rangle \langle \hat{P}_b| ,$$

Stochastic Pauli channel

$$\Lambda : L(\mathcal{H}) \rightarrow L(\mathcal{H}) \qquad 0 \leq p_a \leq 1 ,$$

$$\Lambda(\rho) = \sum_{a \in \Gamma} p_a P_a \rho P_a , \qquad \sum_{a \in \Gamma} p_a = 1 ,$$

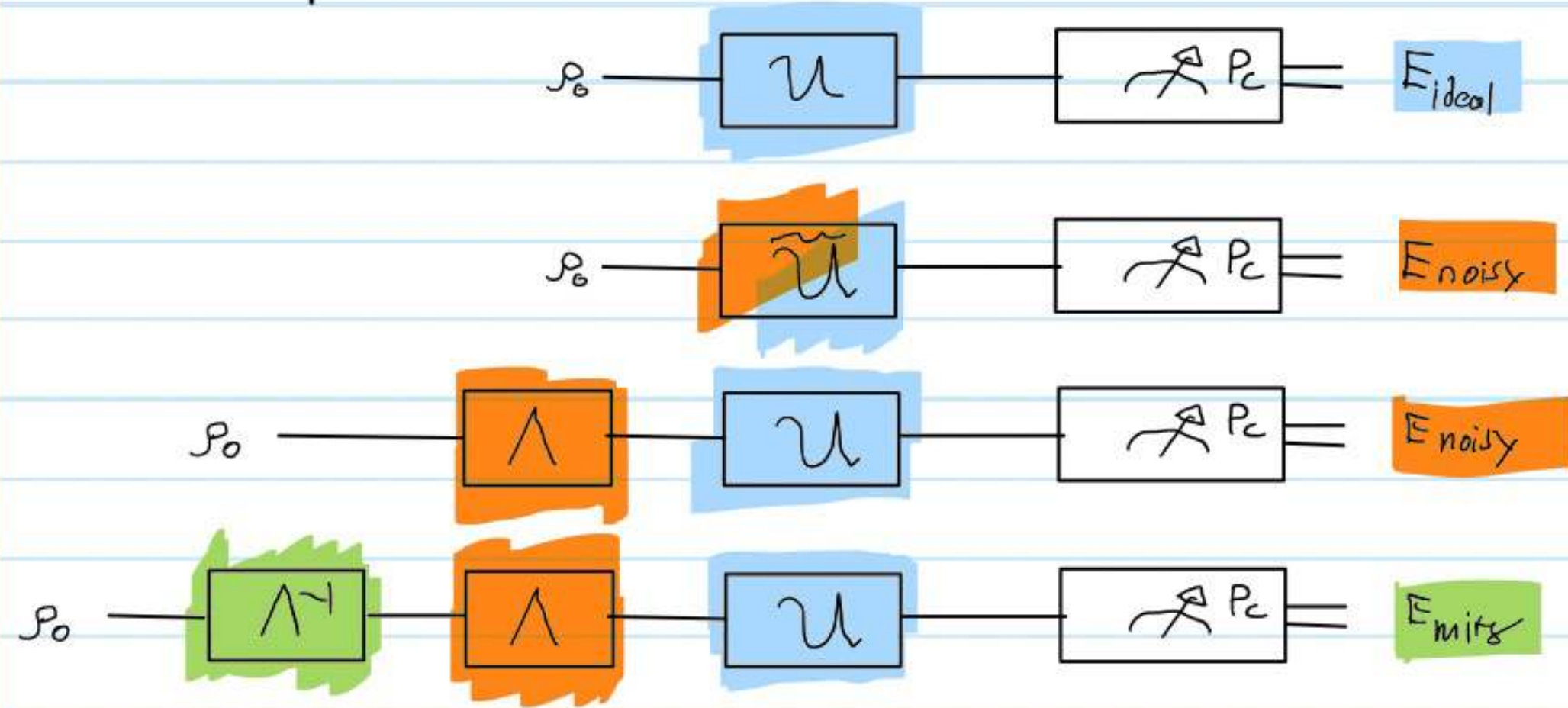
$$\Lambda(\rho) = \frac{1}{\sqrt{|\Gamma|}} \sum_{b \in \Gamma} f_b \operatorname{Tr}(P_b \rho) P_b = \sum_{b \in \Gamma} f_b \frac{|P_b\rangle\rangle \langle\langle P_b|}{\langle\langle P_b | P_b \rangle\rangle} ,$$

$$\Lambda(P_b) = f_b P_b , \quad \forall a \in \Gamma ,$$

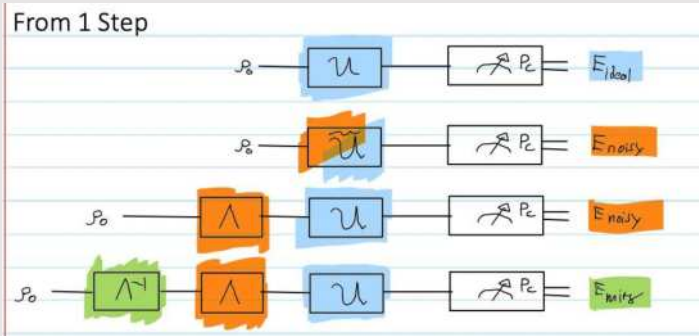
$$-1 \leq f_b \leq 1 \quad f_I = 1 ,$$

Probabilistic error cancellation: Derivation

From 1 Step



Probabilistic error cancellation: Derivation



Channel definitions

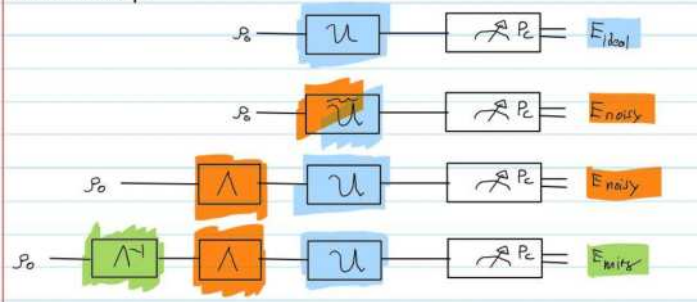
$$U = U \cdot U^\dagger$$

$$\Lambda \approx \sum_a f_a |P_a\rangle\langle P_a|$$

$$= \sum_b c_b P_a$$

Probabilistic error cancellation: Derivation

From 1 Step



Channel definitions

$$U = U \cdot U^\dagger$$

$$\langle \hat{P}_C \rangle(\cdot) = \langle \langle P_C | \cdot \rangle \rangle \quad \text{Need}$$

$$\begin{aligned} \Lambda &= \sum_a f_a |P_a\rangle \langle P_a| \\ &= \sum_b c_b P_a \end{aligned}$$

$$-1 \leq f_a \leq 1$$

$$c_b \geq 0, \sum_b c_b = 1$$

$$c_b = \frac{1}{2^n} \sum_a (-1)^{\langle a, b \rangle_{sp}} f_a$$

$$\vec{c}_b = W^{-1} \vec{f}_a \quad \vec{f}_a = W c_b$$

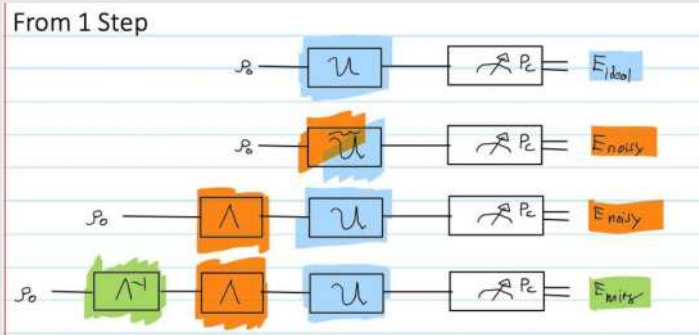
$$\begin{aligned} \Lambda^{-1} &= \sum_a f_a^{-1} |P_a\rangle \langle P_a| \\ &= \sum_b c_b^{\text{inv}} P_b \end{aligned}$$

$$\vec{c}_b^{\text{inv}} = W \vec{f}_a^{-1}$$

$$c_b^{\text{inv}} = \frac{1}{2^n} \sum_a (-1)^{\langle a, b \rangle_{sp}} f_a$$

$$c_b^{\text{inv}} \in \mathbb{R}$$

Probabilistic error cancellation: Derivation



Channel definitions

$$U = U \cdot U^\dagger$$

$$\langle \hat{P}_c \rangle(\cdot) = \langle P_c | \cdot \rangle \quad \text{Need}$$

$$\Lambda \approx \sum_a f_a |P_a\rangle \langle P_a|$$

$$-1 \leq f_a \leq 1$$

$$= \sum_b c_b P_a$$

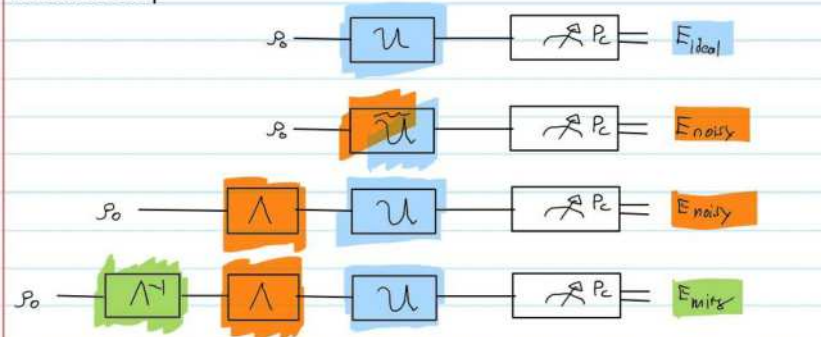
$$c_b \geq 0, \sum_0 c_b = 1$$

$$c_b = \frac{1}{2^n} \sum_a (-1)^{f^{a,b}} f_a$$

$$\vec{c}_b = W^{-1} \vec{f}_a \quad \vec{f}_a = W c_b$$

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |P_a\rangle\langle P_a| \\
 &= \sum_b c_b P_b \\
 \Lambda^{-1} &= \sum_a f_a^{-1} |P_a\rangle\langle P_a| \\
 &= \sum_b c_b^{-1} P_b
 \end{aligned}$$

$$\begin{aligned}
 \langle \hat{P}_c \rangle(\cdot) &= \langle P_c | \cdot \\
 -1 \leq f_a &\leq 1 \\
 c_b \geq 0, \sum_b c_b &= 1 \\
 c_b &= \frac{1}{2^n} \sum_a (-1)^{f_a \cdot b} f_a \\
 \vec{c}_b &= W^{-1} \vec{f}_a \quad \vec{f}_a = W c_b \\
 \vec{c}_b^{-1} &= W \vec{f}_a^{-1} \\
 c_b^{-1} &= \frac{1}{2^n} \sum_a (-1)^{c_a \cdot b} f_a \\
 c_b^{-1} &\in \mathbb{R}
 \end{aligned}$$

Need to introduce:

- Super-op. hermit. unitaries
- Pauli channels P & $P^\dagger$
- super-op. matrices
- unit
- Pauli clac. representation
- SG
- CG

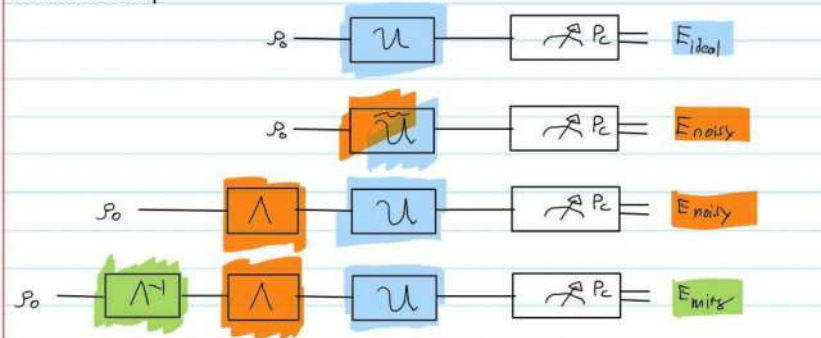
$$E_{\text{ideal}} = \langle \hat{P}_c \rangle_{\rho_0}$$

$$= \langle P_c | U | \rho_0 \rangle$$

ideal exp value with noiseless unitary

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |P_a\rangle\langle P_a| \\
 &= \sum_b c_b P_b \\
 \Lambda^{-1} &= \sum_a f_a^{-1} |P_a\rangle\langle P_a| \\
 &= \sum_b c_b^{-1} P_b
 \end{aligned}$$

$$\begin{aligned}
 \langle \hat{P}_c \rangle(\cdot) &= \langle P_c | \cdot \\
 -1 \leq f_a &\leq 1 \\
 c_b \geq 0, \sum_b c_b &= 1 \\
 c_b &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a \\
 \vec{c}_b &= W^{-1} \vec{f}_a \quad \vec{f}_a = W c_b \\
 \vec{c}_b^{-1} &= W \vec{f}_a^{-1} \\
 c_b^{-1} &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a^{-1} \\
 c_b^{-1} &\in \mathbb{R}
 \end{aligned}$$

Need to introduce:

- Super-op. bra-ket notation
- Pauli elements P_i & PTM
- Super-op notation
- Vectors
- Pauli trace representation
- SVD
- Complex

$$E_{ideal} := \langle \hat{P}_c \rangle_U$$

$$= \langle P_c | U | P_0 \rangle$$

ideal exp value with noiseless unitary

$$E_{noisy} := \langle \hat{P}_c \rangle_{\tilde{U}}$$

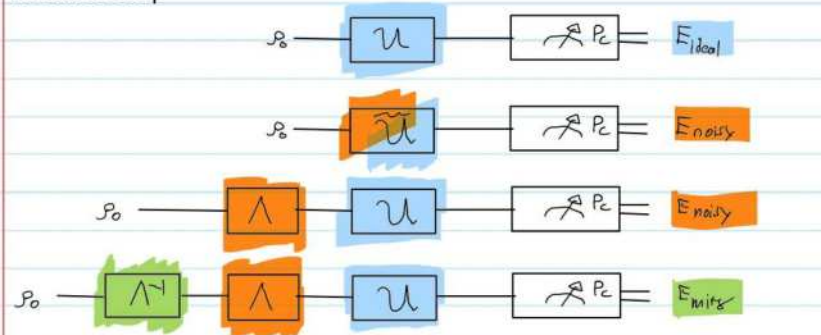
$$= \langle P_c | U \Lambda | P_0 \rangle$$

$$= \langle P_c | \tilde{U} | P_0 \rangle$$

noisy-gate expectation value

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |p_a\rangle\langle p_a| \\
 &= \sum_b c_b p_b \\
 \Lambda^{-1} &= \sum_a f_a^{-1} |p_a\rangle\langle p_a| \\
 &= \sum_b c_b^{inv} p_b
 \end{aligned}$$

Need to introduce:

- Super-op. based notation
- Pauli elements $\mathbb{P}$ & $\mathbb{P}^\dagger$
- super-op notation
- Pauli trace representation
- SG
- CG

$$\begin{aligned}
 \langle \hat{P}_c \rangle &= \langle P_c | \cdot \\
 -1 \leq f_a &\leq 1 \\
 c_b \geq 0, \sum c_b &= 1 \\
 c_b &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a \\
 \vec{c}_b &= W^{-1} \vec{f}_a \quad \vec{f}_a = W c_b \\
 \vec{c}_b^{inv} &= W \vec{f}_a^{-1} \\
 c_b^{inv} &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a^{-1} \\
 c_b^{inv} &\in \mathbb{R}
 \end{aligned}$$

$$E_{ideal} := \langle \hat{P}_c \rangle_U$$

$$= \langle\langle P_c | U | p_0 \rangle\rangle$$

ideal exp value with noiseless unitary

$$E_{noisy} := \langle \hat{P}_c \rangle_{\tilde{U}}$$

$$= \langle\langle P_c | U \Lambda | p_0 \rangle\rangle$$

$$= \langle\langle P_c | \tilde{U} | p_0 \rangle\rangle$$

noisy-gate expectation value

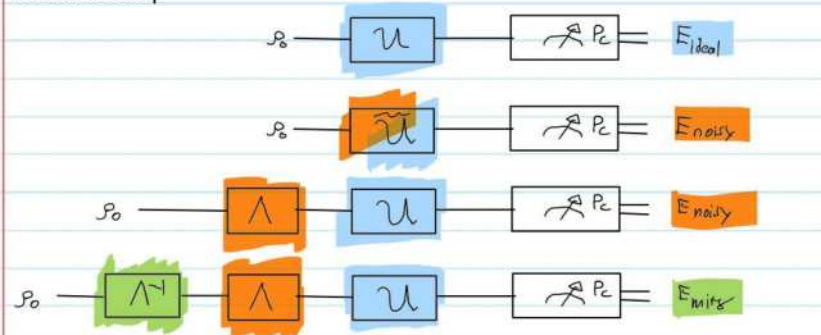
$$E_{mitg} := \langle\langle P_c | U \Lambda \Lambda^{-1} | p_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_a f_a^{-1} |p_a\rangle\langle p_a| \right) | p_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_b c_b^{inv} p_b \right) | p_0 \rangle\rangle$$

Circuit expectation values

From 1 Step



Channel definitions

$$\begin{aligned}
 U &= U \cdot U^\dagger \\
 \Lambda &= \sum_a f_a |P_a\rangle\langle P_a| \\
 &= \sum_b c_b P_b \\
 \Lambda^{-1} &= \sum_a f_a^{-1} |P_a\rangle\langle P_a| \\
 &= \sum_b c_b^{inv} P_b
 \end{aligned}$$

Need to introduce:

- Super-op. based notation
- Pauli channels: $P \in \{I, X, Y, Z\}$
- super-op notation
- Pauli trace representation
- SG to $\mathbb{R}^4$
- $c \in \mathbb{R}$

$$\begin{aligned}
 \langle \hat{P}_c \rangle(\cdot) &= \langle P_c | \cdot \\
 -1 \leq f_a &\leq 1 \\
 c_b \geq 0, \sum_b c_b &= 1 \\
 c_b &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a \\
 \vec{c}_b &= W^{-1} \vec{f}_a \quad \vec{f}_a = W \vec{c}_b \\
 \vec{c}_b^{inv} &= W \vec{f}_a^{-1} \\
 c_b^{inv} &= \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a^{-1} \\
 c_b^{inv} &\in \mathbb{R}
 \end{aligned}$$

$$E_{ideal} := \langle \hat{P}_c \rangle_U$$

$$= \langle\langle P_c | U | P_0 \rangle\rangle$$

ideal exp value with noiseless unitary

$$E_{noisy} := \langle \hat{P}_c \rangle_{\tilde{U}}$$

$$= \langle\langle P_c | U \Lambda | P_0 \rangle\rangle$$

$$= \langle\langle P_c | \tilde{U} | P_0 \rangle\rangle$$

noisy-gate expectation value

$$E_{mitg} := \langle\langle P_c | U \Lambda \Lambda^{-1} | P_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_a f_a^{-1} |P_a\rangle\langle P_a| \right) | P_0 \rangle\rangle$$

$$= \langle\langle P_c | U \Lambda \left(\sum_b c_b^{inv} P_b \right) | P_0 \rangle\rangle$$

$$= \sum_b c_b^{inv} \langle\langle P_c | U \Lambda P_b | P_0 \rangle\rangle$$

sum of trajectories with weight $c_b^{inv} \in \mathbb{R}$

$$= \underbrace{\sum_b c_b^{inv}}_{\text{classical post process}} \underbrace{\langle P_c \rangle(\tilde{U}, b)}_{\text{quantum}}$$



$$E_{\text{mitg}} := \langle \langle P_c | U \Lambda U^\dagger | \rho_0 \rangle \rangle$$

$$= \langle \langle P_c | U \Lambda \left(\sum_a f_a^{-1} | \rho_a \rangle \langle \rho_a | \right) U^\dagger | \rho_0 \rangle \rangle$$

$$= \langle \langle P_c | U \Lambda \left(\sum_b c_b^{\text{inv}} P_b \right) U^\dagger | \rho_0 \rangle \rangle$$

$$= \sum_b c_b^{\text{inv}} \langle \langle P_c | U \Lambda P_b U^\dagger | \rho_0 \rangle \rangle$$

sum of trajectories
with weight $c_b^{\text{inv}} \in \mathbb{R}$

$$= \underbrace{\sum_b c_b^{\text{inv}}}_{\text{classical post process}} \underbrace{\langle P_c \rangle(\tilde{u}, b)}_{\text{quantum}}$$

classical
post process



Quantum circuit we can execute on fHW
and find exp. value from.

$\therefore$ To find noise-free val all we have to do is to compute exp-val
of all 4^n b-modified circuits! This would give us ideal
exp value.

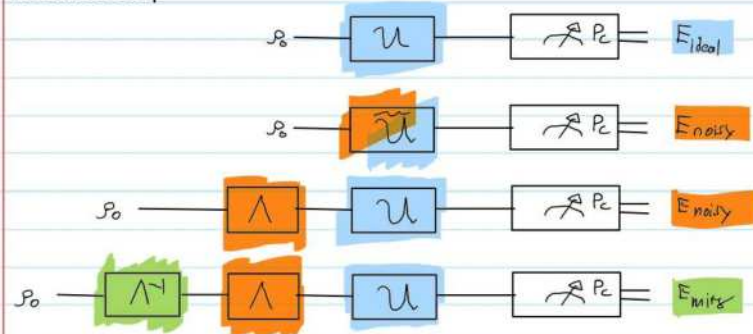
However, $|\{b\}| = 4^n$ grows exponentially, hence, infeasible.

but what if we could sample from it to approximate

full sum. But... can't sample directly from c_b^{inv} which

does not form a valid prob-distribution. let's solve:

From 1 Step



Channel definitions

$$U = U \cdot U^\dagger$$

$$\langle \hat{P}_c \rangle(\cdot) = \langle \langle P_c | \cdot \rangle \rangle$$

Need to introduce
• Super-op. based notations
• Pauli elements σ_i & PTM
• Super-op. notation
• self
• Pauli class representation
• SQ to $u^\dagger$
• $C \circ A B$

$$\Lambda = \sum_a f_a | \rho_a \rangle \langle \rho_a |$$

$$-1 \leq f_a \leq 1$$

$$c_b \geq 0, \sum_b c_b = 1$$

$$c_b = \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a$$

$$\vec{c}_b = W^{-1} \vec{f}_a \quad \vec{f}_a = W \vec{c}_b$$

$$\Lambda^{-1} = \sum_a f_a^{-1} | \rho_a \rangle \langle \rho_a |$$

$$\vec{c}_b^{\text{inv}} = W \vec{f}_a^{-1}$$

$$c_b^{\text{inv}} = \frac{1}{2^n} \sum_a (-1)^{a \cdot b} f_a^{-1}$$

$$c_b^{\text{inv}} \in \mathbb{R}$$

$$= \sum_b c_b^{\text{inv}} P_b$$

Quasi probability distribution

c_b^{inv} can be outside $[0,1]$

$\sum_b c_b^{inv} =: \gamma \geq 1$ generally, for Λ not unitary

eg b4 th chapter

$$\Lambda = (1-p)[I + pX \cdot X]$$

$$\Lambda^{-1} = \left(1 + \frac{p}{1-2p}\right) I + \frac{-p}{1-2p} X \cdot X$$

$$c_{\uparrow}^{inv} = 1 + \frac{p}{1-2p}$$

$b = (0,0)$
choice vector

$$c_{\downarrow}^{inv} = -\frac{p}{1-2p}$$

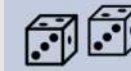
$b = (1,0)$

Turn into probability

$$c_b^{inv} = \underbrace{\text{sgn}(c_b^{inv})}_{\text{sign} \in \{-1, +1\}} \underbrace{\frac{|c_b^{inv}|}{\gamma}}_{\substack{\text{prob} \in [0,1] \\ \text{scale}}} \gamma$$

$$\bar{c}_b^{inv} := \frac{|c_b^{inv}|}{\|c_b^{inv}\|_1} \rightarrow \|c_b\|_1 = \sum_b |c_b^{inv}| \Leftrightarrow L_1 \text{ norm}$$

$$= \frac{|c_b^{inv}|}{\gamma}$$

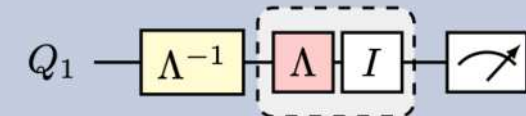


probability

$$P_I = |1 - q|/\gamma$$

$$P_X = |q|/\gamma$$

$$P_I + P_X = 1$$



||

sign

$$I \quad s_I = \text{sign}(1-q)$$

+

$$X \quad s_X = \text{sign}(q)$$

E mitigated

$$E_{\text{mitg}} = \sum_b c_b^{\text{inv}} \langle \hat{P}_c \rangle(\tilde{u}, b)$$

$$= \sum_b \text{sgn}(c_b^{\text{inv}}) \frac{|c_b^{\text{inv}}|}{\gamma} \gamma \langle \hat{P}_c \rangle(\tilde{u}, b)$$

$$= \gamma \sum_b \underbrace{\text{sgn}(c_b^{\text{inv}})}_{\text{scale}} \underbrace{\bar{c}_b^{\text{inv}}}_{\text{classical post-processing}} \underbrace{\langle \hat{P}_c \rangle(\tilde{u}, b)}_{\text{valid QC circuit can run & find value on HW}}$$

scale classical post-processing. valid QC circuit can run & find value on HW

In this form, the decomposition of the error error mitigated expectation value is simply a sum over expectation values of Pauli-gate modified circuits, whose value can be obtained from direct quantum computer execution, weighted by a probability, c bar b inverse, and the sign. The elements that perform the weighing and rescaling can all be done in classical post-processing.

c_b^{inv} can be outside $[0,1]$

$\frac{1}{\gamma} c_b^{\text{inv}} = \gamma \geq 1$ generally, for Λ not unitary

eg b4 chapter $\Lambda = (1-p)I + pX \cdot X$
 $\Lambda^\dagger = (1+p/2p)I + \frac{-p}{1-2p} X \cdot X$

$c_{\frac{1}{2}}^{\text{inv}} = 1 + \frac{p}{1-2p}$ $c_X^{\text{inv}} = \frac{-p}{1-2p}$
 $\frac{1}{2} \leftarrow b=(0,0)$ $\frac{1}{2} \leftarrow b=(1,0)$
 choose $\frac{1}{2}$ $\frac{1}{2}$

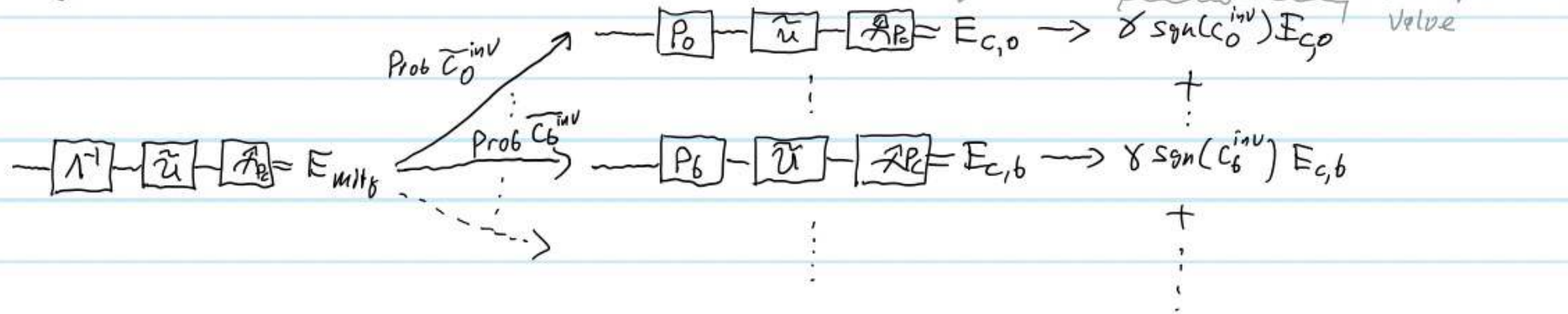
Turn into probability

$$c_b^{\text{inv}} = \underbrace{\text{sgn}(c_b^{\text{inv}})}_{\text{sign} \in \{-1,1\}} \underbrace{\frac{|c_b^{\text{inv}}|}{\gamma}}_{\substack{\text{prob} \in [0,1] \\ \text{scale}}} \gamma$$

$$\bar{c}_b^{\text{inv}} := \frac{|c_b^{\text{inv}}|}{\|c_b^{\text{inv}}\|_1} \xrightarrow{\text{abs inv}} \|c_b\|_1 = \sum_b |c_b^{\text{inv}}| \Leftrightarrow L_1 \text{ norm}$$

$$= \frac{|c_b^{\text{inv}}|}{\gamma}$$

In other words



Quasi-Probability Distribution

c_k^{inv} can be outside $\{0,1\}$

$\sum_k c_k^{inv} = 1$ generally, for A not unitary

eg. for the Hadamard $A = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$

$$c_k^{inv} = \frac{1}{\gamma} \frac{P_k}{\text{prob}(k)}$$

Turn into probability

$$c_k^{inv} = \frac{\text{sgn}(c_k^{inv})}{\gamma} \frac{|c_k^{inv}|}{\text{prob}(k)} \gamma$$

$$\tilde{c}_k^{inv} = \frac{|c_k^{inv}|}{\gamma} \frac{1}{\text{prob}(k)} \gamma \rightarrow \tilde{c}_k^{inv} = \frac{1}{\gamma} \frac{P_k}{\text{prob}(k)}$$

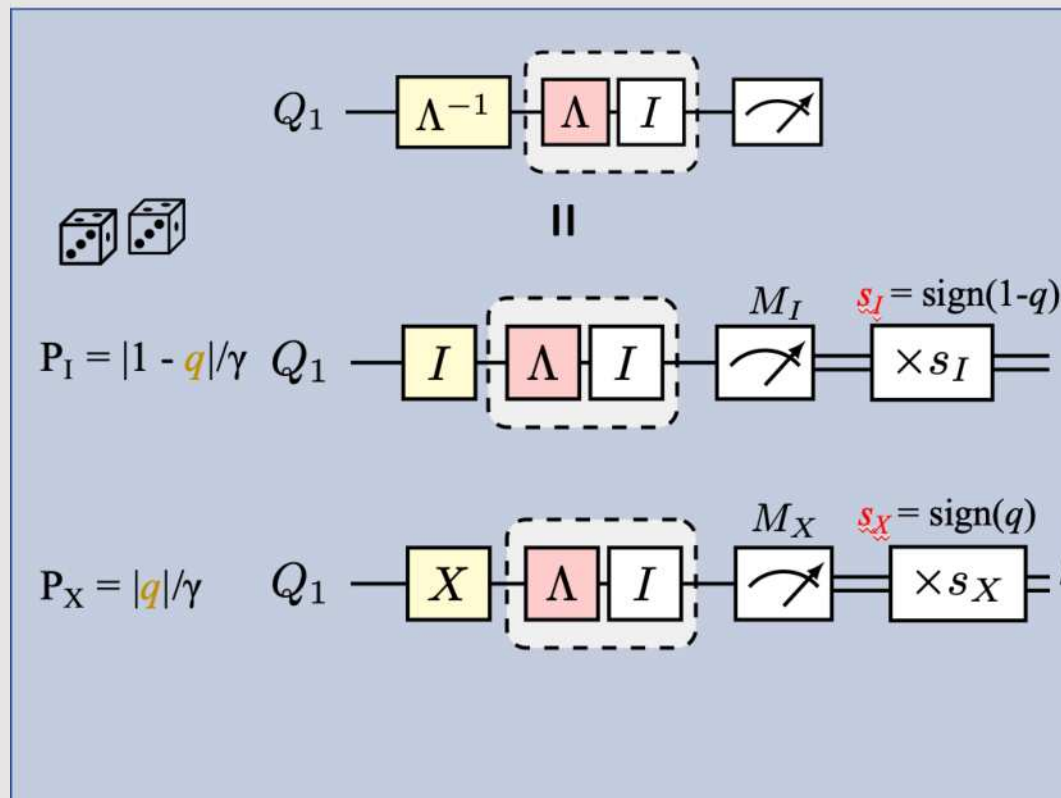
$$E_{mit,b} = \sum_k c_k^{inv} \langle \tilde{U}_b \rangle (U, b)$$

$$= \sum_k \frac{\text{sgn}(c_k^{inv})}{\gamma} \frac{|c_k^{inv}|}{\text{prob}(k)} \gamma \langle \tilde{U}_b \rangle (U, b)$$

$$= \gamma \sum_k \text{sgn}(c_k^{inv}) \tilde{c}_k^{inv} \langle \tilde{U}_b \rangle (U, b)$$

scale classical prob. value by C gate inverse, and the sign

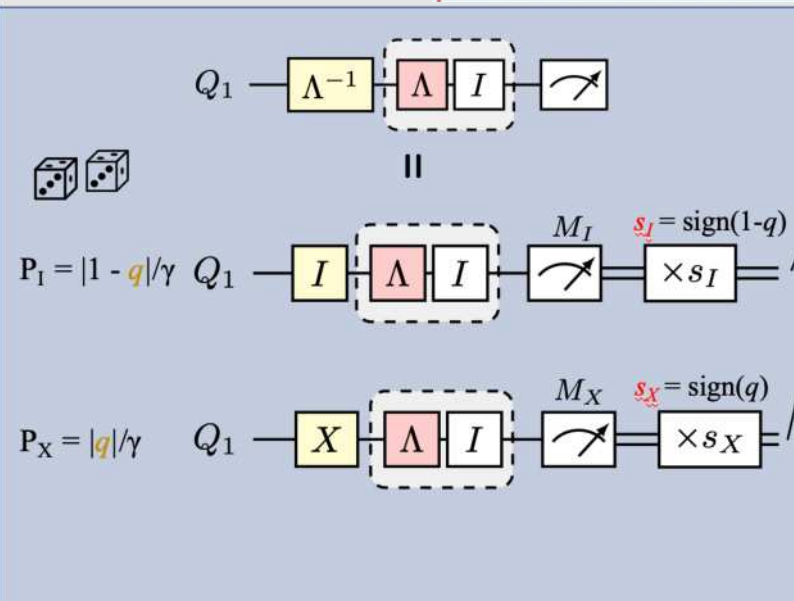
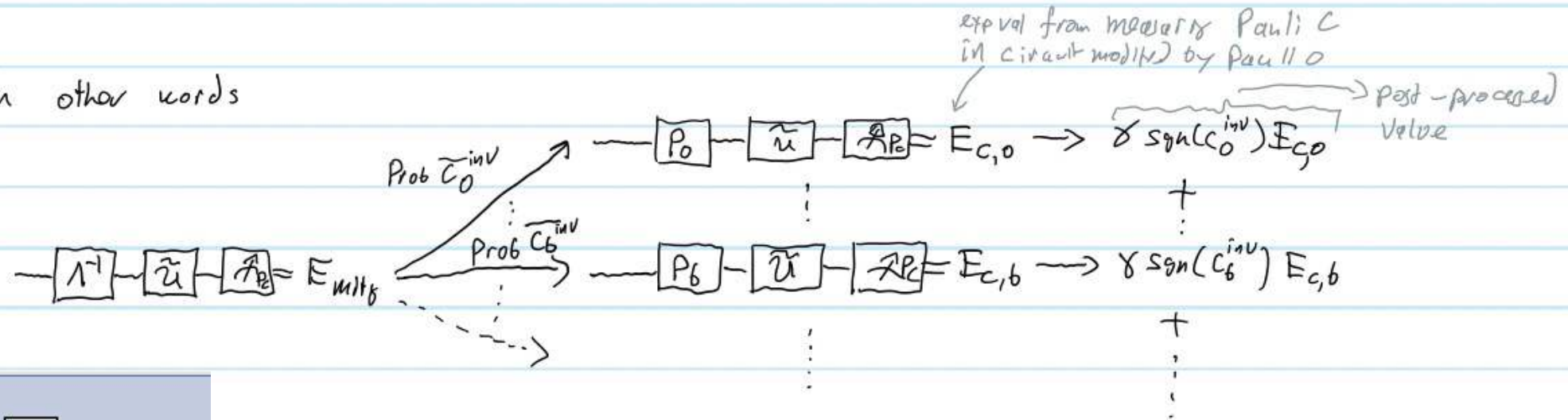
In this form, the decomposition of the error error mitigated expectation value is simply a sum over expectation values of Pauli gate modified circuits, whose value can be obtained from direct quantum computer execution, weighted by a probability, c bar inverse, and the sign. The elements that perform the weighing and rescaling can all be done in classical post-processing.



Estimator

post processing.

In other words



$$E_{c, \text{mitg}} = \sum_b \gamma \text{sgn}(C_b^{\text{inv}}) E_{c,b}$$

mitigated value for Pauli C obtained from the quasi-prob distribution

From above, we know this is an unbiased estimator,
but what about the error and sampling

mitigated value for Pauli C obtained from the quasi-prob distribution

From above, we know this is an unbiased estimator,
but what about the error and sampling

Estimator, Sampling, and Error Bounds

Sample circuit of the form

$$\left\{ \underbrace{\bar{C}_b^{\text{inv}}}_{\text{prob}} : \underbrace{-[P_b] - [X] - [XP_b]}_{\text{Pauli observable single-shot outcome ie e-val}} = Y \in \{-1, +1\} \right\} \rightarrow X = \text{sign}(\bar{C}_b^{\text{inv}}) Y$$

post processed value X
 $X \in \{-1, +1\}$

Let's say we sample M instances, randomly sample a single value for b and obtain one-shot value on the QC, ie one random instance of $Y = 1$ or $Y = -1$, which we then post-process.

The results are thus the classical random variables

$$\{X_1, X_2, \dots, X_M\} \quad \text{or} \quad \{X_m : m \in 1, \dots, M\}$$

Quasi-Probability Distribution

C_b^{inv} can be outside $\{0, 1\}$

$\bar{C}_b^{\text{inv}} = \text{sign}(C_b^{\text{inv}})$ generally for A not unitary

eg. for the Hadamard $A = \frac{1}{\sqrt{2}}(I + \sigma_x)$

$$C_b^{\text{inv}} = \frac{1}{\sqrt{2}}(I + \sigma_x) \quad C_b^{\text{inv}} = \frac{1}{\sqrt{2}}(I + \sigma_x)$$

Turn into probability

$$\bar{C}_b^{\text{inv}} = \text{sign}(C_b^{\text{inv}}) \frac{|C_b^{\text{inv}}|}{\sum_i |C_b^{\text{inv}}|} \gamma$$

$$\bar{C}_b^{\text{inv}} = \frac{1}{\sum_i |C_b^{\text{inv}}|} \sum_i |C_b^{\text{inv}}| \gamma_i \rightarrow \bar{C}_b^{\text{inv}} = \frac{1}{\sum_i |C_b^{\text{inv}}|} \sum_i |C_b^{\text{inv}}| \gamma_i$$

$$E_{\text{avg}} = \sum_b \bar{C}_b^{\text{inv}} \langle \hat{R} \rangle(\bar{C}_b^{\text{inv}}, b)$$

$$= \sum_b \text{sign}(C_b^{\text{inv}}) \frac{|C_b^{\text{inv}}|}{\sum_i |C_b^{\text{inv}}|} \gamma_i \langle \hat{R} \rangle(\bar{C}_b^{\text{inv}}, b)$$

$$= \sum_b \text{sign}(C_b^{\text{inv}}) \frac{|C_b^{\text{inv}}|}{\sum_i |C_b^{\text{inv}}|} \gamma_i \langle \hat{R} \rangle(\bar{C}_b^{\text{inv}}, b)$$

In this form, the decomposition of the error error mitigated expectation value is simply a sum over expectation values of Pauli-gate modified circuits, whose value can be obtained from direct quantum computer execution, weighted by a probability, γ bar inverse, and the sign. The elements that perform the weighing and rescaling can all be done in classical post-processing.

Probabilistic error cancellation: Derivation

where each $X_m \in \{1, -1\}$ and is distributed to model.

Bernoulli distribution with some probability which can be any valid value and can vary from shot-to-shot m .

Our mitigation estimator is then for M shots:

$$E_M := \gamma \frac{1}{M} \sum_{m=1}^M X_m = \frac{1}{M} \sum_{m=1}^M \gamma \text{sgn}(c_{b_m}^{\text{inv}}) \overbrace{\gamma_{m,b}}^{\text{noisy circuit}}(\tilde{U}, b_m)$$

Pauli chosen for m -th shot

There are now 2 random processes:

b_m : which pauli b we pick for shot m

γ_{b_m} : which outcome ± 1 we get for b_m circuit of shot m

Unbiased Estimator of the Ideal, noise-free circuit expectation

$$\mathbb{E}[E_M] = \frac{1}{M} \sum_{m=1}^M \mathbb{E}[\gamma X_m] \quad \text{iid rand vars}$$

Unbiased estimator

$$\mathbb{E}[F_M] = \frac{1}{M} \sum_{m=1}^M \mathbb{E}[\gamma X_m] \quad \text{iid rand vars}$$

$$= \mathbb{E}[\gamma X_m] \quad \text{no } X_m \text{ is different}$$

$$= \mathbb{E}[\gamma \text{sgn}(c_{b_m}^{\text{inv}}) Y_{b_m}(\tilde{u}, b_m)] \quad \text{where rand var is } b_m \text{ now, not just } m, \text{ so}$$

$$\approx \mathbb{E}_{b_m}[\gamma \text{sgn}(c_b^{\text{inv}}) Y_b(\tilde{u}, b)] \quad \text{drop notation} \quad \text{Prob}[b] = \bar{c}_b^{\text{inv}}$$

$$= \sum_b \mathbb{E}_Y[\gamma \text{sgn}(c_b^{\text{inv}}) Y_b] \text{Prob}[b]$$

$$= \sum_b \underbrace{\gamma \text{sgn}(c_b^{\text{inv}})}_{\text{post-process of outcome}} \underbrace{\bar{c}_b^{\text{inv}}}_{\text{sample prob}} \underbrace{\mathbb{E}_Y[Y_b]}_{\text{rand outcome}} \longrightarrow$$

$$\approx \sum_b \gamma \text{sgn}(c_b^{\text{inv}}) \frac{\bar{c}_b^{\text{inv}}}{\gamma} \langle\langle P_C | \mathcal{U} \wedge \mathcal{P}_b | \mathcal{P}_0 \rangle\rangle$$

$$\text{note: } \langle\hat{P}_C\rangle(\tilde{u}, b) = \langle\langle P_C | \mathcal{U} \wedge \mathcal{P}_b | \mathcal{P}_0 \rangle\rangle = \mathbb{E}[Y_b]$$

$\mathbb{E}$ Rand variable ± 1 for output of the b -th pauli circuit.

$\therefore$ for some classical func of b $f(b)$ which does not depend on the value Y_b but only on the label b :

$$\mathbb{E}[f(b) Y_b] = f(b) \langle\hat{P}_C\rangle(\tilde{u}, b)$$

Probabilistic error cancellation: Derivation

$$= \langle \langle P_c | U \Lambda \left(\sum_b c_b^{-1} P_b \right) | P_0 \rangle \rangle$$

$$= \langle \langle P_c | U \Lambda \Lambda^{-1} | P_0 \rangle \rangle$$

$$= \langle \langle P_c | U | P_0 \rangle \rangle$$

$$= \langle \hat{P}_c \rangle (U_{\text{ideal}}) \quad \text{without noise!}$$

unbiased estimator of the true noise-free, ideal value of the circuit



the value y_b but only on the label b :

$$\mathbb{E}[f(b) y_b] = f(b) \langle \hat{P}_c \rangle (U, b)$$

(Optional step) Variance

Variance of E_M

$$\underbrace{V_{\{b_m, x_m\}}}_{\text{with respect to}} [E_M] = \frac{\gamma^2}{M^2} \sum_{m=1}^M V[x_m]$$

$$= \frac{\gamma^2}{M} V[x(U, b)]$$

x_m iid
can drop subscript m
and emphasize b and value x

note the same variance is just
rescaled by γ^2 due to γ

Generalizing: Raveling trajectories with quasiprobabilities

Channel we *want*
to implement

CPTP operation we
can implement

$$\mathcal{C}(\cdot) = \sum_i a_i \mathcal{F}_i(\cdot)$$

Real coefficients, turn
into *quasi-probability*

Putting the following techniques all on the same footing

Technique

Prob. error cancelation (PEC)

Circuit cutting (knitting) of gates

Circuit cutting of wires

Classical sim. algorithms (QP)

Channel $\mathcal{C}$

noise inverse

non-local gate

large unitary

unitary

Quantum Error Mitigation for Non-Equilibrium Quantum Dynamics

Lecture 1

Big picture

Why quantum computers?

Status and outlook

Why error mitigation?

Noise in quantum computers

Overview of error mitigation

Mitigation fundamentals

Probabilistic error cancelation (PEC)

Introduction

One qubit example

General derivation

Next lectures

Learning noise

State-of-art PEC experiments

Key techniques: Twirling

T-REX mitigation

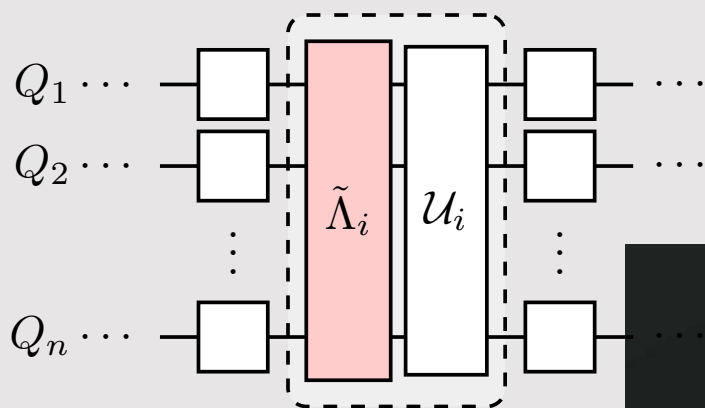
...

State-of-art experiments at the 100Q+, depth 50+: uncovering local integrals of motion

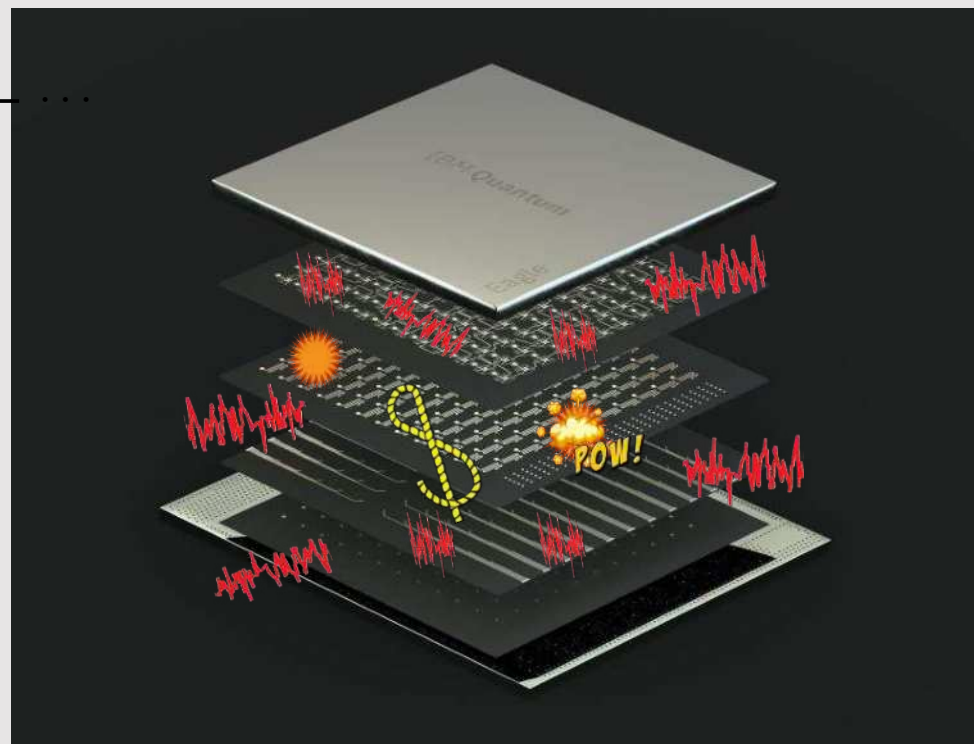
...



Is it possible to learn the noise with accuracy, efficiency, and scalability?



Energy relaxation T_1
Dephasing T_2
Coherent errors ZZ
Classical crosstalk
Quantum crosstalk
State preparation error
Measurement correlated errors
...

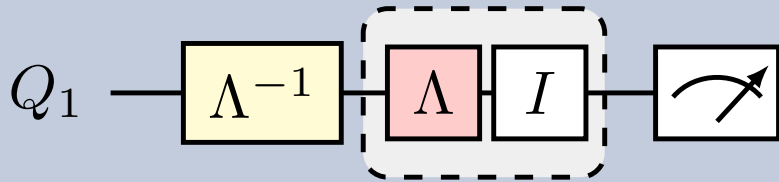


Control errors
Photon shot noise
1/f charge noise
1/f flux noise
Nonequilibrium quasiparticles
Leakage
Cosmic rays
...

PEC: Nice, **but** why hasn't worked so far for experiments?

Practical challenges

Small scale



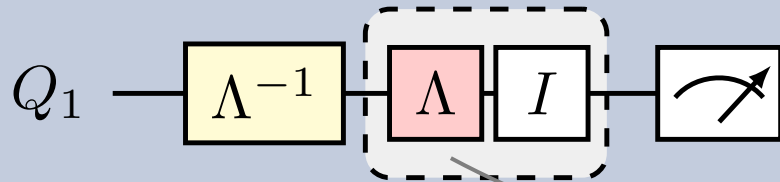
Critically hinges on knowing the full noise near perfectly

Despite the method's theoretical appeal (1-10), practical challenges have limited its demonstration to the one and two-qubit level (2, 3)

1. S. Endo, S. C. Benjamin, Y. Li, Physical Review X 8, 031027 (2018).
2. C. Song, et al., Science Advances 5, arXiv:2109.04457(2019).
3. S. Zhang, et al., Nature Communications 11, 587 (2020).
4. C. Piveteau, D. Sutter, S. Woerner, arXiv:2101.09290 (2021).
5. S. Endo, et al., J. Physi Soc. of Japan 90, 032001 (2021).
6. C. Piveteau, et al., arXiv:2103.04915 (2021).
7. R. Takagi, Phys. Rev. Research 3, 033178 (2021).
8. R. Takagi, S. Endo, S. Minagawa, M. Gu, arXiv:2109.04457 (2021).
9. Y. Guo, S. Yang, arXiv preprint arXiv:2201.00752 (2022).
10. ...

PEC: Nice, **but** why hasn't worked so far? Challenges

Small scale



2 qubits

255 parameters

10 qubits

10^{12} parameters

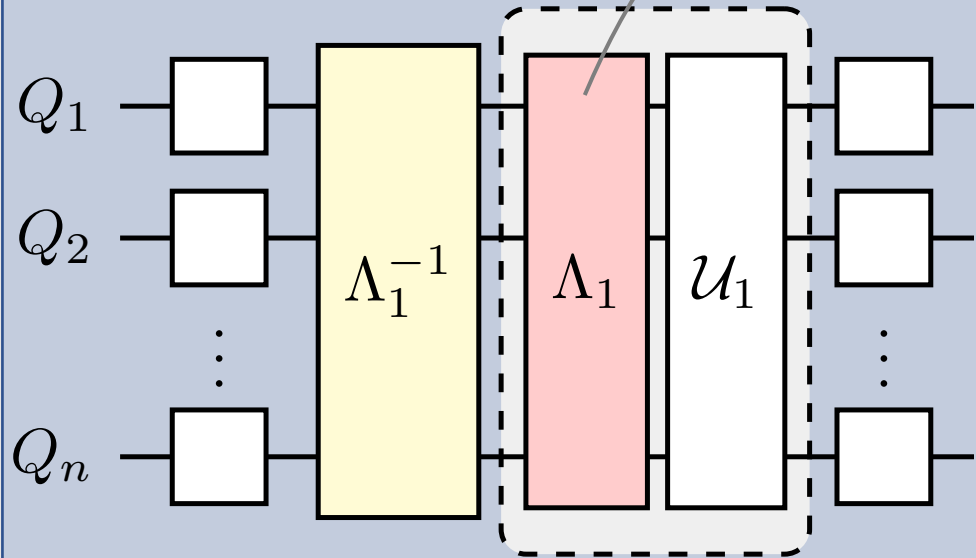
50 qubits

10^{60} parameters

noise param values $10^{-2} - 10^{-5}$

additive error sampling cost ($>10^2 - 10^{10}$)

Large scale



Challenges

learning complexity

- efficient
- scalable
- accurate
- compact, tractable representation

noise in full device

- cross-talk
- correlated errors
- parallel gates

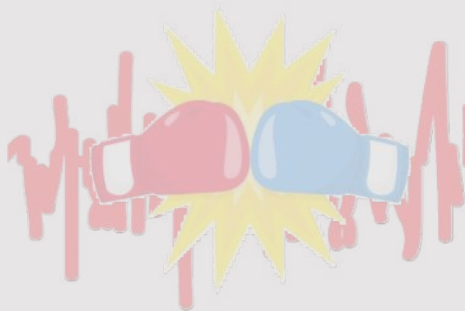
Outline



Idea

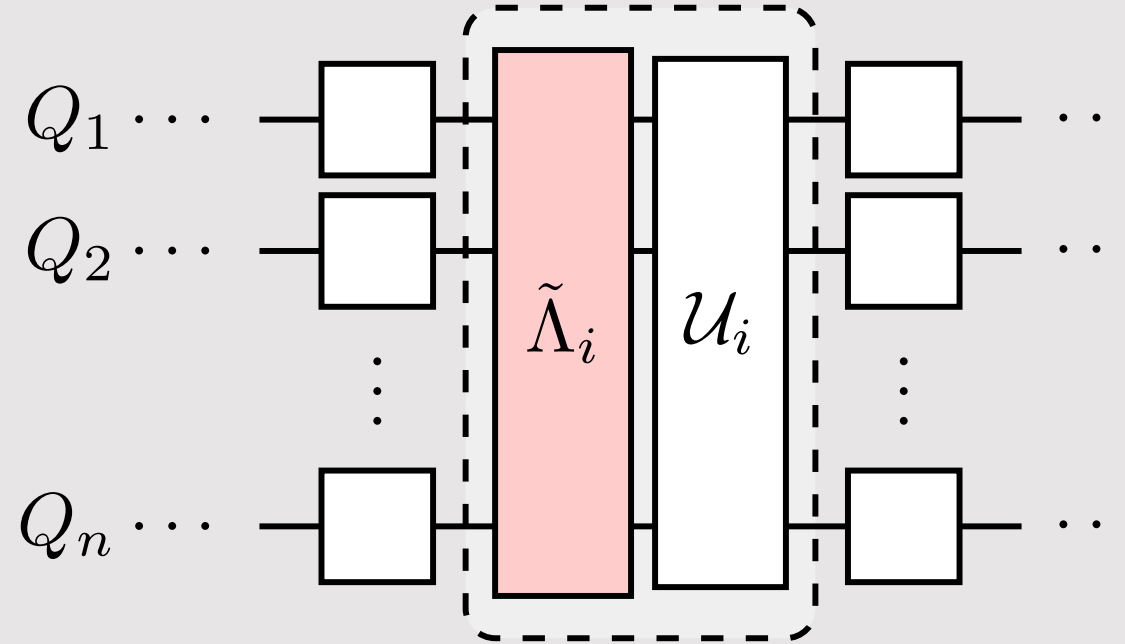


Learn



Cancel
(realization)

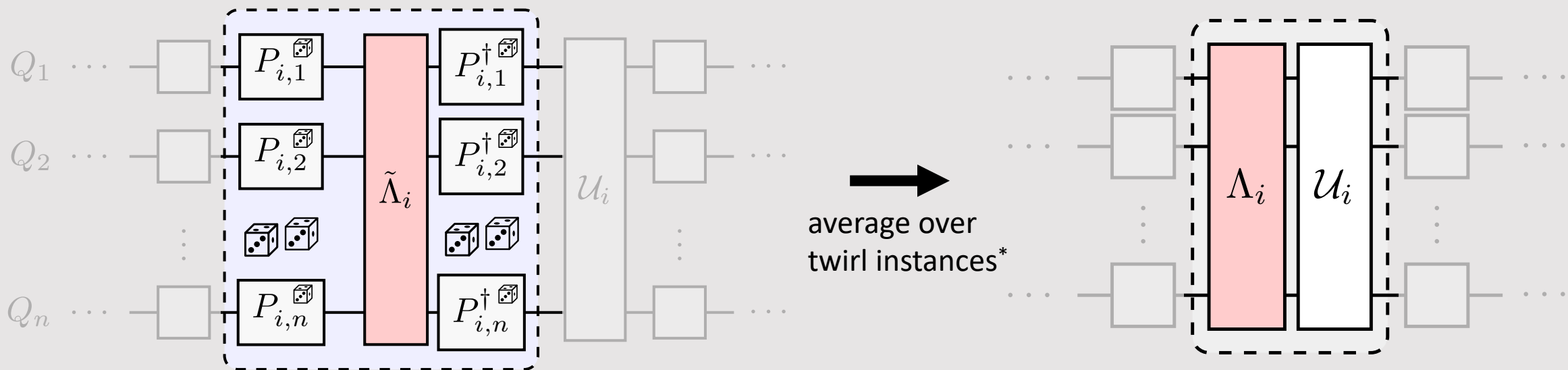
Step 1: Simplify the noise



noise that includes cross-talk errors, etc.
characterized by some $4^n \times 4^n$ matrix

Step 1: Simplify the noise: twirl

twirl reduces to noise $4^n \times 4^n$ matrix to diagonal one with 4^n entries in Pauli basis



Twirling references

1. C. H. Bennett, G. Brassard, S. Popescu, B. Schumacher, J. A. Smolin, **Tutorial** W. K. Wootters, et al., Phys. Rev. Lett. 76, 722 (1996).
2. E. Knill, arXiv:0404104 (2004).
3. O. Kern, G. Alber, D. L. Shepelyansky, EPJ D 32, 153 (2005).
4. M. R. Geller, Z. Zhou, Physical Review A 88, 012314 (2013).
5. J. J. Wallman, J. Emerson, Physical Review A 94, 052325 (2016)
6. Hashim *et al.*, Phys. Rev. X 11, 041039 (2021)
7. Tutorial: zlatko-minev.com/blog/twirling (2022)
8. ...



Stochastic Pauli channel

$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

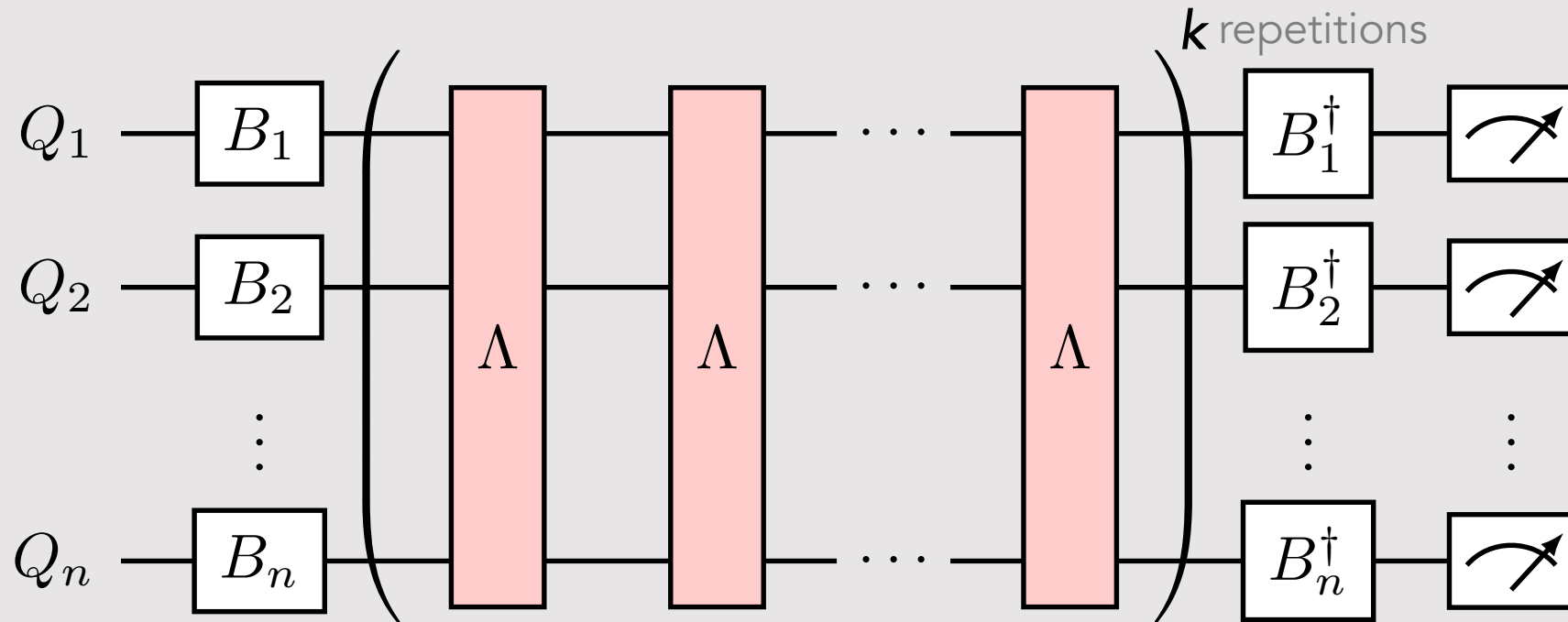
$$\Lambda(P_a) = f_a P_a \quad \text{eigenvecs are Paulis}$$

* some sub-Clifford twirl group (use Paulis)

Zlatko Minev, IBM Quantum (69)

Step 2 wish: amplify noise

for the i -th layer



prepare circuit in pre-determined Pauli basis

Since diagonal channel will amplify eigenvalues learn with multiplicative precision

measure circuit in same pre-determined Pauli basis

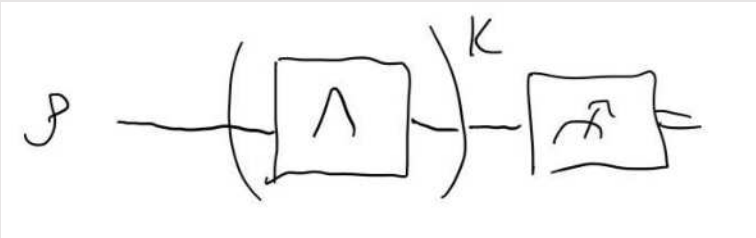
Akin to:

RB, Cycle RB, K-body noise reconstruction, ...

S.T. Flammia and J.J. Wallman ACM Trans QC 1, 3 (2020), ...

For something of a review of protocols, see Helsen, *et al.*, *A general framework for randomized benchmarking* (arXiv:2010.07974)

Notes



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

$$\Lambda(P_a) = f_a P_a$$

For a qubit

$$\rho = \frac{1}{2} \left(\hat{I} + \rho_X \hat{X} + \rho_Y \hat{Y} + \rho_Z \hat{Z} \right)$$

In general:

$$\rho = \frac{1}{2^n} \sum_b \rho_b \hat{P}_b$$

$$\Lambda(\rho) = \rho'$$

$$\rho' = \frac{1}{2^n} \sum_b \rho'_b \hat{P}_b$$

Pauli decomposition of a density matrix is a powerful tool - offering a versatile representation. It expresses a density matrix as a linear combination of Paulis, often referred to as the Pauli basis.

Posterior state: Action of the channel on the input state

Pauli decomposition of posterior state

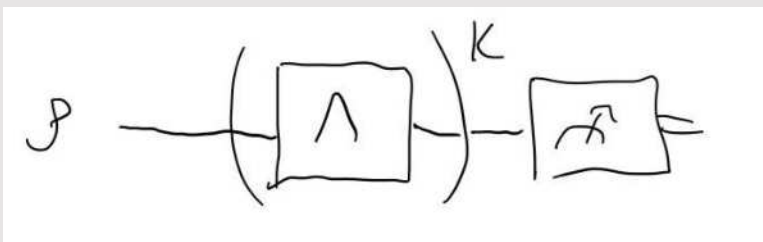
Since the channel is linear

$$\Lambda(a \hat{A} + b \hat{B}) = a \Lambda(\hat{A}) + b \Lambda(\hat{B})$$

$$\Lambda(\rho) = \Lambda\left(\frac{1}{2^n} \sum_b \rho_b \hat{P}_b\right)$$

$$= \frac{1}{2^n} \sum_b \rho_b \underbrace{\Lambda(\hat{P}_b)}$$

Just need to know action on P_b



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

$$\Lambda(P_a) = f_a P_a$$

or written another way:

$$\begin{aligned} \rho_c' &= \sum_b \rho_b \frac{1}{2^n} \text{Tr}(\hat{P}_c^\dagger \Lambda(\hat{P}_b)) \\ &= \sum_b \rho_b \frac{\langle\langle P_c | \Lambda | P_b \rangle\rangle}{\langle\langle P_c | P_c \rangle\rangle} \end{aligned}$$

$$\Lambda(\rho) = \Lambda\left(\frac{1}{2^n} \sum_b \rho_b \hat{P}_b\right)$$

$$= \frac{1}{2^n} \sum_b \rho_b \underbrace{\Lambda(\hat{P}_b)}_{\text{just need to know action on } P_b}$$

$$= \rho'$$

$$= \frac{1}{2^n} \sum_c \rho_c' \hat{P}_c$$

since $\text{Tr}(\hat{P}_a^\dagger \hat{P}_b) = \delta_{ab} 2^n$
orthogonal

Solve for ρ_c' by equate & orthogonal case:

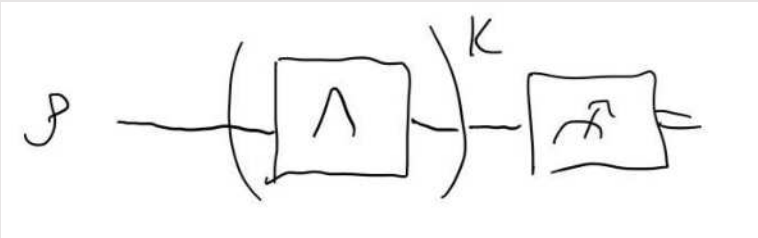
$$\text{Tr}\left(\hat{P}_c^\dagger \frac{1}{2^n} \sum_b \rho_b \Lambda(\hat{P}_b)\right) = \text{Tr}\left(\hat{P}_c^\dagger \frac{1}{2^n} \sum_{c'} \rho_{c'}' \hat{P}_{c'}\right)$$

$$\sum_b \rho_b \text{Tr}(\hat{P}_c^\dagger \Lambda(\hat{P}_b)) = \sum_{c'} \rho_{c'}' \text{Tr}(\hat{P}_c^\dagger \hat{P}_{c'})$$

$$= \sum_{c'} \rho_{c'}' \delta_{cc'} 2^n$$

$$\frac{1}{2^n} \sum_b \rho_b \text{Tr}[\hat{P}_c^\dagger \Lambda(\hat{P}_b)] = \rho_c' \quad \checkmark$$

Notes



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

$$\Lambda(P_a) = f_a P_a$$

The measurement outcome then is (say measured in the $\hat{P}_c$ basis):

$$\langle \hat{P}_c \rangle = \text{Tr}(\hat{P}_c \rho')$$

For a single step

$$= \text{Tr}(\hat{P}_c \sum_{c'} \frac{1}{2^n} \rho_{c'} \hat{P}_{c'})$$

$$= \sum_{c'} \rho_{c'} \frac{1}{2^n} \text{Tr}(\hat{P}_c \hat{P}_{c'}) \rightarrow 2^n \delta_{cc'}$$

$$= \rho_c$$

now sub in

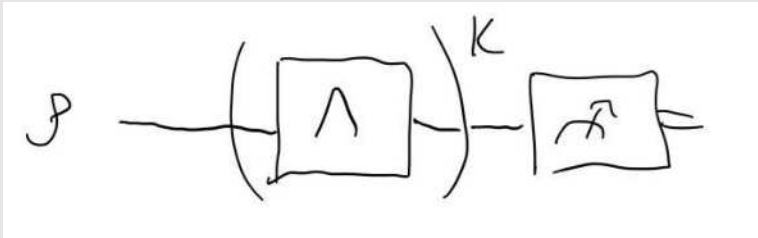
$$= \sum_b \rho_b \frac{\langle\langle P_c | \Lambda | P_b \rangle\rangle}{2^n}$$

→ for a Pauli channel $\langle\langle P_c | \Lambda | P_b \rangle\rangle = f_c \delta_{bc} 2^n$

$$= f_c \rho_c \leftarrow \text{initial state coefficient}$$

fidelity c of Λ

Notes



$$\Lambda_i(\rho) = \sum_{a=0}^{4^n-1} c_{ia} P_a \rho P_a^\dagger$$

$$\Lambda(P_a) = f_a P_a$$

What is ρ_c for an eigenstate of $\hat{P}_c \rightarrow 1$.

e.g. for $c=z$: $\hat{P}_c = \hat{Z}$ and $\rho = |0\rangle = \frac{1}{2}(I + Z)$

$$\Rightarrow \rho_{c=z} = \text{Tr}(Z \rho) = +1$$

For repeating the channel K times



$$\rho' = \Lambda^K \rho$$

$$\Rightarrow \rho'_c = \sum_b \rho_b \frac{\langle\langle P_c | \Lambda^K | P_b \rangle\rangle}{2^n}$$

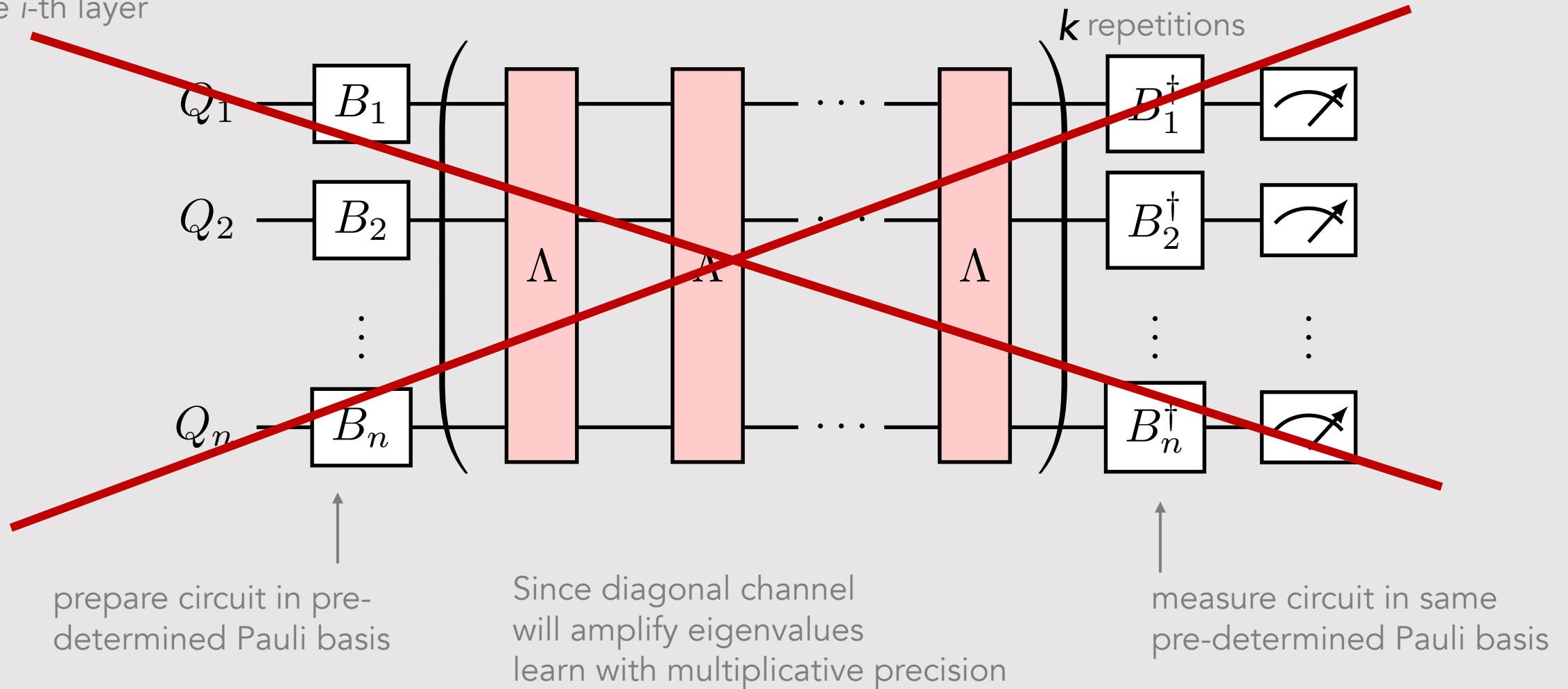
$$= \sum_b f_b^K \rho_b$$

$$\Rightarrow \langle \hat{P}_c \rangle = f_c^K \rho_c$$

Can include all diagonal SPAM

Step 2 wish: amplify noise

for the i -th layer



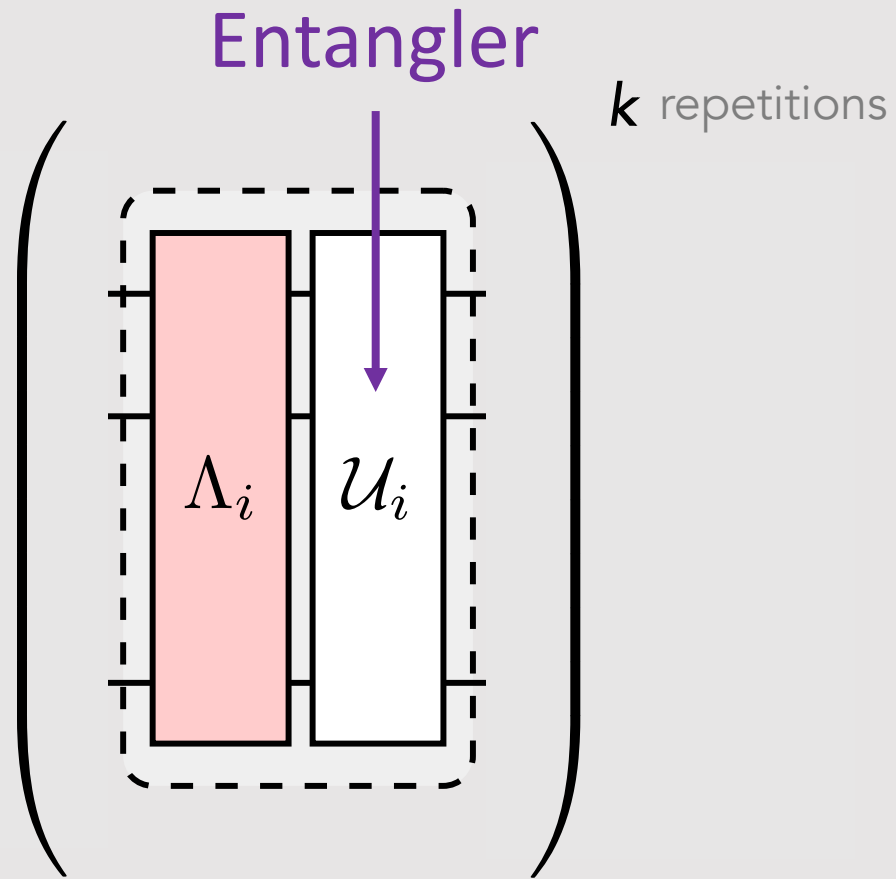
Akin to:

RB, Cycle RB, K-body noise reconstruction, ...

S.T. Flammia and J.J. Wallman ACM Trans QC 1, 3 (2020), ...

For something of a review of protocols, see Helsen, *et al.*, *A general framework for randomized benchmarking* (arXiv:2010.07974)

Step 2: Ideally, amplify the noise and learn



Ideally wish

~~$$\Lambda_i^k(P_a) = f_{ia}^k P_a$$~~

Akin to:

RB, Cycle RB, K-body noise reconstruction, ...

S.T. Flammia and J.J. Wallman ACM Trans QC 1, 3 (2020), ...

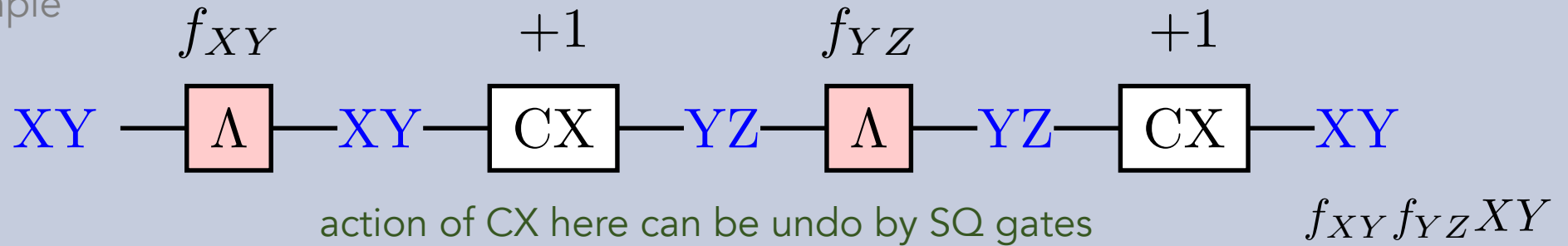
Erhard *et al.*, arXiv:1902.08543; Ferracin *et al.*, arXiv:2201.10672, ...

For something of a review of protocols, see Helsen, *et al.*, arXiv:2010.07974

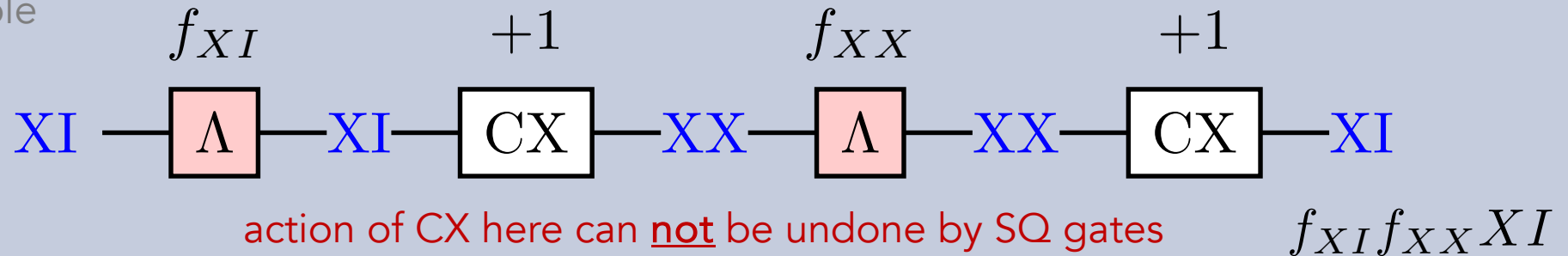
Let's see how the amplification works with gates: no-go theorem

$$\Lambda(P_a) = f_a P_a$$

2Q example



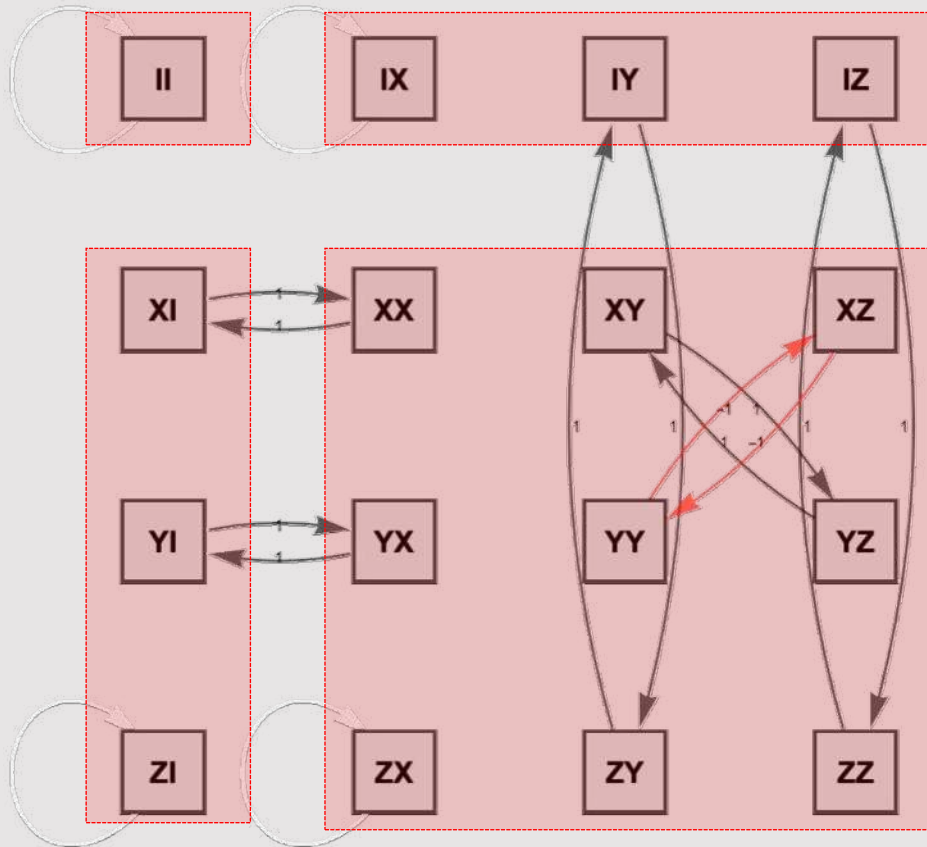
2Q example



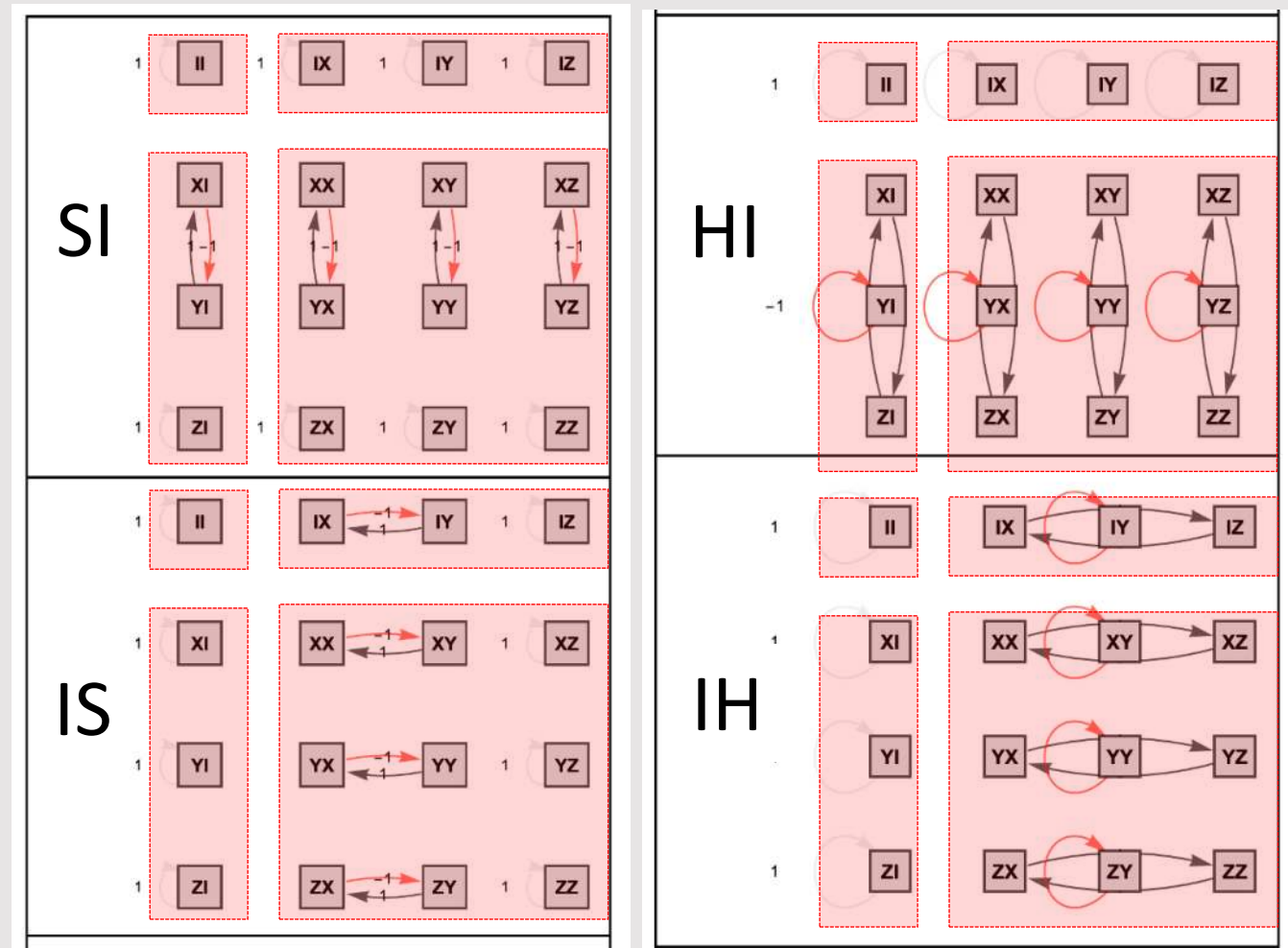
fundamental degeneracy – can not undo some non-local – need entangling operation

How to gates move state Paulis around?

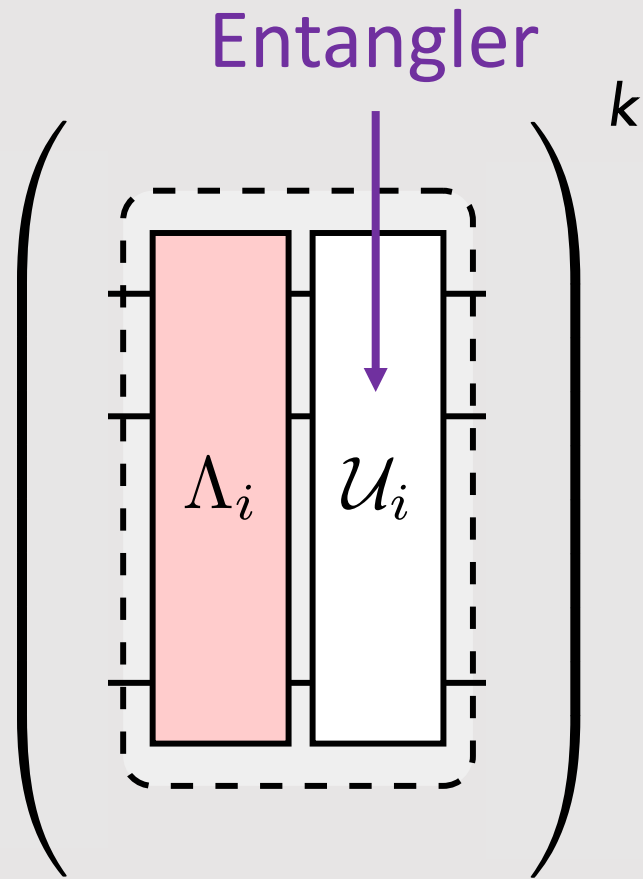
cNOT



Example single qubit gates



Not so simple to learn noise of entangling gates



Ideally wish

$$\Lambda_i^k(P_a) = f_{ia}^k P_a$$

Fundamental **no-go theorem on learning**



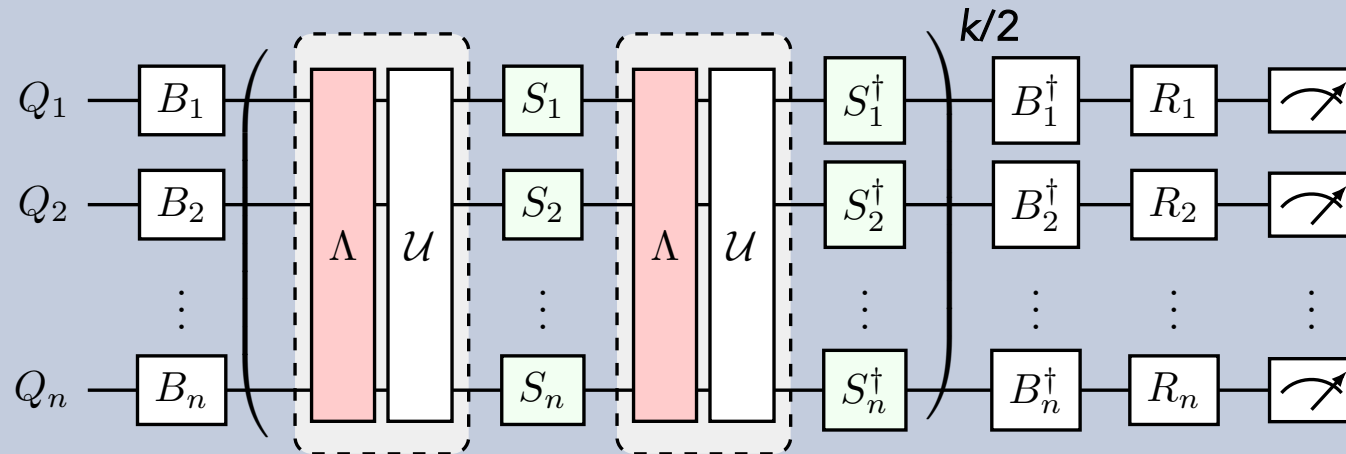
For general and in-depth

Senrui Chen, Y. Liu, M. Otten, A. Seif, B. Fefferman, L. Jiang
arXiv:2206.06362 (2022)

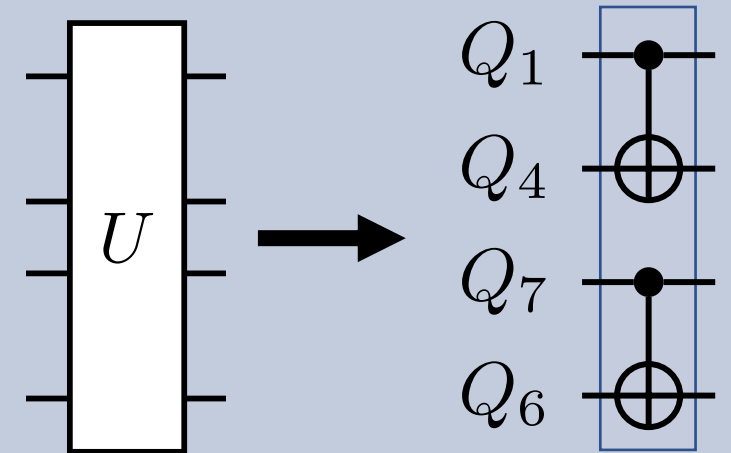
or supplement of our paper for qubit version and work by
S. Flammia, S Benjamin, and teams.

Solution: Custom protocol + weak assumption

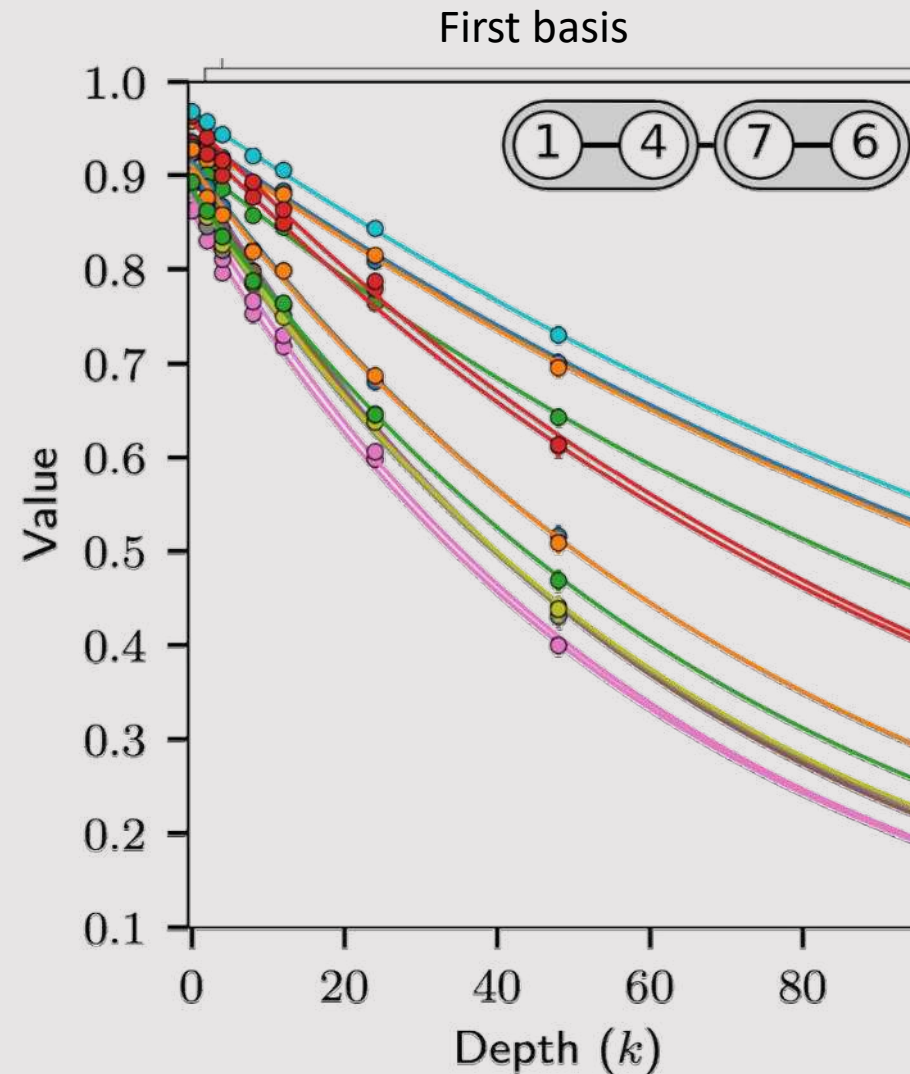
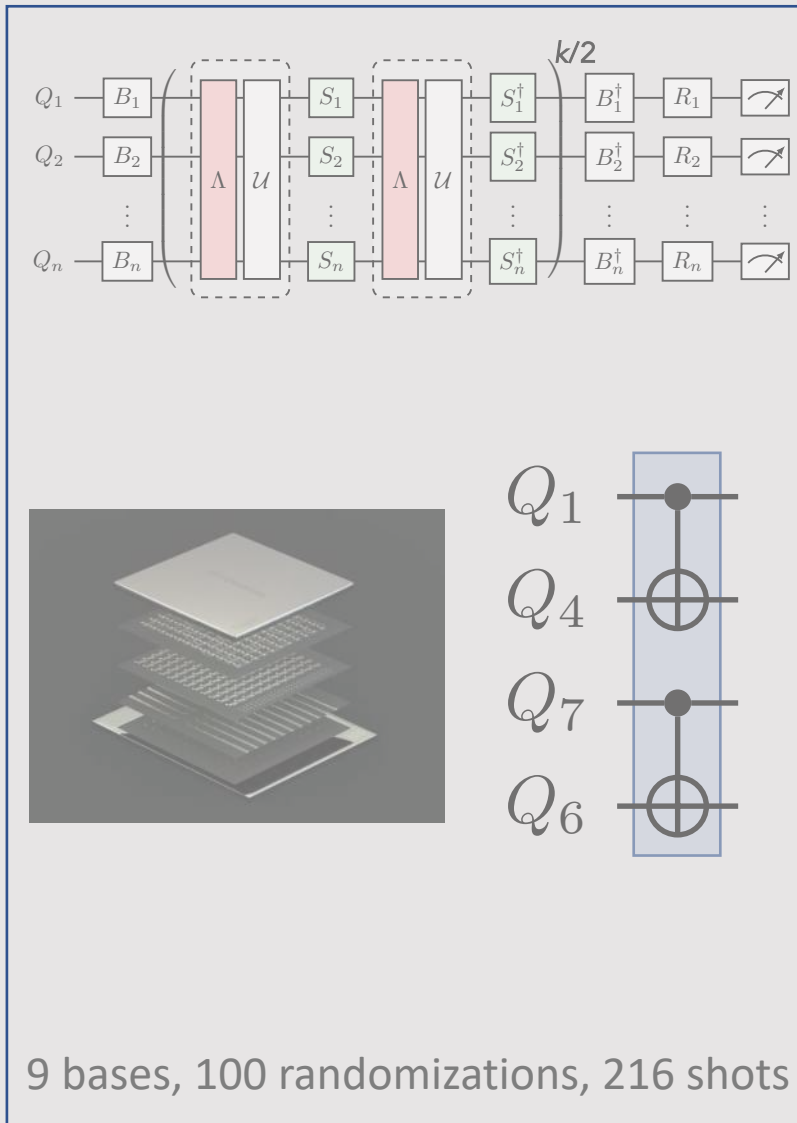
Learning circuits



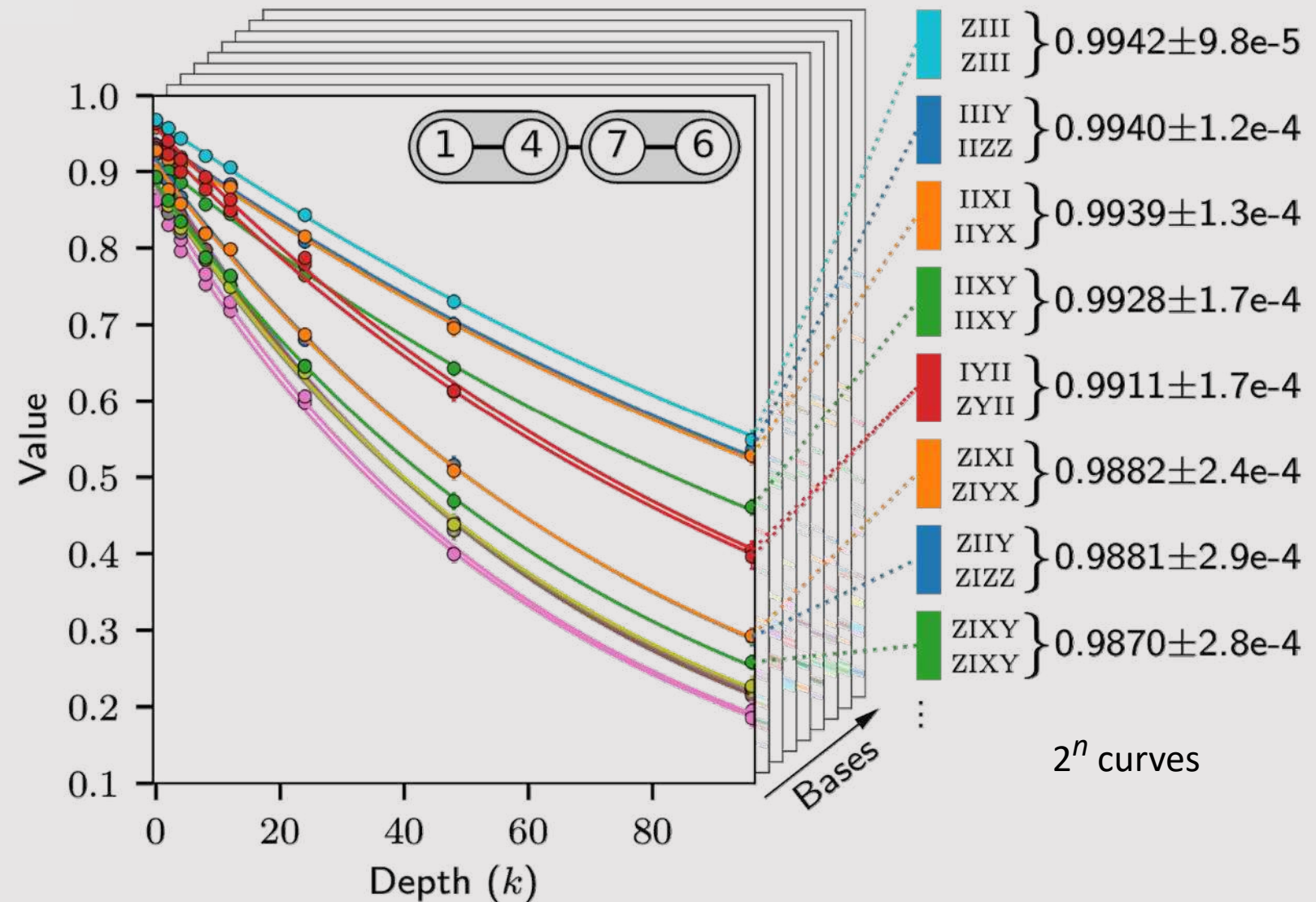
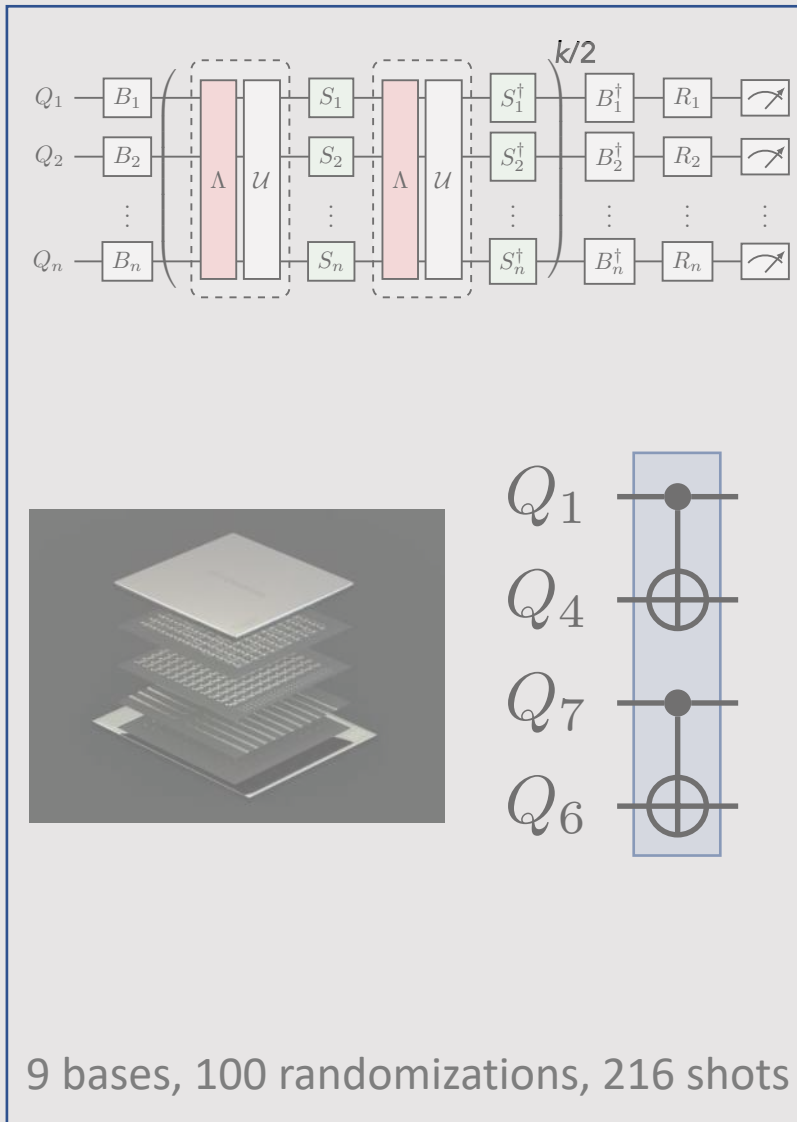
Example



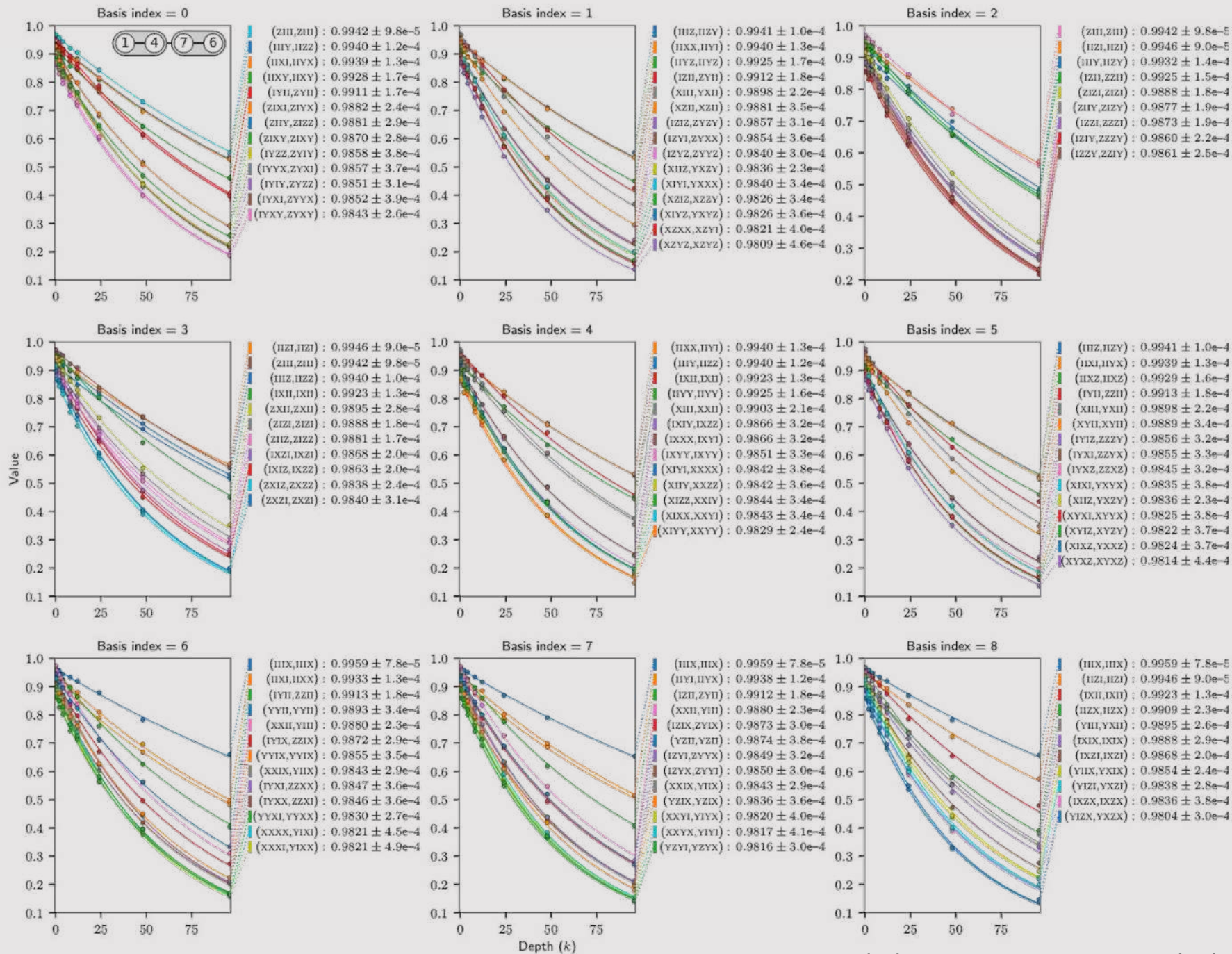
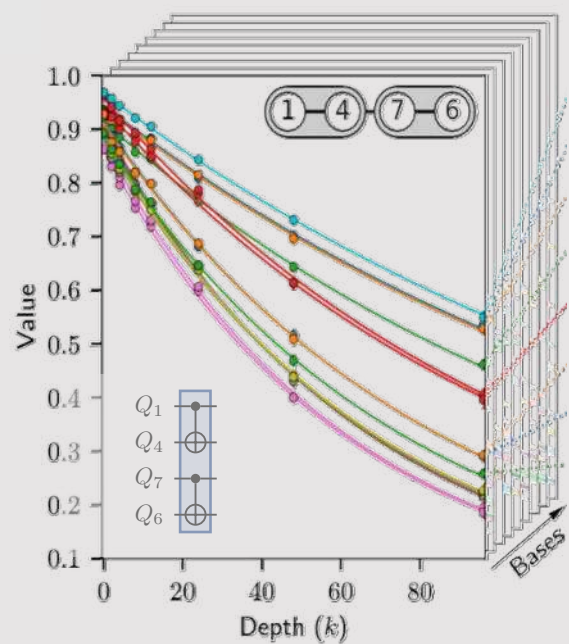
Learning the noise: raw data



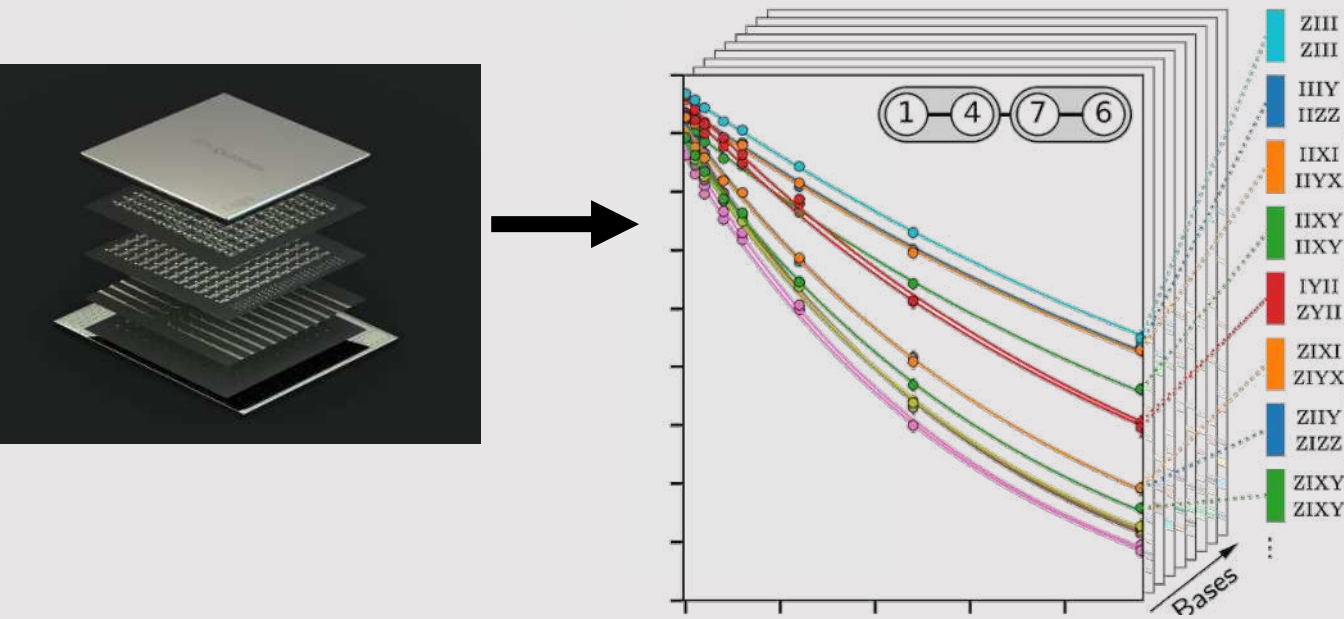
Learning the noise: raw data



Raw data



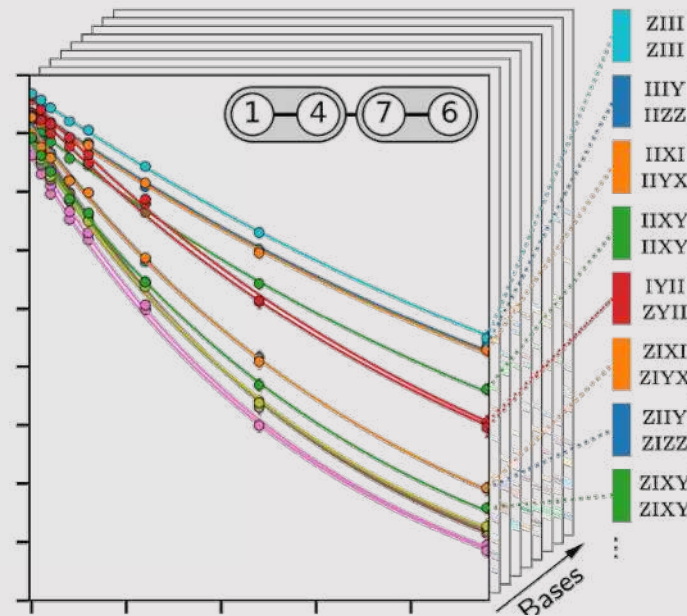
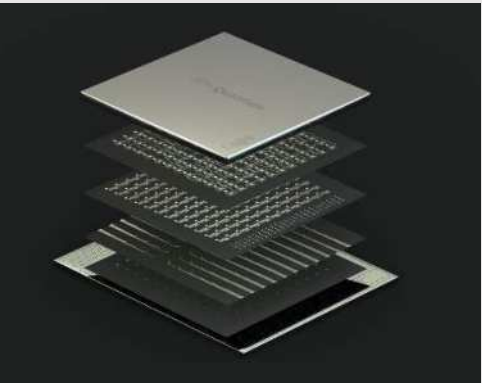
Reconstructing quantum channel from measurement data



$$\Lambda(\rho) = \sum_{a=0}^{4^n-1} c_a P_a \rho P_a^\dagger$$

Still 4^n

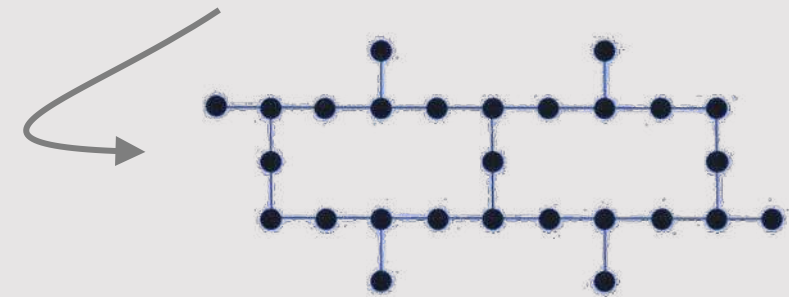
Sparse Pauli-Lindblad model



$$\Lambda(\rho) = \sum_{a=0}^{4^n-1} c_a P_a \rho P_a^\dagger$$

$$\Lambda(\rho) = \exp[\mathcal{L}](\rho)$$

$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k (P_k \rho P_k - \rho)$$



Magic



icon: Eucalyp

Highlight: Ewout van den Berg

Notes

Dissipator for a given Pauli

$$\begin{aligned}\mathcal{D}[\rho]_P &= P P^\dagger - \frac{1}{2}(P P^\dagger + P^\dagger P) \rho \\ &= P P^\dagger - \rho \\ &= (P \cdot P - I) \rho\end{aligned}$$

$$e^{\hat{\mathcal{L}}} = \prod_a \exp(\gamma_a \mathcal{D}[\hat{P}_a])$$

$$= \prod_a \Lambda[\rho_a]$$

$$\begin{aligned}\hat{\mathcal{L}} &= \sum_a \gamma_a \left(\hat{P}_a - \frac{1}{2} I \right) \\ &= \sum_a \gamma_a \mathcal{D}[\hat{P}_a]\end{aligned}$$

$$\text{but } [\mathcal{D}[\rho_a], \mathcal{D}[\rho_b]] = 0$$

Each sub channel

$$\begin{aligned}
 \Lambda[\rho_a] &= \exp(\gamma \mathcal{D}[\rho]) \\
 &= \exp(\gamma (\hat{p}_a + \hat{I})) \\
 &= \exp(\gamma \hat{p}_a) \exp(\gamma \hat{I}) \exp(\cancel{\frac{-1}{2} \gamma^2 [\hat{p}_a, \hat{I}]}) \dots \\
 &= \exp(\gamma \hat{p}_a) \exp(\gamma \hat{I})
 \end{aligned}$$

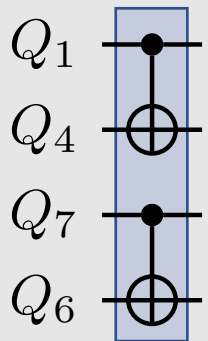
$$\begin{aligned}
 \hookrightarrow \sum_n \frac{\gamma^n \hat{I}^n}{n!} &= \sum_n \frac{\gamma^n}{n!} \hat{I} \\
 &= \exp(\gamma) \hat{I} = \exp(\gamma) \hat{I} \cdot \hat{I}
 \end{aligned}$$

Each sub channel

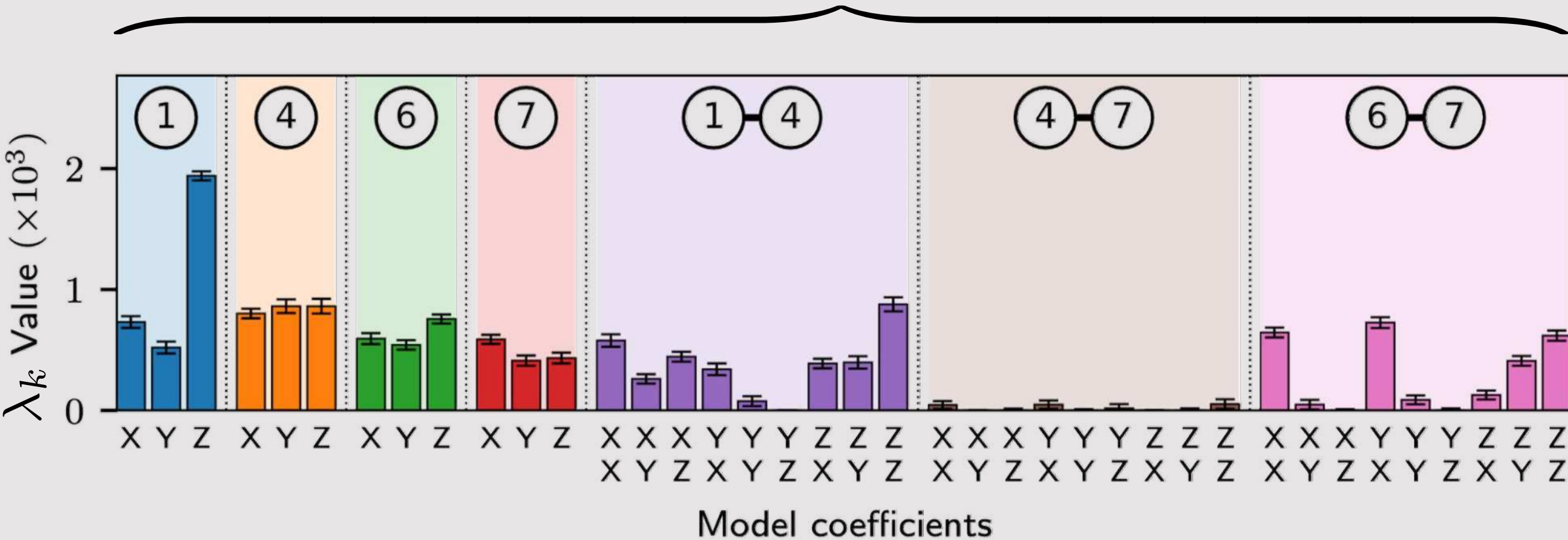
$$\begin{aligned}
 &= \exp(\gamma \hat{P}_a) \\
 &\quad \hookrightarrow \sum_{n=0}^{\infty} \frac{\gamma^n}{n!} (\hat{P}_a)^n \quad \text{note} \quad \begin{aligned} \hat{P}_a^{n-\text{even}} &= \hat{I} \\ \hat{P}_a^{n-\text{odd}} &= \hat{P}_a \end{aligned} \\
 &= \left(\sum_{\substack{n=0 \\ \text{even}}}^{\infty} \frac{\gamma^n}{n!} \right) \hat{I} + \left(\sum_{\substack{n=1 \\ \text{odd}}}^{\infty} \frac{\gamma^n}{n!} \right) \hat{P}_a \\
 &= \cosh(\gamma) \hat{I} + \sinh(\gamma) \hat{P}_a
 \end{aligned}$$

$$\Lambda[\hat{P}_a] = \cosh(\gamma_a) \hat{I} + \sinh(\gamma_a) \hat{P}_a$$

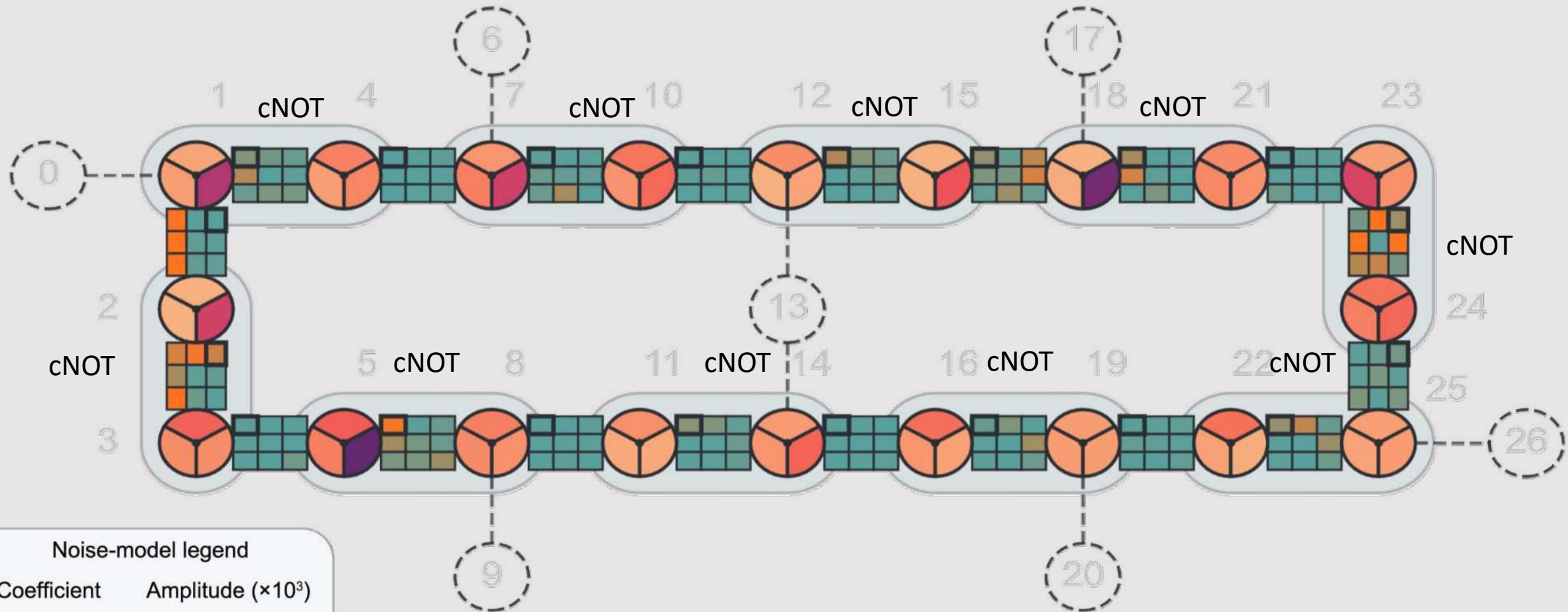
Sparse Lindblad tomogram



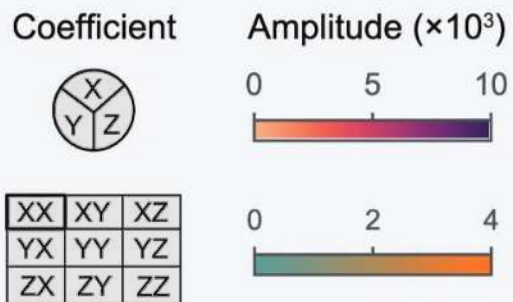
$$\mathcal{L}(\rho) = \sum_{k \in \mathcal{K}} \lambda_k \left(P_k \rho P_k^\dagger - \rho \right)$$



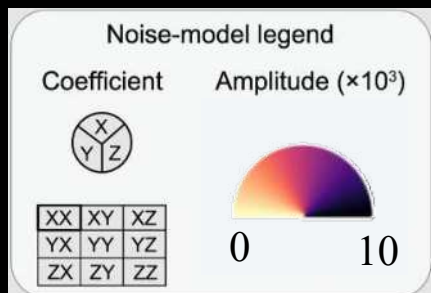
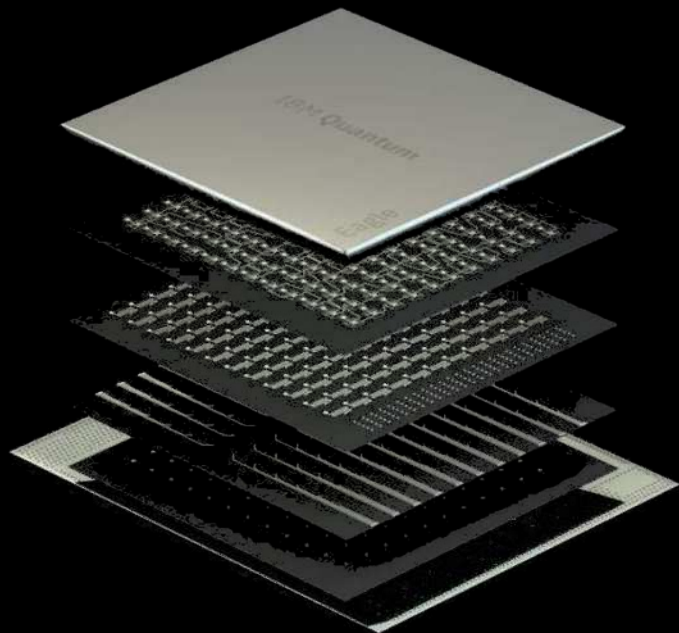
Noise tomogram for 20Q Ising-ring Trotter layer



Noise-model legend



Noise tomogram for 127Q Trotter layer



Same number of learning circuits as for 4Q

